AdAstra: A Web Platform for Astronomical Image Detection, Object Classification, and Knowledge Retrieval

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A project submitted to the Department of Computer Science and Engineering
in partial fulfillment of the requirements for the degree of
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Declaration

It is hereby declared that

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3. The project does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
4. We have acknowledged all main sources of help.

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Abstract

The current astronomical surveys produce huge image data sets that require automated analysis methods. Deep learning models are accurate at classification, but are not necessarily interpretable and do not perform well in other observational contexts. This work introduces AdAstra, an interactive web platform that integrates convolutional neural networks (CNNs) with retrieval-augmented generation (RAG) to deliver precise astronomical image classification and human-readable explanations based on astronomical knowledge.

The platform classifies images into five astronomical object types (constellation, galaxy, nebula, planet, star) with a test accuracy of around 83% on SpaceNet and 86% on Galaxy Zoo morphology. The CLIP embeddings are used to represent the semantic content of images, and then fused with a curated astronomy knowledge base using the Gemma language model to produce multi-level explanations (beginner, intermediate, advanced).

The web platform is accessible on all devices and is able to process multiple simultaneous requests with load-balanced server architecture. Users upload images, get classifications and confidence scores, and get explanations grounded in knowledge and referencing astronomy facts. The combination of classification, semantic embeddings, and knowledge retrieval fills important gaps in existing astronomical image analysis tools, providing both accuracy and interpretability.

Keywords: Astronomical image classification, convolutional neural networks (CNN), retrieval-augmented generation (RAG), explainable AI, CLIP embeddings, large language models, Galaxy Zoo, SpaceNet, knowledge-grounded explanation generation, web-based deep learning platform.

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Chapter 1

Introduction

1.1 Background

The paradigm of modern astronomy is changing because of the large-scale sky surveys and high throughput telescopes like the Sloan Digital Sky Survey (SDSS), Gaia, and new observatories like the Vera C. Rubin Observatory. The observational data generated by these projects is enormous, ranging from high resolution images to photometric measurements, spectra and multi wavelength data for billions of astronomical objects. This has led to the failure of manual inspection and rule-based classification methods to be reliable and scalable.

Astronomical surveys have also helped to shape the conception of scientific archives for the storage and coherent access to large scale observational data. With the rapid growth of the amount of data gathered from surveys, new repositories, accessible to the search engine and to computers, have taken on more significance in astronomical research. For instance, the ALMA Science Archive offers users cutting-edge query interfaces, extensive metadata, visualization tools, and programmatic access to ALMA observations in the archive, enabling the efficient retrieval and exploitation of the huge astronomical datasets (Stoehr et al., 2022). Similar is the case in Gaia Data Release 3 (Gaia DR3): catalogues of large size facilitate automated analysis and validation. Employing Gaia DR3 photometric data, Marton et al. (2023) tested variable young stellar object candidates for about 80,000 objects, showcasing the power of using large machine generated astronomical catalogues for automated classification, validation and large scale astrophysical studies.

This paper highlights the classification of astronomical images using two main datasets:

1. SpaceNet contains approximately 12,900 samples across five astronomical object classes (constellation, galaxy, nebula, planet, and star), integrating lower-to-higher resolution conversion with standard augmentations and diffusion-based synthetic sample generation to increase generalization performance.

2. Galaxy Zoo which focuses on galaxy morphology classification with 61,578 labelled images (spiral and elliptical). These datasets enabled vigorous and effective training and evaluation of image-classification models for wide ranged astronomical

object recognition and morphological analysis.

The initial astronomical object classification algorithms were based on manually crafted characteristics, including brightness, shape descriptors, surface brightness profiles and concentration indices, grouped using statistical or shallow machine learning models. White (1997) suggested a scalable separation of star-galaxy based on morphological parameters and normalisation techniques that were developed to minimise the plate-to-plate variation in photographic surveys. To enhance object differentiation, Zhang and Zhao (2004) applied support vector machines and neural networks to multi-wavelength catalogue features, such as colour ratios and flux measures. Likewise, Shamir (2009) introduced an automated galaxy morphology classification based on a large number of manually designed image descriptors (WND-CHARM). These techniques yielded interpretable results, but were very sensitive to noise, image resolution, and observing conditions, and were not very effective on faint or distant astronomical objects.

The recent developments in machine learning and deep learning have significantly transformed the classification of astronomical images from hand crafted features to end-to-end learning. The classification of star-galaxy-quasars and the recognition of galaxy morphology from large survey data sets, such as SDSS and Galaxy Zoo, have been shown to achieve state of the art results with convolutional neural networks (CNNs) (Martinazzo et al., 2022; Ma et al., 2022; Dalvi, 2022). Large natural-image datasets have made it possible to transfer learning to smaller astronomical datasets, and self-supervised learning techniques have been developed to learn representations from unlabelled multi meter survey images (Martinazzo et al., 2021). Recently, vision transformers (ViTs) and attention-enhanced CNNs have demonstrated better sensitivity to small, distant and morphologically ambiguous objects (Bhavanam et al., 2024; Pani, 2024).

Interpretability is as vital as classification accuracy in astronomical image recognition. A deep learning model may be used to make a prediction (class label) with a confidence value, but this does not provide an explanation for the prediction. To overcome this, the proposed task includes an explanation stage that uses retrieval to provide an explanation after classification. At this stage, it seeks out appropriate astronomical and morphological information related to the class of object to be predicted and combines it with the model output to produce a more human-readable and interpretable explanation of the model output.

1.2 Rational of the Study or Motivation

A number of significant gaps remain despite the positive outcomes of machine learning in the classification of astronomical objects. The motivation for this study arises from the following key observations:

- Low signal-to-noise ratio (SNR) of characters in both weak and ambiguous conditions: classification performance is significantly impacted when using

faint or compact objects or similar characters at low signal-to-noise ratios, which is typical of large astronomical surveys (Ma et al., 2022; Dalvi, 2022).

- Weak or ambiguous conditions: low signal to noise ratio (SNR) for faint/compact objects, and class imbalance often encountered in large astronomical surveys, significantly hamper the performance of classification.

- Diversity and domain shift of surveys: Large-scale surveys include a diversity of filter systems, resolution, sky coverage and observing conditions. Surveys trained on one survey do not necessarily generalise to another, and a model can perform very poorly due to domain shift (Martinazzo et al., 2021; Martinazzo et al., 2022).

- Weaknesses in robustness and confidence calibration: modern deep learning models are often overconfident, even when they are wrong, especially when there is distribution shift. The lack of calibration hinders the trustworthiness of automatic pipelines as scientific discovery tools (Guo et al., 2017).

- Limitations of environmental modelling: The effect of the environment or the atmosphere is well studied, for instance, in the context of seeing prediction, noise modelling, spatial clustering etc, but not directly incorporated into the astronomical object classification frameworks (Ferdosi et al., 2010; Gebhard et al., 2024).

- Scalability and interpretability: Next generation astronomical data will be too big to be analysed manually and will need scalable and interpretable automation. Predictions should not be just a label and a confidence score, they should also contain knowledge to support the prediction for the user to understand.

1.3 Problem Statement

The scale of astronomical images is increasing to the point where traditional feature based classification is starting to become impractical and manual inspection becomes increasingly difficult. Deep learning models can classify astronomical images with very high accuracy, but they typically only return a label and a level of confidence; and the prediction itself is not an explanation of the decision. This lack of interpretability is a significant issue in scientific applications, and knowing "why" a model gives a certain prediction is crucial for establishing the reliability of the prediction. One of the challenges is that the classes are imbalanced, the images are not of the same quality, and the images of astronomical objects, such as stars, planets, nebulae, galaxies, and constellations, are similar. Such influences may lead to predictions with high confidence that are not correct, thus diminishing confidence in the automated classification. Moreover, the accuracy of various CNN architectures may differ based on the datasets and object categories, which makes it challenging to decide which CNN is the most accurate for classification. Existing approaches, however, have to deal with the problem of accurate large scale classification, and they lack an automated way to compare the performance of models and to give knowledge based explanations of model predictions in an interpretable manner.

1.4 Objective

This study is carried out with the objectives of:

- To build an automatic astronomical image classification system with deep learning models based on convolutional neural networks (CNN).
- To categorize SpaceNet astronomical images by the five classes: constellation, nebula, galaxy, planet, and star.
- Assign galaxy images to a set of pre defined morphological classes based on a set of probability based labels generated from volunteer classifications in the Galaxy Zoo images.
- Train and evaluate the models: ResNet18, ResNet50, and EfficientNet-B0, on standard performance metrics such as accuracy, precision, recall, F1 score, and confusion matrices.
- To compare the evaluated CNN architectures in terms of prediction accuracy, size of the models and inference cost to identify an appropriate CNN model for deployment.
- To apply the chosen classification model as the basis for a retrieval augmented explanation stage.
- To enhance the predictions of CNN with a retrieval-augmented generation (RAG) layer to provide relevant astronomical or morphological knowledge that is retrieved and then used to describe the CNN prediction in natural language.
- To evaluate correct and incorrect prediction cases to determine strengths, weaknesses and typical error patterns for the classification and explanation aspects.

1.5 Methodology in Brief

This study employs two sets of astronomical images following a systematic experimental approach.

Dataset: The classification tasks are based on two datasets acquired independently from the raw NASA/SDSS archives, with SpaceNet containing five classes after dropping the classes of asteroid, black hole and comet due to high class imbalance.

Sample size: constellation, 1,552; galaxy, 3,984; nebula, 1,192; planet, 1,472; and star, 3,269 and Galaxy Zoo containing 61,578 labelled images split into two classes: elliptical (25,860) and spiral (35,718). In both data sets, there are some classes that are under-represented, but the number of images per class (hundreds to thousands per SpaceNet class, tens of thousands per Galaxy Zoo class) is sufficiently large that excluding or synthetically rebalancing these classes is feasible for supervised training. The resulting split in the image set into Galaxy Zoo train / Galaxy Zoo validation / Galaxy Zoo test (44,489 / 7,852 / 9,237) is consistent with the same imbalance ratio found in the full image set, as is the class imbalance ratio for the

training set (25,806 spiral vs. 18,683 elliptical), shown in Figure 1.1. All the images were converted to RGB, resized to 224x224, normalized, and converted to tensors, and were horizontally flipped and rotated during their training, before training and comparing ResNet18, ResNet50, and EfficientNet-B0.

```
============================================================
STEP 4: VALIDATE IMAGES & CREATE TRAIN/VAL/TEST SPLIT
============================================================

Validating images...
✓ Valid images: 61578

Train set: 44489
Val set:   7852
Test set:  9237

Train class counts:
label
spiral        25806
elliptical    18683
Name: count, dtype: int64
```

Figure 1.1: Train/Test/Split

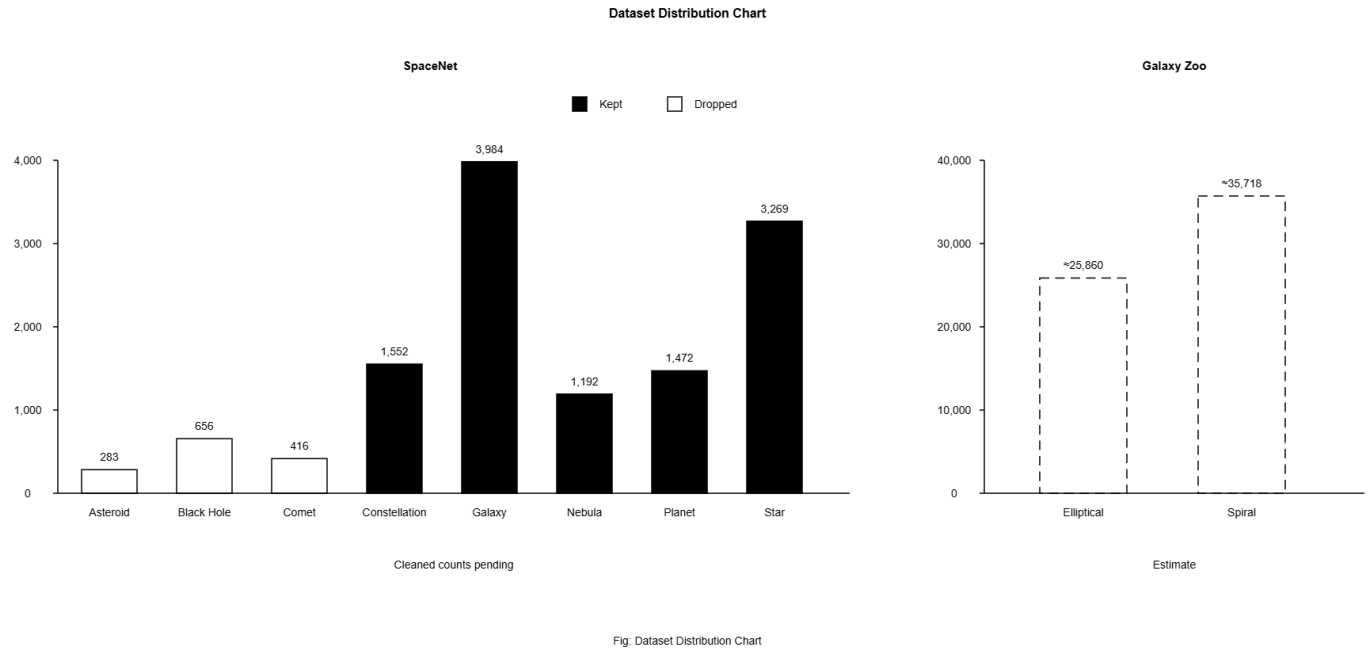


Figure 1.2: Dataset Distribution Chart

Model Architecture: The choice of model architecture started with the comparison between the Convolutional Neural Networks (CNNs) and the Vision Transformers (ViTs). CNNs are designed to learn spatial inductive biases, like locality or translation invariance, directly from moderately-sized and domain-specific image data sets, which allows them to efficiently learn features from these studies. In

contrast, ViTs take images as a sequence of patches, and use self-attention to learn global relationships. This enables the model of ViTs to capture long-range visual dependency well, but they generally require large-scale pre-training and more computational power to obtain a good performance. Given the large scale and specific type of the datasets analyzed in this paper and the computational and deployment needs of the proposed web-based platform, a CNN based approach was chosen as a more suitable architectural basis. This decision was driven by the data efficiency, low computation requirement, and the proven suitability for image classification under low to moderate data conditions (IDC) of CNNs.

RAG Architecture: The RAG stage follows classification of the images and generates an understandable explanation based on external astronomical expertise. After the CNN model makes a prediction and its confidence score, it uses the prediction to search a curated knowledge base of 113 astronomy related Wikipedia articles (6,222 chunks) using vector search with FAISS to obtain relevant information. The retrieved context is then fed to the large language model in the form of a LangChain LCEL pipeline, which covers all three levels: beginner, intermediate, and advanced explanations. The explanations provided are backed up by references to sources, and the final output is not just a class prediction, but contextual and knowledge-based information, which can be rendered for the user via the web interface.

1.6 Scopes and Challenges

This study focuses on the problems of astronomical image classification, comparative benchmarking of potential CNN architectures, and knowledge-aware explanation of the model's predictions. Within the proposed framework, environmental crowding, cross-survey standardisation between observatories and differences in telescope observing conditions are not discussed explicitly.

One of the key problems of astronomical image classification is the consistency and ambiguity of the ground truth labels. Large-scale human annotation is time-consuming, and astronomical labels may be uncertain, subjective, or heterogeneous. This uncertainty makes supervised learning challenging and encourages the application of probabilistic and uncertainty aware methods instead of assuming that all labels are the absolute truth (Walmsley et al., 2020).

Another challenge arises from class imbalance and gradual morphological variation. There are often unequal numbers of astronomical objects in each category and morphological features can grade into one another, with no sharp boundaries between classes. These properties can have a negative impact on conventional model training and evaluation techniques. Therefore, more general solutions to this issue have been explored, such as hierarchy aware and imbalance sensitive methods (Ma et al., 2022).

Another limitation is the domain shift between astronomical surveys. Models based on one survey can have lower performance on observations taken under other filters, other photometric characteristics, other instruments, or other imaging conditions. This means that reliable cross-survey generalisation is still a key challenge in astro-

nomical image analysis (Martinazzo et al., 2021).

Model robustness is also influenced by the presence of noise, uncertainty and image processing artefacts. Even if the image is not deliberately manipulated, acquisition noise, errors at the pixel level, and artefacts from the storage and pre-processing of the image can cause the predictions to be unstable. These limitations highlight the importance of robustness evaluation, uncertainty aware modelling, noise stress testing, and domain adaptation techniques (Csurka, 2017; Wang and Deng, 2018).

Last, prediction confidence does not necessarily equal prediction correctness. However, modern neural networks can yield a very high level of confidence even in cases where the prediction is wrong, especially when presented with data that is different from the distribution used during training. This calibration problem is particularly important for scientific image-analysis pipelines, and could be misinterpreted as a measure of the reliability of the model (Guo et al., 2017).

Chapter 2

Literature Review

2.1 Preliminaries

Astronomical image classification is a well established field of observational astronomy that classifies and categorizes astronomical objects and their morphology completely automatically, in layers of sky survey images, with ever expanding volume. Early classification methods usually adopted the paradigm of detect-measure-classify, where astronomical sources were first detected, descriptive photometric or morphological features were measured, and statistical or machine learning methods were used to classify them based on those features (White, 1997; Zhang and Zhao, 2004). While these methods enabled extensive automatic analysis automatically, they proved highly susceptible to the quality of extracted features and had a significant limit on their performance when acquiring low signal to noise or compact objects (White, 1997; Shamir, 2009).

Associated with the rapid development of astronomical surveys has also been the growth of astronomical databases, catalogues, and services available through networks. Establishing part of the infrastructure for modern data intensive astronomy, Andernach (1998) pointed out the increasing significance of digital astronomical resources for enabling professional astronomical research. With the growing size of observational data, automated computational techniques have been crucial to glean insight from these resources.

Classical feature based classification has a major drawback: the features that are computed from an astronomical source can be quite sensitive to the observational and instrumental conditions. Apparent brightness, morphology and source structure can be affected by the effects of atmospheric seeing, sky background, signal to noise ratio, instrumental point spread function, instrumental noise, and/or instrumental filter systems. These effects can be especially important for faint or distant sources, and can diminish the separation between visually similar classes of objects, e.g., stars and compact galaxies (White, 1997; Shamir, 2009). Thus, astronomical image classification should not be just considered a task of visual pattern recognition, but also the problem affected by physical and instrumental properties of astronomical observations.

Recent studies have been directed towards end-to-end deep learning, where feature

representations are learned directly from image data without being manually designed. Convolutional neural networks (CNNs) and more recently vision transformers (ViTs) can learn hierarchical visual representations directly from the pixel-level input, without relying on handcrafted descriptors (Martinazzo et al., 2021; Martinazzo et al., 2022; Ma et al., 2022; Dalvi, 2022).

Information from several catalogues and wavelengths can also be useful for astronomical classification. For example, Ita et al. (2010) used AKARI infrared all-sky survey data, 2MASS and Hipparcos catalogues to study stellar populations and to search for infrared-excess sources. These studies show that the astronomical classification can be enhanced by additional information from a context or catalogue other than a single image source.

Despite these developments, most of the current literature focuses on classification performance, representation learning, robustness or domain adaptation as individual research problems. Less attention has been paid to building a comprehensive astronomical image-analysis system that systematically compares different CNN architectures, chooses a model that is suitable for deployment requirements, and then links the selected CNN-based classifier to a knowledge-based explanation mechanism The central research question that the present study centers on is the separation between accurate classification and human-interpretable, astronomy-based, explanations.

2.2 Review of Existing Research

2.2.1 Classical Astronomical Image Classification

As the number of observations in astronomical surveys increased, manual classification of sources became increasingly impractical and automated techniques were developed to process very large observational datasets. Initial studies thus focused on classification pipelines explicitly designed on morphological and photometric measurements. In the context of star-galaxy separation, White (1997) showed that the morphological parameters and statistical normalisation could minimize the impact of plate-to-plate variation in early sky surveys. The study also revealed, however, that the separability of the morphology decreases with decreasing brightness of the sources, and hence the classification performance decreases.

Later studies extended these techniques to include data other than morphology. Using the flux ratios and colour information from cross-matched X-ray, optical, and infrared catalogues, Zhang and Zhao (2004) showed that machine learning techniques, such as support vector machines and neural networks, could be used to enhance the separation of stars, galaxies, and active galactic nuclei. However, these methods were still reliant on the completeness, consistency and reliability of manually defined catalogue features. Observational noise, systematic effects of the survey, and missing and uncertain measurements may therefore affect their performance.

Shamir (2009) proposed a framework called WND-CHARM for galaxy morphology classification based on a large set of handcrafted image descriptors that encode texture and structural properties. This work demonstrated scaleable automated galaxy

morphology classification, but revealed some practical constraints. The method was time consuming and depended on the image preprocessing and image quality, and relatively small variations in the preparation procedures could affect the classification results.

The classical astronomical classification thus set the stage for large-scale automated analysis and at the same time revealed the shortcomings of manually designed features. These techniques can provide relatively interpretable decision variables, but they often struggle with weak sources, changing observing conditions, and sensitivity to preprocessing (White, 1997; Shamir, 2009). These limitations spurred the use of deep learning methods that can learn representations directly from astronomical images.

However, the use of handcrafted features introduced another limitation.The more feature extraction was automated, the less transparent the internal decision processes of classification systems were. herefore, performance gains in prediction did not necessarily result in gains in interpretations.This compromise between a good representation learning and a limited explanatory power is particularly relevant to the problem of the present research.

2.2.2 Deep Learning Approaches

The transition from hand-designed astronomical features to learned representations directly from image data is a major step forward in deep learning. CNNs have been shown to perform well in various applications, including star-galaxy separation and galaxy morphology classification, in surveys like SDSS and datasets related to Galaxy Zoo (Martinazzo et al., 2022; Ma et al., 2022). For instance, Ma et al. (2022) tackled the typical challenges of astronomical classes with similar visual features and imbalanced data by proposing hierarchical classification and imbalance-aware learning strategies.

Transfer learning has also gained popularity due to the fact that astronomy-specific labelled datasets are often smaller than computer vision datasets. Dalvi (2022) showed that CNNs trained on ImageNet can be adapted to classify stars, galaxies, and quasars in SDSS images. This can minimize the need for astronomy-specific training data. However, transfer learning relies on the assumption that the representations learned from natural images are still applicable to astronomical images, which can be compromised if there are significant differences in the instrumentation, observational conditions, or survey design.

Another way to alleviate the need for labeled data is through self-supervised learning. Martinazzo et al. (2021) employed magnitude prediction as a pretext task to acquire useful representations from a vast amount of unlabeled astronomical images. The representations worked well in each survey, but not as well when moved between surveys that had different filter systems and photometric properties. The results show that learned astronomical representations can be sensitive to domain shift even in the absence of explicit class labels in the representation learning process.

The focus of recent research has shifted to the use of attention mechanisms, multimodal information fusion, and transformer architectures for modelling the global visual context and subtle astronomical structures (Bhavanam et al., 2024; Pani, 2024). They are approaches that enhance the representational ability of the traditional CNN, but their assessment has been largely focused on single surveys and/or controlled experimental datasets, and the issues of transferability and deployment remain open.

In total, the presented works show the ability of CNNs and any other deep learning algorithm to perform well on astronomical image classification. Many of the papers, however, focus on boosting classification performance or on learning representations, and not on the question of how competing CNN architectures should be compared when the deployment requirements for the resulting system are also of concern. A good performer in an experimental environment may not be the best architecture for a real-time, web-based system that has to make inferences, optimize parameters, and respond quickly.

The present study thus compares the ResNet18, ResNet50, and EfficientNet-B0 models on the astronomical classification tasks considered. The different architectures offer varying depths and computational compromises and allow the study to analyze predictive performance in conjunction with practical deployment concerns. The comparison is not intended to suggest that the biggest or deepest network is to be deployed, but should be used to select a backbone to be integrated into the total AdAstra pipeline.

This architecture-choice process only tackles one aspect of the overall research gap. Even if an efficient and accurate CNN is discovered, the resulting model still mostly just outputs a class label as well as a confidence score. These outputs tell us what the network predicts but do not give much information about what the predicted astronomical category means or if they are related to existing knowledge in astronomy. The second large component of the present study is motivated by this: retrieval-augmented explanation.

2.2.3 Explainability and Retrieval-Augmented Explanation

One of the key drawbacks of deep learning-based astronomical classification is interpretability. CNNs are capable of producing a predicted label and a confidence score, but the numbers do not directly convey a scientific context of the prediction. The problem is especially pertinent when the visual appearance of objects is confounded or when a model provides a high-confidence false alarm. Confidence value should not then be viewed as an explanation of the decision per se term.

One way of enhancing the usability of such outputs is through retrieval-augmented explanation. In this method, firstly, classification is accomplished and later on, pertinent external knowledge in association with the predicted class will be recalled. Such information can subsequently be modeled into an explanation of the predicted astronomical object or morphology in a natural-language manner. Critically, retrieval phase does not modify the initial prediction of the classifier but it enriches

the initial prediction with the contextual information which is aimed at enhancing the human perception of the system output.

Retrieval-augmented systems are highly dependent on the relevance of the information retrieved. Mortaheb et al. (2025) demonstrated that traditional similarity-based retrieval can yield irrelevant content and explored relevance-oriented re-ranking as a way to enhance the retrieved context and mitigate the risk of hallucinated answers. They conclude that retrieval quality is a key factor in reliable retrieval-augmented generation.

The value of evidence in multimodal retrieval systems has been explored as well. Luo et al. (2026) introduced information-theoretic approach where evidence retrieved by the model is scored based on the information provided for model output, which strikes the work that most semantically similar evidence does not necessarily provide the most useful evidence for downstream reasoning.

Likewise, Zhang et al. (2026) explored retrieval and reasoning over information with both textual and visual elements, highlighting the applicability of retrieval mechanisms beyond textual evidence.

There must also be a clear assessment of the reliability of retrieved evidence and generated responses in case it were a retrieval-augmented system. In 2025, Mortaheb et al. proposed RAG-Check, a method to assess separately the retrieval relevance and response correctness. Important distinction because when the context is only loosely related to the original task, the response can seem internally coherent despite the lack of connection with the original task.

These are studies that have shown that there has been significant improvement in retrieval quality, multimodal evidence selection, and RAG evaluation. The reviewed work, however, is mainly focused on general retrieval and multimodal RAG reliability, which is not the case with the integration of retrieval-augmented explanation and astronomical image classifiers. This distinction is key to the research question that AdAstra is tackling. Current astronomical classification studies are mainly concerned with the task of visual prediction, while RAG research mainly deals with information retrieval and language generation. In the reviewed literature, the joint research direction of the above two research directions in the context of a single astronomical classification workflow is found to be rather less explored.

The proposed system retrieves after CNN makes its prediction. The predicted astronomical category or galaxy morphology is then used to acquire relevant domain knowledge, and then a human-readable explanation is generated. The goal is not to have the language model classify the image, but to distinguish between prediction and explanation: the CNN is used for visual recognition, and the retrieval-augmented component provides the context of astronomical knowledge that matches the output of the image classifier.

The separation is of methodological importance as it also shows a limit to the explanation mechanism. The retrieval stage can retrieve correct information regarding

the wrong class even if the classifier does not predict the correct class, and explain the retrieved information in a fluent manner. The explanatory account that this generates is accordingly knowledge-grounded with respect to the prediction, but not, of itself, in verification of the veracity of the prediction. The limitation is taken into account in the evaluation of correct and incorrect cases in the present study.

Thus, it is not the intention of the present work to provide a solution to explainable AI in general.In contrast, the work investigates a pragmatic way of post-classification interpretability in which a CNN prediction is associated with retrieved astronomical knowledge and subsequently translated into a text.This directly helps the identified gap of automated classifications and domain-oriented interpretation.

2.2.4 Environmental and Observational Effects

In parallel with the development of image classification, another stream of astronomical research has focused on the environmental and observational conditions that impact astronomical measurements. For instance, Ye (2011) investigated the numerical prediction of atmospheric seeing and cloud cover, and found that the local atmospheric conditions can significantly affect the observational quality. This work has been mostly used for observation planning and telescope scheduling and not for downstream image-classification pipelines.

In planetary and atmospheric science, machine learning is being used to model physical properties in the presence of uncertainty in observations. Plaschzug et al. (2025) presented machine learning models based on three-dimensional global climate models for exoplanets, and Gebhard et al. (2024) developed noise-aware atmospheric retrieval models that explicitly model the uncertainty and incorporate verification procedures. While these approaches show the potential of modelling with noise and context, these methods are still conceptually distinct from traditional astronomical image-classification pipelines.

However, observational conditions have a significant impact on astronomical image characteristics. For faint and compact objects, the apparent morphology, brightness and detectability of an object can vary depending on seeing, sky background, point spread function and detector noise (White, 1997; Shamir, 2009). Ye (2011) also showed that some observational parameters can be explicitly measured and modelled instead of being considered as uncontrolled variation.

There is another literature that has defined relations between galaxy properties and physical environment. Studies of observations have demonstrated that the morphology, star-formation activity, clustering, and local density are systematically related among galaxies (Blanton et al. 2005; Kauffmann et al. 2004; Peng et al. 2010). Correlations between galaxy properties and environment have also been found at large scales over large redshift ranges (Scoville et al., 2013; Darvish et al., 2015; Zehavi et al., 2011).

Although these factors are scientifically relevant, environmental and observational modelling is not a main part of the present study. AdAstra's areas of interest

include image-level classification, comparison of CNN architectures, and retrieval-augmented explanation generation Therefore, environmental measurements, telescope conditions, and cross-observatory standardization are not the main focus of this study and are left as areas for future development.

This is significant for the correct identification of the research gap The present study does not focus on addressing environmental domain adaptation or observation-aware classification.Rather, these factors are used to pinpoint constraints that any deployed astronomical classifier may eventually have to overcome. The novelty of the present work is the integration of efficient image classification with domain grounded interpretability, with the recognition of the observational context as an open extension.

2.2.5 Robustness and Domain Shift

When astronomical classifiers are used outside of the exact data sets they were trained on, robustness is an important consideration. Astronomical surveys vary in instrumentation, point spread function, filter systems, survey depth, and observing conditions, which can lead to changes in data distribution that can impact model performance. The importance of domain shift was shown by Martinazzo et al. (2021) who showed that representations learned on one survey can transfer poorly to another survey with different photometric properties. Class imbalance and subtle morphological differences can also cause uneven errors across object categories within individual surveys, which can drive imbalance-aware learning strategies (Ma et al., 2022).

Model calibration is another important factor to consider.However, it has been shown that modern neural networks can be poorly calibrated, that is, high confidence does not translate into high predictive accuracy (Guo et al., 2017). This restriction is particularly important when a model is presented with data with a distribution that is not the same as the one it was trained on. Therefore, caution is advised when considering the confidence scores, especially for ambiguous or marginal astronomical objects.

These concerns set a valuable fence around the contribution that the current study can make. The AdAstra pipeline outputs the model's level of confidence and explanatory context; an explanation is not necessarily a guarantee that the prediction is accurate. Likewise, the present study does not attempt to make the claim of universal cross-survey robustness. Robustness, calibration and domain adaptation instead are important limitations with which the proposed system should be understood in terms of performance and interpretation.

This separation further emphasizes the primary research need to be addressed by this study: if user interaction with the astronomical AI system is to be meaningful, it must be accompanied by an element of confidence. The retrieval-augmented explanation has a different aim: to place the prediction in terms of understand astronomical context, whereas the calibration research is concerned with whether or not the confidence is a good reflection of the prediction likelihood. Both functions

are complementary, but must not be equated with each other.

2.2.6 Research Gap and Positioning of the Present Study

A literature review shows that significant advances have been made in astronomical image classification. Initial feature-based methods proved the feasibility of automated classification, but were very reliant on the manually defined photometric and morphological measurements (White, 1997; Zhang and Zhao, 2004; Shamir, 2009). End-to-end representation learning and performance on large astronomical datasets, using CNNs (Dalvi, 2022), transfer learning (Martinazzo et al., 2021; Martinazzo et al., 2022), self-supervised learning (Ma et al., 2022), attention mechanisms, and transformer-based models (Bhavanam et al., 2024; Pani, 2024) were then introduced, corresponding to ever more complex visual classification problems.

However, this progress has not yet been fully reflected in practical applications, particularly in moving from classification toward interpretable deployment.The majority of the astronomical image-classification literature has either been dedicated to enhanced predictive power, to learning representations, to addressing imbalance, or to enhanced robustness. By contrast, relatively little work has been devoted to a consistent 'one-stop' approach, where several CNN designs are consistently compared, a backbone is chosen for deployment based on both predictive and computational metrics, and the chosen classifier is then integrated with an external astronomical knowledge-base that can provide human-readable explanations.

A body of second research shows that galaxy properties strongly depend on the physical environment and observational context (Blanton et al., 2003, Kauffmann et al., 2004, Cooper et al., 2005, Peng et al., 2010, Zehavi et al., 2011, Scoville et al., 2013, and Darvish et al., 2015). However, this research is still largely independent of image-classification pipelines. The present study does not seek to directly address this environmental modelling gap; rather, it recognizes it as an important gap and direction for future extension.

In a similar vein, recent research has explored retrieval relevance (Mortaheb et al., 2025), evidence selection (Luo et al., 2026), multimodal reasoning (Zhang et al., 2026), and response verification (Zhang et al., 2026); however, the incorporation of these advancements into existing astronomical CNN classification pipelines, as reviewed here, is not common practice. The research opportunity thus is not just to increase the classification accuracy or to add another stand-alone language model. Instead, it is to explore the possibility of using visual classification and domain-grounded explanation as complementary steps of a single astronomical analysis system.

The present study aims at filling this gap up in the pipeline called AdAstra. ResNet18, ResNet50, EfficientNet-B0 are trained and tested in the classification tasks of SpaceNet and Galaxy Zoo. Common classification measures are used to compare the effectiveness of their performance, along with practical considerations pertinent for deployment. Then a suitable CNN backbone is used as a production classifier, in a

web based inference pipeline. After classification, the class and the classification confidence is then forwarded to a retrieval-augmented explanation stage, where appropriate astronomical knowledge is retrieved and provided to a language model to generate a source-based, readable description of the object/morphology of the predicted class.

The central idea is thus the fusion of three elements, which are usually discussed independently from each other: astronomical classification using CNN, selection of the architecture for deployment, and explanation of retrieved knowledge with the help of knowledge. The proposed system does not claim that the CNN can already provide clear interpretations of its learned features.In contrast, it offers a functional post-classification explanation layer which associates a prediction with a relevant astronomical knowledge, but leaves the difference between prediction and explanation intact.

2.3 Summary of Key Findings

The literature shows that the classification of astronomical images has evolved from handcrafted morphological and photometric descriptors to more and more complex deep learning techniques. CNNs, transfer learning, self-supervised learning, and attention-based architectures have greatly improved automated classification, particularly when working with individual astronomical datasets.There are however, many challenges with respect to class imbalance, domain shift, observational variability, robustness, uncertainty, and predictive calibration.

The review also reveals that the development of astronomical classification and advances in retrieval-augmented generation have largely been pursued independently. While astronomical deep learning research often focuses on improving classification accuracy, RAG research generally emphasizes retrieval quality, evidence relevance, and the reliability of generated responses.The literature reviewed thus shows that there is a need for a system that can link these capabilities without mixing them up.

Based on this, this study compares ResNet18, ResNet50, and EfficientNet-B0 to be deployed as backbones on the astronomical image-classification tasks of SpaceNet and Galaxy Zoo; selects the best backbone to deploy based on predictive and computational considerations; and adds a retrieval-augmented explanation layer to the selected backbone. This design helps the system to generate an explanation along with a model output and predict a class with related confidence score.

The resulting research gap can thus be summarized as follows: While significant advances have been made in the predictive performance of astronomical image classification research, relatively little work has been done to combine systematic CNN architecture comparison, deployment-friendly model selection, and retrieval-augmented, knowledge-grounded explanation into a single astronomical image-analysis platform.

AdAstra is designed to address this need. It is not just another astronomical classifier; it is not just appending a language model following prediction.Rather, the system investigates the potential of there being an efficient visual classifier that can be determined experimentally and then attached to "weeded" astronomical knowledge to produce efficient but more user-friendly outputs.

Table 2.1: Overview of Literature Synthesis

Reference	Theme	Research gist	Limitations	Relevance to present study
White (1997); Zhang & Zhao (2004); Shamir (2009)	Classical feature-based classification	Handcrafted morphological/photometric features with statistical/ML classifiers enable early large-scale classification.	Sensitive to noise, preprocessing and observing conditions; poor performance for faint/compact objects.	Establishes classical baselines and motivates end-to-end learning.
Dieleman et al. (2015); Domínguez Sánchez et al. (2018); Martinazzo et al. (2022); Ma et al. (2022); Dalvi (2022)	CNN-based morphology and object classification	CNNs learn hierarchical image features and outperform classical methods on SDSS/Galaxy Zoo tasks.	Require large labelled datasets; limited robustness evaluation across surveys.	Supports EfficientNet-B0 as the primary lightweight classification backbone.
Ackermann et al. (2018); Domínguez Sánchez et al. (2018); Martinazzo et al. (2021)	Transfer learning and self-supervised learning	Transfer/SSL reduces labelling cost and improves performance within surveys.	Performance degrades under cross-survey domain shift.	Motivates robustness and domain-shift analysis.
Bhavanam et al. (2024); Pani (2024)	Attention mechanisms and vision transformers	Attention and ViTs capture global context and improve classification of faint/ambiguous objects.	Data-hungry; often evaluated on small or survey-specific datasets.	Justifies evaluating ViTs alongside CNNs in future work.
Guo et al. (2017); Ciprijanović et al. (2021); Martinazzo et al. (2021)	Robustness, perturbations and calibration	Neural networks are overconfident and brittle under noise and distribution shift; domain adaptation improves robustness.	Robustness not routinely integrated into pipelines.	Supports calibration and reliability evaluation.

Table 2.1: Overview of Literature Synthesis (Continued)

Reference	Theme	Research gist	Limitations	Relevance to present study
Blanton et al. (2005); Kauffmann et al. (2004); Peng et al. (2010); Zehavi et al. (2011)	Environmental effects on galaxy properties	Galaxy colour, SFR and clustering correlate strongly with environment.	Environment definition depends on scale and estimator.	Motivates environment-aware analysis as future work.
Cooper et al. (2005); Scoville et al. (2013); Darvish et al. (2015); Aragón-Calvo et al. (2010)	Environment measurement techniques	Density, clustering and cosmic-web measures quantify environment across redshift and scale.	Affected by survey geometry and redshift uncertainty.	Guides selection of clustering/density methods for future work.
Ye (2011); Gebhard et al. (2024); Plaschzug et al. (2025)	Observational and noise-aware modelling	Observational conditions and noise are measurable and can be modelled.	Often disconnected from classification pipelines.	Inspires integrating observation-aware reliability checks.
Andernach (1998); Stoehr et al. (2022); Marton et al. (2023); Ita et al. (2010)	Astronomical data infrastructure and catalogues	Growth of online astronomical databases, science archives and large automatically classified catalogues.	Coverage and interface conventions vary widely across archives and catalogues.	Motivates future retrieval from live astronomical literature and catalogue services.
Mortaheb et al. (2025, x2); Luo et al. (2026); Zhang et al. (2026)	Retrieval quality and reliability in multimodal RAG	Re-ranking, utility-based evidence selection and mixed-modal retrieval improve RAG relevance and reduce hallucination.	Evaluated on general-domain multimodal benchmarks, not astronomy.	Positions the study's dense FAISS retrieval against more advanced alternatives for future work.

Chapter 3

Requirements, Impacts and Constraints

This chapter specifies the functional and non-functional requirements of the proposed AdAstra astronomical image classification and retrieval-augmented explanation system, and the societal, environmental, ethical, technical, economic and project-management issues that arise during the development of the system. The chapter emphasizes the needs and limitations of the final prototype, and does not duplicate the experimental methodology elsewhere in the report. The system is used as an experimental academic prototype throughout the system to show the integration of astronomical image classification, model comparison, knowledge retrieval, natural-language explanation and web-based interaction. This scope aligns with the original project specification which outlined the work in terms of classification, explanation, practical constraints and responsible prototype deployment.

3.1 Final Specifications and Requirements

The AdAstra system is composed of two main astronomical classification tasks, a common explanation and a web delivery pipeline. The first one is a classification of SpaceNet images into five classes of astronomical objects: constellation, galaxy, nebula, planet, and star. The second task is to classify galaxy images with probability-based labels from volunteer classifications, into a set of selected morphological categories, using the Galaxy Zoo dataset.

Three convolutional neural network architectures, ResNet18, ResNet50, and EfficientNet-B0, were trained and evaluated for the classification experiments. The aim of comparing these models was to assess their predictive performance and to identify the most suitable model architecture for deployment, given the need for a trade-off between classification accuracy and computational efficiency. The architecture that was selected for the production backbone was EfficientNet-B0 because of its good balance of predictive performance, parameter efficiency and inference cost among the architectures tested. This is especially relevant since the chosen model is integrated into a web-based inference pipeline that is interactive, and the complexity of the computation can impact responsiveness and scalability.

Images are validated and normalised when they are inputted before inference. The pipeline also handles 16-bit image handling with intensity stretching as necessary and offers a SHA256 identifier for input tracking. The image is classified by the deployed EfficientNet-B0 model (ONNX), and the predicted class, confidence and abstention decision are returned.

A CLIP-based gate (ViT-B/32, ONNX) is used to determine if the uploaded image is astronomical or not, before classification. The object classifier predicts "galaxy" and then a galaxy morphology classifier is run. This study does not cover the analysis of galaxy environment (cluster, group or field), which is considered future work.

After classification, AdAstra uses FAISS to perform a vector retrieval over a curated knowledge base of astronomy-related reference material, such as selected 113 Wikipedia articles to retrieve relevant astronomical knowledge. The retrieved information is then fed into a LangChain LCEL pipeline, which feeds it into the Gemma 4 language model, which generates knowledge-grounded explanations at beginner, intermediate, and advanced levels. Source references are kept to enable generated explanations to be linked to the retrieved material and not given as free-form answers.

The system provides a structured JSON response via a FastAPI backend and is presented on a web interface created with React. The platform also enables free-form, RAG-based interaction with the same knowledge base, enabling users to ask for more astronomy-related explanations while maintaining source attribution for the information provided.

3.1.1 Functional Requirements

The system should be able to perform the following functions:

- Allow astronomical image upload via web interface and validate before inference.
- Resized and transformed the input images to the size and shape of the deployed model (224×224 pixels).
- Classify SpaceNet images into the five types of astronomical objects: constellation, galaxy, nebula, planet, and star.
- Label images from Galaxy Zoo as one of the galaxy morphology classes that the models were trained and tested on.
- Return the predicted class and a confidence score.
- Use confidence and margin criteria where relevant, to ensure that predictions that are not certain do not need to be made as a full classification.
- After comparing ResNet18, ResNet50 and EfficientNet-B0, support inference with the selected production model.

- Use similarity search with FAISS to retrieve astronomical or morphological knowledge from the curated domain knowledge base.
- Use LangChain LCEL and Gemma 4 to create natural language explanations from the retrieved information.
- Include explanation levels for beginner, intermediate and advanced users.
- Maintain source information for generated explanations to enable users to backtrack to the knowledge used to support the response.
- Provide model predictions, confidence information, retrieved evidence, explanation content, warnings, provenance information, and relevant processing information, as appropriate, in the backend response.
- Develop a user interface using React to upload images, display classification results and read explanations generated.
- Support RAG-based conversational interaction over the same astronomy knowledge base.
- The system was also designed to record the significant results of the experiments such as the weight values of the trained model, the class mapping by the model, the training history, the evaluation results etc. in order to use the model again and reproduce the results if necessary. This was found during the initial prototyping stage as the requirement of retaining model weights, class names, training histories, and result files was deemed to be a critical feature for the entire system.

3.1.2 Non-Functional Requirements

The system should meet a number of quality and deployment requirements as well as being correct.

The first is computational efficiency. The model was developed using limited GPU resources and the final model should be lightweight enough to be used in practical inference in a web-based environment. The original development requirements also included efficient execution with limited notebook resources and repeatability via saved models and outputs.

Reproducibility is the second requirement. Previous experiments should be able to be reviewed or repeated without having to retrain the model after each development session by saving model weights, class definitions, training records, and evaluation outputs.

The third is interpretability. AdAstra is not intended to merely give a black and white classification label. The retrieval-augmented explanation component is thus added to provide relevant astronomical context related to the model prediction. This explanation should be conceptually different from prediction verification, however, in that the RAG component explains the predicted category, but does not validate

the correctness of the classifier.

The system should also be maintainable and modular. The classification, knowledge retrieval, explanation generation, API handling and front-end presentation are implemented as separate units, and the redesign of any unit does not necessitate the redesign of the entire system.

The architecture should also be elegant. If an optional component fails or is not present, an appropriate warning should be generated, if possible, but the entire inference pipeline should not fail. This is especially important for validation modules like CLIP gate.

Finally, the deployed system should be capable of processing unseen images repeatedly without retraining or changing model parameters as specified in the original problem statement, in which test images are to be processed by already-trained models.

3.2 Societal Impact

The project is educational and scientific because modern astronomical surveys generate image collections at a scale where manual inspection becomes increasingly challenging. Automated classification can be useful for repetitive image analysis tasks, and can also be used as an easy-to-understand example of the power of modern deep learning techniques applied to real astronomical images. The original project specification also stated that students, researchers and astronomy learners could be potential beneficiaries of such a system.

The explanation layer brings this value into the realm of automation. A label such as galaxy or nebula, without any context, provides little educational value. AdAstra can offer a structured explanation by accessing relevant astronomical information and making the classification results more user-friendly for those who may not be astronomically knowledgeable.

The system should be regarded, however, as an assistive and educational prototype rather than as an authoritative scientific decision making tool. It is only useful when the underlying classifier is accurate and the retrieved information is relevant. Thus, explanations produced should be used to help the user understand, not replace expert analysis.

3.3 Environmental Impact

The environmental impact of the project is primarily associated with the computational resources needed to train the CNN and for inference. The training was done with a limited number of GPUs and experiments were done for a limited time because of practical notebook and time constraints. Model outputs and weights

were not saved to repeat unnecessary experiments. This development practice was already incorporated into the original resource-management strategy.

The EfficientNet-B0 selection also offers computational efficiency during deployment. A light weight production model consumes less computational resources for repeated inference than larger candidate architectures. However, in this study, energy use was not measured directly and no quantitative claim is made regarding the actual reduction in electricity use or environmental impact.

If more AdAstra were deployed, the frequency of requests, the server infrastructure, the execution hardware of the models, and the workload of the knowledge generation would affect energy consumption. The most appropriate environmental solution would then be to not retrain unnecessarily, to retain reusable model artefacts and to use computationally efficient inference components with still acceptable predictive performance.

3.4 Ethical Issues

The primary ethical concern in AdAstra is reliability and communicating uncertainty. A neural network may be able to classify a sample with a fairly high level of confidence, but still misclassify it. When such a prediction is fed into a retrieval-augmented generation system, the RAG system can generate correct information about the mispredicted class and generate a fluent and convincing explanation. This behavior was identified as a major limitation in the prototype testing.

This is a key difference in how to understand AdAstra's work. The retrieval component is not intended to verify or correct the CNN prediction, but rather to explain and put the prediction in context. Thus, a factually correct explanation for the predicted class may be given even though the original classification is wrong.

The generated explanations should thus be understood as contextual information linked to the output of the classifiers and not as evidence of the scientific correctness of the prediction. The system should not be offered as an independent scientific body and the interface and documentation should clearly state this. This was also highlighted in the original requirements, which required the explanation to be related to the predicted category, but not to prove the correctness of the prediction.

The ethical concerns of personal privacy are relatively small, since the data sets used in this work consist of astronomical images and not personal information. However, the provenance of the datasets should be respected. Galaxy Zoo classifications are partly based on contributions from volunteers, and astronomical images, model architectures, source data and literature need to be appropriately acknowledged.

3.5 Standards

Throughout the development of the models, standard machine learning and computer vision techniques were used. Images were resized to 224 × 224 pixels, normalised and split into suitable training and evaluation sets. Random flipping and rotation was used to enhance generalisation during training. The accuracy, precision, recall, F1-score, confusion matrix, and normalised confusion matrix were used to assess the performance of the models, enabling the comparison of the architectures beyond the overall accuracy.

Reproducibility was not a choice of reporting step, but it was a requirement of the system. We saved model weights, class mappings, training histories, result graphs and other artefacts of an experiment to enable it to be verified or used later.Public repositories were used as convenient entry points to datasets, and the initial academic research and data sources were viewed as the scientific sources of knowledge on the work.

The application is separated into model-serving and user-interface components for deployment. The backend sends prediction and explanation results in structured JSON format via FastAPI, and the React frontend displays the results to the user. This separation makes the system more maintainable and enables the classification and knowledge-retrieval parts to change without having to redesign the entire web interface. Input validation is also considered as technical standard of the prototype. The supported image formats are verified prior to inference, and the preprocessing is done uniformly according to the needs of the trained classifier.

3.6 Project Management Plan

The project was designed in a step-by-step fashion starting from the selection and preparation of the data sets. The projects were chosen based on their astronomical classification goals, and the necessary files were retrieved from publicly available repositories, SpaceNet and Galaxy Zoo. Images were resized, normalised, filtered from the dataset and augmented for training. The three candidate architectures (ResNet18, ResNet50, and EfficientNet-B0) were then trained and tested on the chosen classification tasks. After each experimental stage, model weights, class names, histories, confusion matrices, performance reports and training curves were saved.

After comparing the different options, EfficientNet-B0 was chosen as the backbone for deployment. The selected classifier was then combined with the knowledge-retrieval and explanation parts. The final RAG architecture involves using FAISS to fetch relevant information from the curated astronomy knowledge base and then sending the retrieved context to LangChain LCEL for the generation of explanations by Gemma 4.

The final inference pipeline was then exposed via FastAPI and linked to the React UI. The test cases were both successful and unsuccessful prediction cases, allowing

the behaviour of the explanation layer to be studied in both correct and incorrect CNN outputs. This is the same development sequence as in the original one, where the chosen EfficientNet-B0 model was linked to the explanation stage and tested with representative prediction examples.

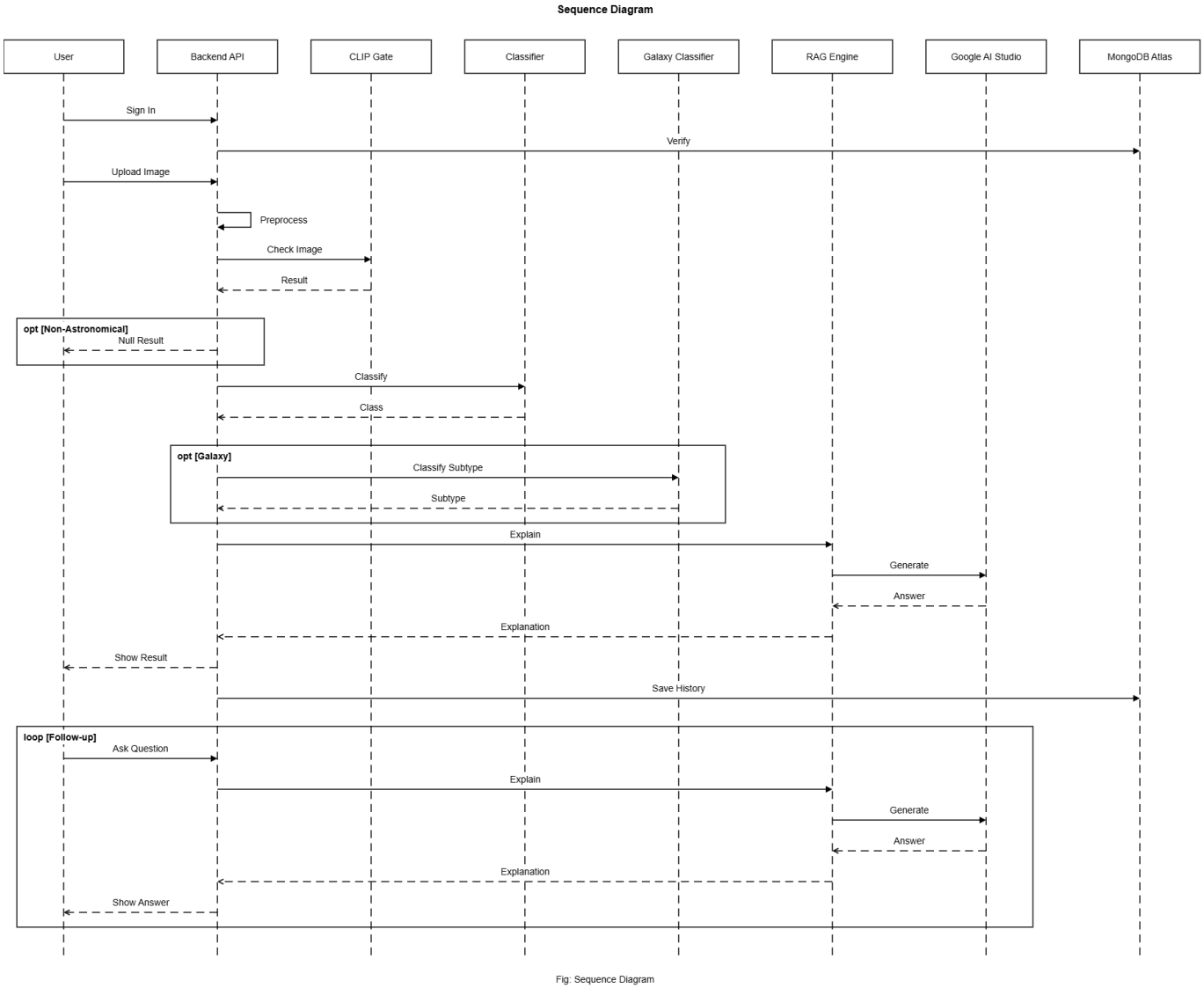


Figure 3.1: Sequence Diagram

3.7 Risk Management

Table 3.1: Summarises the main risks identified during the project and how each one was handled.

Risk	**Mitigation**
Kaggle session resets could wipe saved outputs.	All weights, class names, training histories and graphs were written to the Kaggle working/output directory during each session, so results could be downloaded, recovered and reused.
The models could overfit the training data.	Validation accuracy and loss curves were monitored, and data augmentation with random flips and rotations was applied.
Class imbalance in the datasets could distort performance figures.	Evaluation relied on macro F1-score, weighted F1-score and confusion matrices rather than accuracy alone.
Visually similar classes within the five retained SpaceNet classes could be confused with one another.	Confusion matrices were analysed to identify these patterns, and the limitations are reported openly.
RAG could generate a fluent explanation for a wrong prediction.	The report states clearly that RAG is an explanation layer, not a correction layer, and includes an incorrect example to demonstrate this limitation.
GPU time could run out.	Compact CNN models were employed and each run was capped at 30 epochs, with EfficientNet-B0 being chosen in part to reduce compute costs during the run.
Public dataset repositories might not fully document the scientific background of the data.	Original research papers and data-release papers served as the academic references, with the repositories used only for practical access to the files.

3.8 Economic Analysis

Development of the prototype per se was relatively inexpensive, as the astronomical data were all readily available, and the project's primary software and institutionally accessible/notebook-computing resources were open source and free. During the experimental phase, no purchase of special equipment was required and no special model training software had to be purchased. These were then the main resources used and developer time and run-time of computational resources.

The added cost of a production deployment would be added recurring expenditure. While there are many potential costs, they could involve hosting of the web-server, storage for the astronomy knowledge base, CPU or GPU inference resources, and any external services needed to accommodate larger numbers of concurrent users. The true cost of operation will depend greatly on traffic volume, inference hardware, and the occurrence of the requests to generate classifications and explanations.

In the context of this, it is relevant the selection of EfficientNet-B0, since a classifer that is computationally efficient may require fewer resources for repeated inference

as compared to larger candidate architectures. If the system would be scaled up, decreases in manual intervention (in repetitive image-classification tasks) may also contribute to any potential economic value, though no direct cuts in bill are claimed or computed in the prototype under consideration within this paper.

Overall, AdAstra illustrates the feasibility of building a functioning prototype system equipped with astronomical image classification and knowledge-based explanation on reasonably easily available computational means. However, its reliability still depends on the underlying classifier, and the reliability of the classification component does not rise with the addition of the RAG component, but the contribution of the RAG component is to better interpret the classification meaning in the context. Future operational deployment conclusions must thus be made with this caveat in mind.

Chapter 4

Proposed Methodology

4.1 Design Process or Methodology Overview

This chapter explains the design and construction of that platform and reports the behaviour of the image classifiers deployed in its heart. The project went through five phases: Data preparation, Classifier comparison, Recognition pipeline, Retrieval-augmented explanation, Web implementation.

Data preparation: Two public datasets were gathered and preprocessed: SpaceNet (astronomical object recognition) and Galaxy Zoo (galaxy morphology). They are both resized to one input format and partitioned into training, validation and test sets.

Classifier comparison: Three different convolutional neural networks (CNNs) pre-trained on the ImageNet data sets: ResNet18, ResNet50 and EfficientNet-B0 were retrained on each data set using the same protocol and evaluated on held-out test images.

Recognition pipeline: The selected classifiers were ranked in 2 stages. The first classifies the image as one of five object types (constellation, galaxy, nebula, planet or star). If the answer is Galaxy, the second phase is to be assigned a morphology, either elliptical or spiral. Images that are not astronomical are rejected by a CLIP-based gate in front of both stages and a confidence rule flags predictions that are uncertain.

Retrieval-augmented explanation: A language model is promped with passages from a curated astronomy knowledge base, and the passages are retrieved using the predicted class and the language model writes an explanation, constrained only to the retrieved passages, which is then checked before being displayed.

Web implementation: The entire pipeline was enveloped in a web application that included user accounts, saved analysis history, an astronomy question-and-answer assistant, and a sky explorer and a Moon-phase page.

The key design is unifying the classifier with the explanation stage. The retrieval stage just restates the CNN's decision. Since the explanation is produced based

on the class predicted, an incorrect prediction would be explained just as well as a correct one. Much of the system design is thus geared toward preventing the explanation stage from acting on inputs it is not supposed to believe: CLIP gate, uncertainty rule, limitation of the LM to passages retrieved, citation check.

4.2 Design (Model) Specification

4.2.1 Candidate Classifiers

We used three different CNN architectures to compare how model depth and efficiency affect astronomical image classification. A lightweight baseline of ResNet18 (18 layers) was added. To answer the question of whether deeper nets can separate visually similar classes, ResNet50 is used here (number of layers = 50, bottleneck residual blocks). The goal of EfficientNet-B0 is to achieve competitive accuracy with significantly fewer parameters, a requirement for an application running on a free-tier hosting, by scaling down the depth, width and input resolution of the network. The final classification layer was replaced to ensure the number of classes of this layer was equal to the number of classes of the task (5 classes for SpaceNet and 2 classes for Galaxy Zoo) and all the layers were fine-tuned using the ImageNet weights.

4.2.2 EfficientNet-B0 Architecture

EfficientNet-B0 is used as an object classifier in AdAstra. It was constructed from torchvision implementation of the ImageNet-pretrained weights. The 224×224 input is passed through a stem convolution, seven stages of mobile inverted-bottleneck (MBConv) blocks and a head convolution, and reduced to a 1,280-dimensional vector through global average pooling. The original linear layer of size 1000 was changed to a linear layer whose size matches the number of classes for the task (5 for SpaceNet, 2 for Galaxy Zoo), keeping the same original dropout rate of 0.2. The outputs are then converted to the class scores displayed as percentages on the website by a softmax (Model Architecture Diagram).

4.2.3 Two-Stage Recognition and Confidence Handling

There are two steps to recognition. The object classifier generates five probabilities for each class. If the top class is galaxy, a second classifier, trained on Galaxy Zoo labels, classifies the galaxy as elliptical or spiral, and the result page reports both. If the highest probability is less than 0.55, or if the difference between the two highest probabilities is less than 0.15, then the prediction is considered uncertain. If the result is uncertain, it is not blocked. It is displayed with a warning, the runner-up class is identified and the runner-up is also retrieved so that the explanation can cover both scenarios.

4.2.4 CLIP Gate

The image encoder is a CLIP model (ViT-B/32) exported to ONNX and quantised to 8 bits, with a size of about 88.6 MB. It is used to distinguish astronomical images from ordinary photographs. The pipeline does not force the image into one of the five astronomical classes, but stops and returns no classification if the image is judged not to be astronomical. This is required because a closed set classifier always returns one of its classes, regardless of the input.

4.2.5 Retrieval-Augmented Explanation

The explanation stage is part of a retrieval-augmented generation (RAG) pipeline (RAG Flow Diagram). The knowledge base was created offline by cleaning 113 Wikipedia articles on astronomical objects of all non-content sections (references, external links, etc.) and dividing them into 6,222 chunks of approximately 800 characters that overlap. Every chunk is embedded using the all-MiniLM-L6-v2 sentence-embedding model (384 dimensions, L2-normalised), and stored in a FAISS inner-product index, so that similarity-searching is an exact cosine-similarity search. The index is built once, offline, from Wikipedia; at run-time only the index itself is queried, not Wikipedia.

The predicted class (or the runner up class if the class is not known) is used as the query at run-time. The results are returned up to four passages with a similarity score of at least 0.25. They are given in a single prompt along with the class, the confidence and question status, and the language model, Gemma 4 (gemma-4-26b-a4b-it, accessed via Google AI Studio), which is told to only use facts from the passages, to cite them as [n], and to state when the passages don't have a fact. The user selects a reading level and one model call is made at each level, as needed, with the reading level being beginner (no jargon, one analogy, two paragraphs), intermediate (terms defined, three paragraphs) or advanced (technical, at most four paragraphs). Generation is set to 0.3 and minimal reasoning, as the default setting was discovered to use more than 4,000 hidden tokens and time out. Once generated, citation numbers are eliminated if they do not correspond with one of the passages retrieved, and a warning is added when the text has no source. This retrieval and generation chain also responds to free-form questions in the assistant, based on the previous two utterances of the conversation.

4.2.6 System Architecture

The system is deployed as a web client and a backend API on Vercel (System Architecture diagram). The pipeline is executed in the following order: preprocessing, CLIP gate, object classifier, galaxy classifier, and calling the RAG engine for explanations, all of which are executed in the backend API. The RAG engine is based on the FAISS index and calls Google AI Studio for generation. When the service is cold, the Hugging Face Hub is contacted to download the embedding model. Authentication and history services use MongoDB Atlas to store user accounts and

saved analyses. The design is now no longer fully stateless, as each main endpoint needs to be signed in, and each user's analysis history is kept on the server.

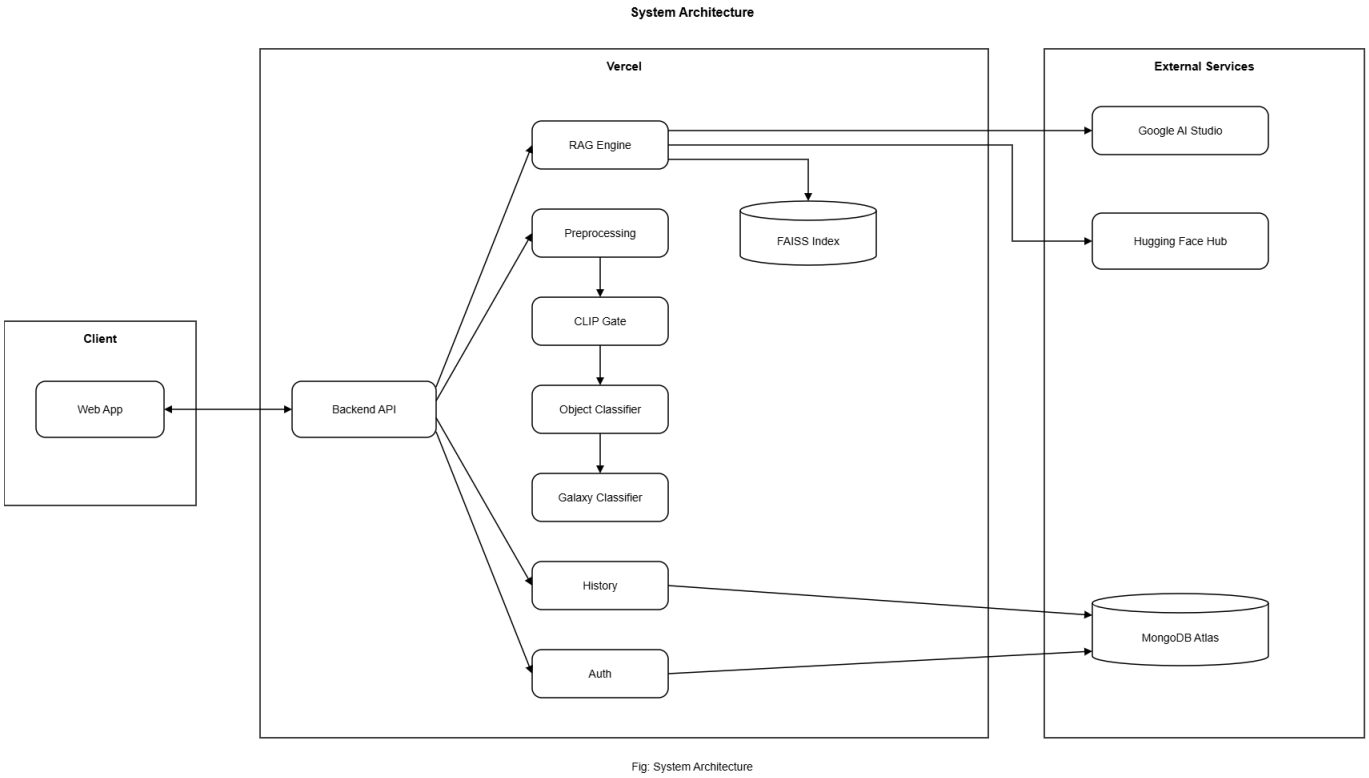


Figure 4.1: System Architecture

4.3 Data Collection Preprocessing

4.3.1 Data Collection

The datasets used were two public datasets. SpaceNet, downloaded from Kaggle, consists of approximately 12,900 images organised into eight folders (asteroid, black hole, comet, constellation, galaxy, nebula, planet and star) where the name of the folder is used as the label. Galaxy Zoo consists of 61,578 labelled galaxy images. The images are classified as elliptical or spiral based on the percentage of votes from the volunteer classifiers for the morphological question "Is the galaxy smooth or has spiral arms?". All 61,578 images were used.

4.3.2 Data Cleaning and Class Selection

SpaceNet was prepared in two steps, while no images were removed from Galaxy Zoo. First, images that were not opened or could not be converted to RGB were eliminated. This only impacted SpaceNet, where 1,376 galaxy images were not readable, leaving the class with 2,608 images. Secondly, the number of classes in the SpaceNet label space was decreased from eight to five. The asteroid, black hole and comet classes were eliminated due to their small size compared to the other classes and the most confusing visual appearance, and the remaining five classes (constellation, galaxy, nebula, planet and star) were kept. The final SpaceNet set after

cleaning is 10,093 images. Galaxy Zoo was used without filtering, with all 61,578 images kept. The resulting class distributions are shown in Dataset Distribution Chart.

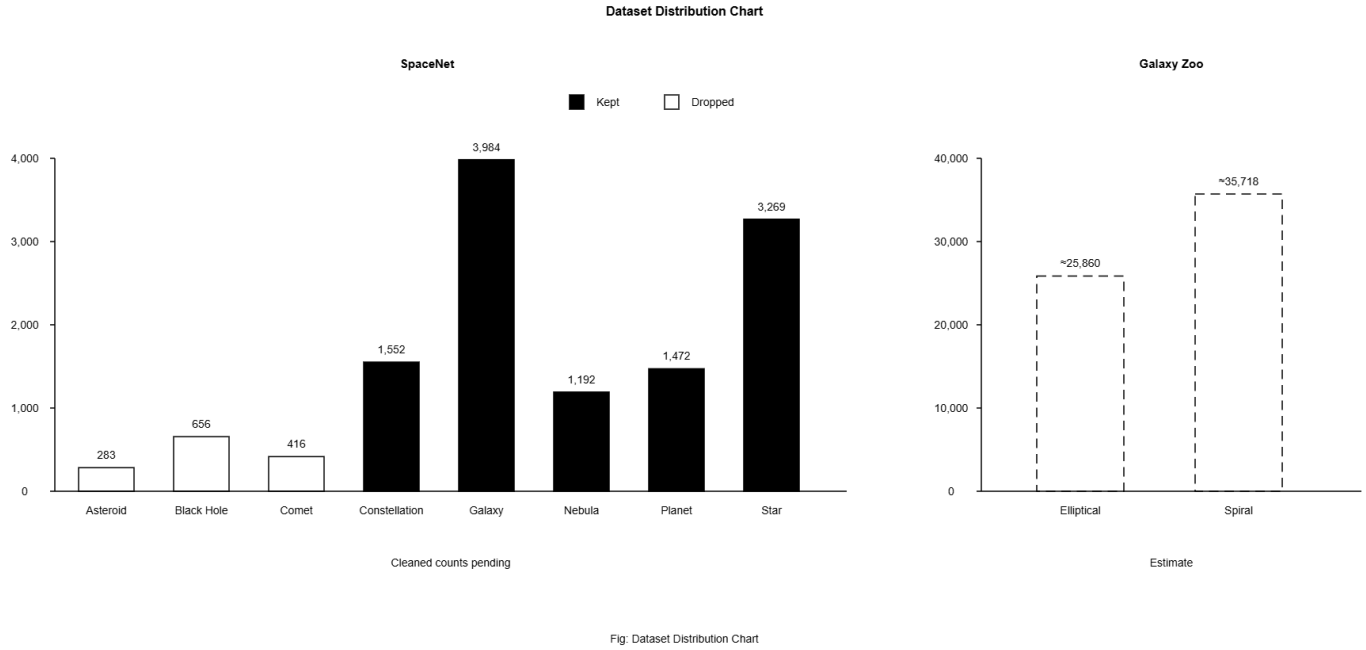


Figure 4.2: Dataset Distribution Chart

4.3.3 Data Partitioning

The cleaned data were divided in to disjoint partitions once, and same partitions were used across all three architectures (Table 4.1). Galaxy Zoo has been split into 44,489 training, 7,852 validation and 9,237 test images. SpaceNet was split into 8,074 training images and 2,019 test images, earmarked for both training and monitoring.

Table 4.1: Summary of the prepared datasets

Property	**SpaceNet**	**Galaxy Zoo**
Task	Object classification (5 classes)	Morphology (elliptical / spiral)
Raw images	About 12,900 (8 classes)	61,578
Valid images used	10,093	61,578
Training / validation / test	8,074 / – / 2,019	44,489 / 7,852 / 9,237
Training class counts	constellation 1,241; galaxy 2,086; nebula 954; planet 1,178; star 2,615	spiral 25,806; elliptical 18,683
Test class counts	constellation 311; galaxy 522; nebula 238; planet 294; star 654	spiral 5,358; elliptical 3,879
Input size	$224 \times 224 \times 3$	$224 \times 224 \times 3$

4.3.4 Transformation and Augmentation

All images were converted to RGB, resized to 224 × 224 pixels, scaled to the range 0 to 1 and normalised with the ImageNet mean (0.485, 0.456, 0.406) and standard deviation (0.229, 0.224, 0.225). Random horizontal flips, vertical flips and small rotations were applied during training to accentuate the diversity of astronomical images, as there is no preferred orientation. No augmentation of validation and test images.

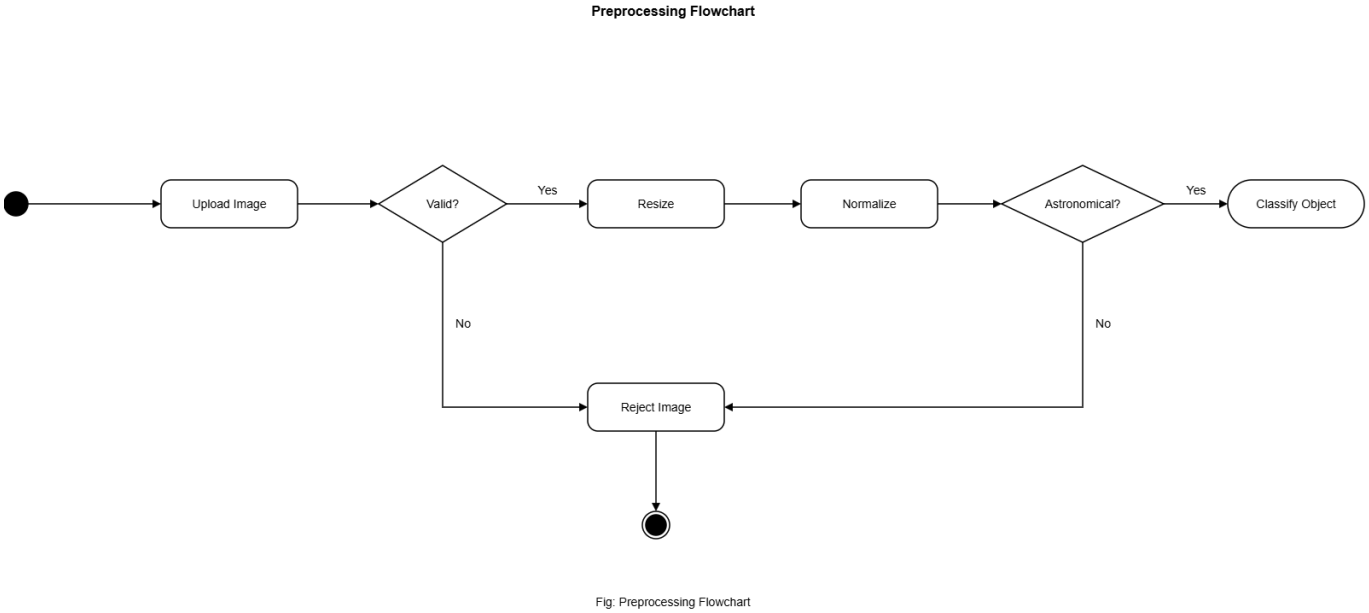


Figure 4.3: Preprocessing Flowchart

4.4 Implementation of Selected Design

4.4.1 Training Configuration

In all six models (three architectures on two datasets) the same training protocol (Table 4.2) was used. Optimiser: AdamW, Loss: cross entropy, Arithmetic: mixed precision on GPU during training. The learning rate was set to 1×10^{-4}, with no scheduled decay (Table 4.2). It was found that the classes were imbalanced, and the classes of the minority needed the full training budget, so early stopping was removed and the patience was set to the number of epochs. For the final evaluation on the test set, the checkpoint with the highest accuracy on the monitoring split was reloaded: the validation set for Galaxy Zoo, and for SpaceNet the test set itself, since SpaceNet has no separate validation partition (Table 4.1). The accuracy of training and testing, macro F1 score, macro recall, per class precision, recall and F1 score per class and confusion matrix were recorded for each model.

Table 4.2: Training configuration

Setting	**Value**
Initialisation	ImageNet-pretrained weights, all layers fine-tuned
Optimiser	AdamW, learning rate 1×10^{-4}, weight decay 1×10^{-4}
Loss function	Cross-entropy
Batch size	32 (SpaceNet), 64 (Galaxy Zoo)
Epochs	30, patience 30
Precision	Automatic mixed precision
Augmentation	Random horizontal and vertical flips, small rotations (training images only)
Checkpoint selection	Highest validation accuracy

4.4.2 SpaceNet Results

Table 4.3 gives the SpaceNet results. The three architectures reached test accuracies between 83.0% and 83.7%, and their training accuracy was between 93.7% and 95.0%.

Table 4.3: SpaceNet test results (2,019 test images)

Model	**Training accuracy**	**Test accuracy**	**Macro F1**	**Macro recall**
ResNet18	93.70%	83.01%	83.22%	83.44%
ResNet50	95.00%	83.51%	83.48%	83.26%
EfficientNet-B0	94.20%	83.66%	83.62%	83.42%

ResNet18

Of the 2,019 test images, ResNet18 was able to correctly classify 1,676. The accuracy of the training set improved over the 30 epochs, and the accuracy of the validation set kept at a low 80% range, while the loss of the validation set fluctuated with the decreasing loss of the training set. Constellation was the worst of the 3, with 234/311 correct, Constellation is weakest (234/311), planet strongest (267/294), and galaxy in between (438/522). The greatest confusions were with the prediction of stars as constellation (46), nebula (33) or galaxy (28), and galaxies predicted as star (44) (Figure 4.4).

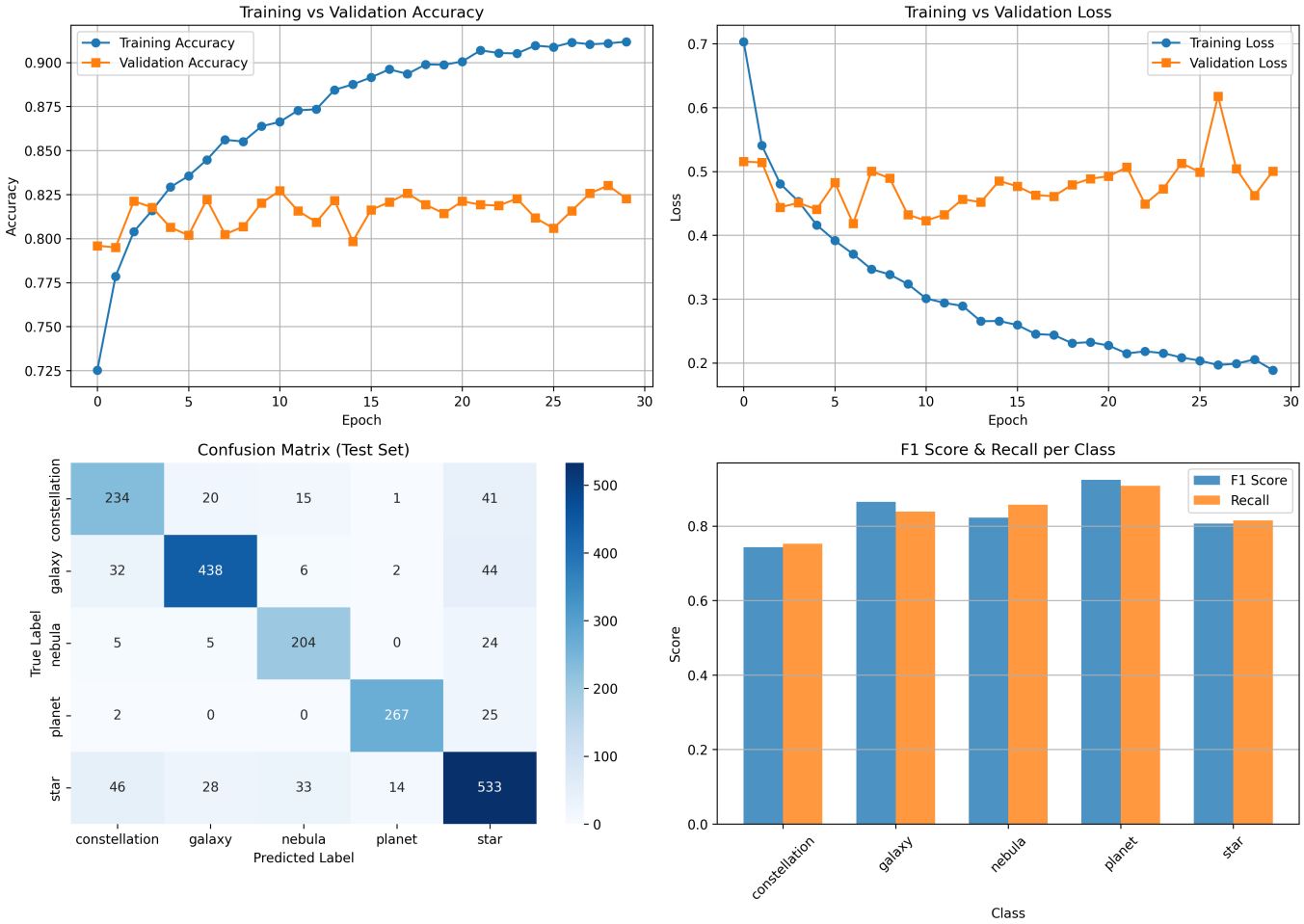


Figure 4.4: SpaceNet ResNet18 training and validation accuracy, loss, test confusion matrix and per-class F1-score and recall

ResNet50

ResNet50 had the highest accuracy on its training set (95.00%) but did not perform this well on the test set with 1686 correctly classified images. The monitoring accuracy stabilized around 82–84% while the monitoring loss increased in later epochs. Recall was about 84% for star (551 out of 654), 91% for planet, and about 73% for constellation. For stars (Figure 4.5), the errors were concentrated on the star class: 30 stars were predicted as constellation and 29 as galaxy.

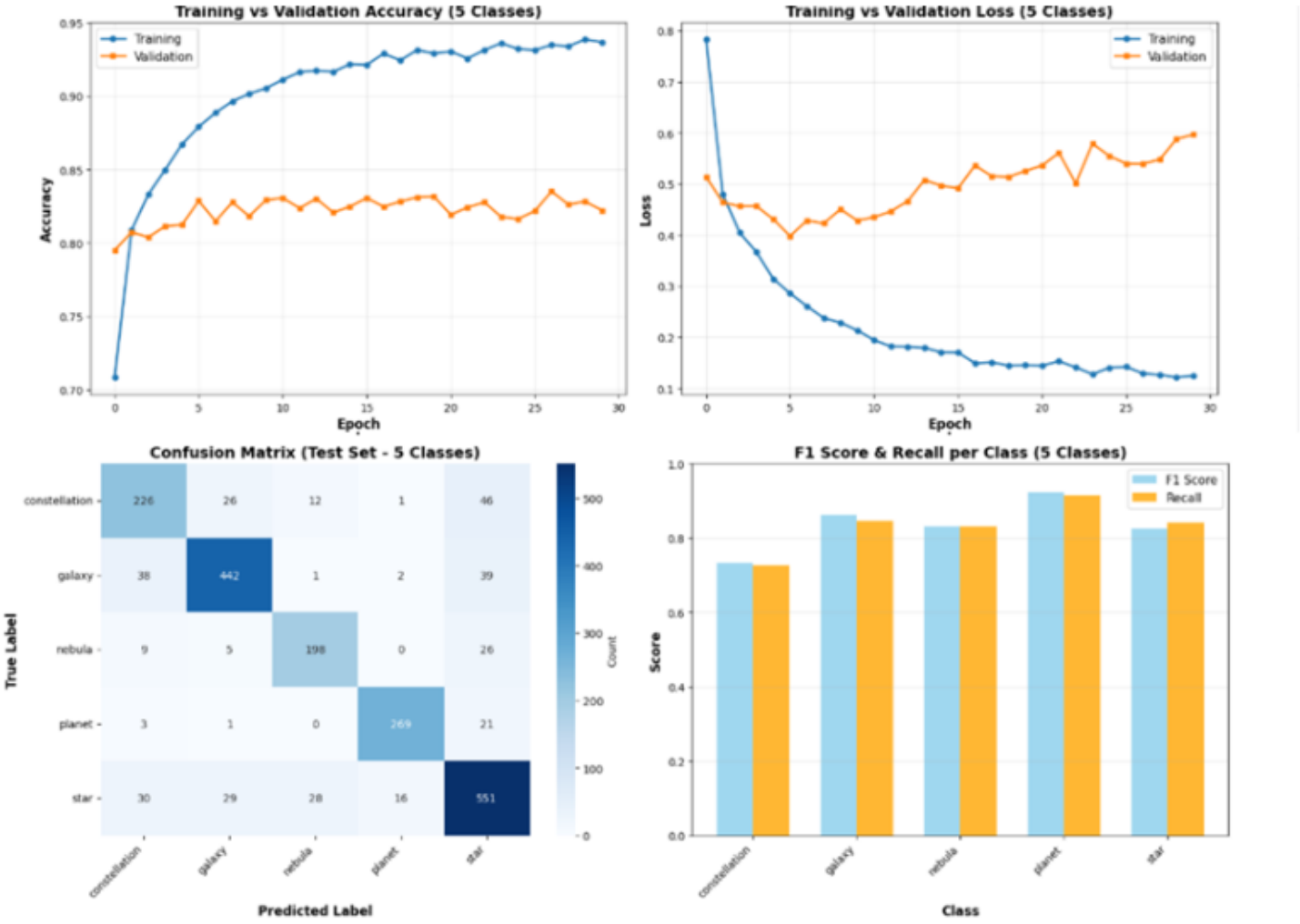


Figure 4.5: SpaceNet ResNet50 training and validation accuracy, loss, test confusion matrix and per-class F1-score and recall

EfficientNet-B0

The best SpaceNet result was obtained for EfficientNet-B0 that correctly classified 1,689 test images and attained a macro F1-score of 83.62% and accuracy score of 83.66%. Planet was the strongest class (271 of 294 correct, F1 = 0.92) and constellation the weakest (229 of 311, F1 = 0.73). The most common errors were 44 stars classified as constellation, 44 constellations classified as star and 34 galaxies classified as star (Figure 4.6 and Table 4.4).

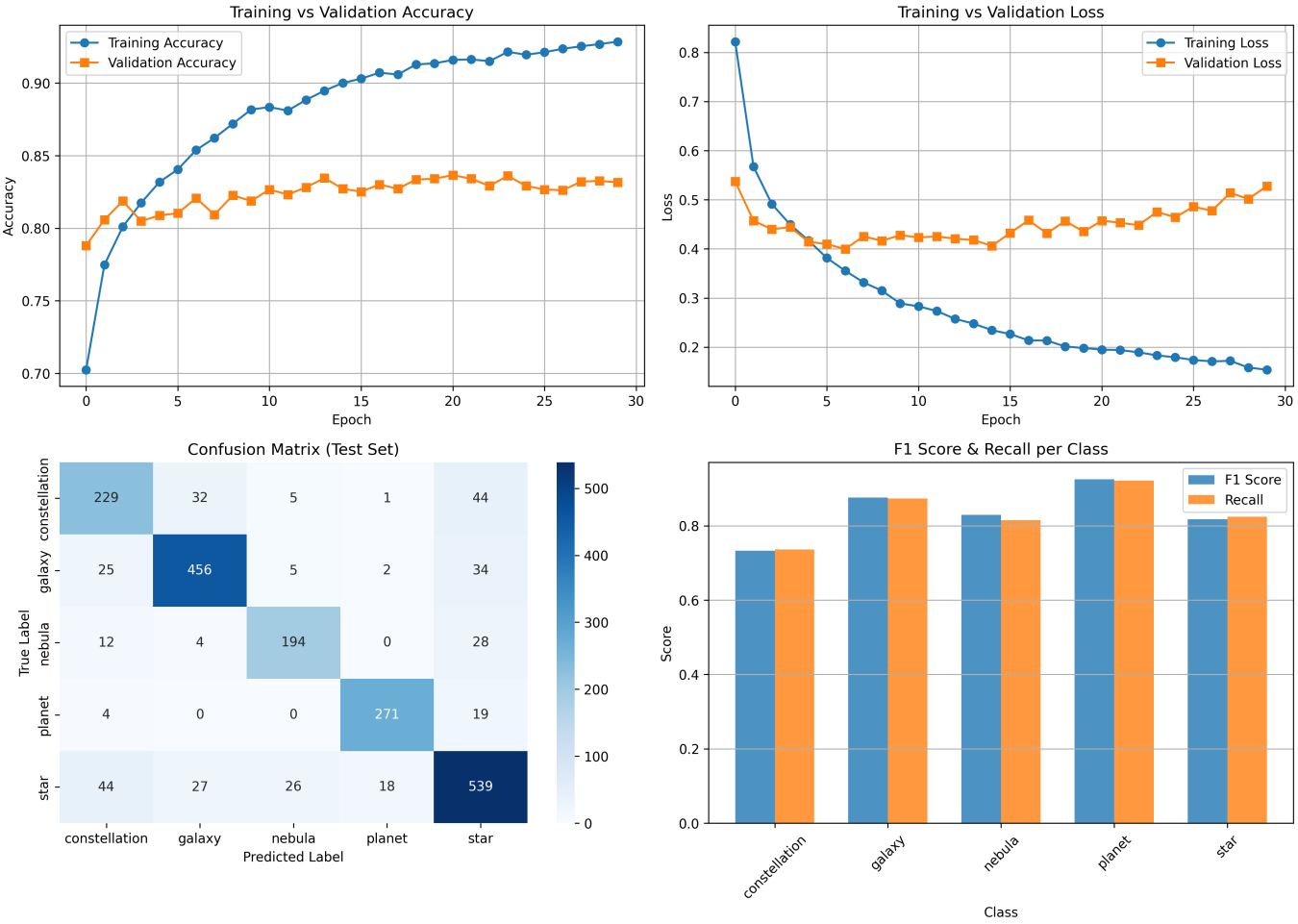


Figure 4.6: SpaceNet EfficientNet-B0 training and validation accuracy, loss, test confusion matrix and per-class F1-score and recall

Table 4.4: SpaceNet per-class F1-score on the test set (from the classification reports)

Class	**Support**	**ResNet18 F1**	**ResNet50 F1**	**EfficientNet-B0 F1**
Constellation	311	0.74	0.73	0.73
Galaxy	522	0.88	0.88	0.88
Nebula	238	0.82	0.83	0.83
Planet	294	0.92	0.92	0.92
Star	654	0.81	0.82	0.82

4.4.3 Galaxy Zoo Results

All three Galaxy Zoo models were evaluated on the same 9,237 test images, of which 5,358 (58.0%) are spiral. Because of this imbalance, macro-averaged scores are reported next to accuracy (Table 4.5).

Table 4.5: Galaxy Zoo test results (9,237 test images)

Model	Training accuracy	Test accuracy	Macro F1	Macro recall	Best val. accuracy
ResNet18	89.07%	86.86%	86.60%	86.88%	86.78%
ResNet50	91.62%	85.95%	85.72%	86.12%	86.23%
EfficientNet-B0	87.12%	86.25%	85.85%	85.75%	86.04%

ResNet18

ResNet18 achieved the highest Galaxy Zoo test accuracy, classifying 8,023 of 9,237 images correctly. Training took 162.1 minutes on a Google Colab GPU. Its errors were evenly spread: 502 elliptical galaxies were labelled spiral and 712 spirals were labelled elliptical, giving recalls of about 87% for both classes (Figure 4.7).

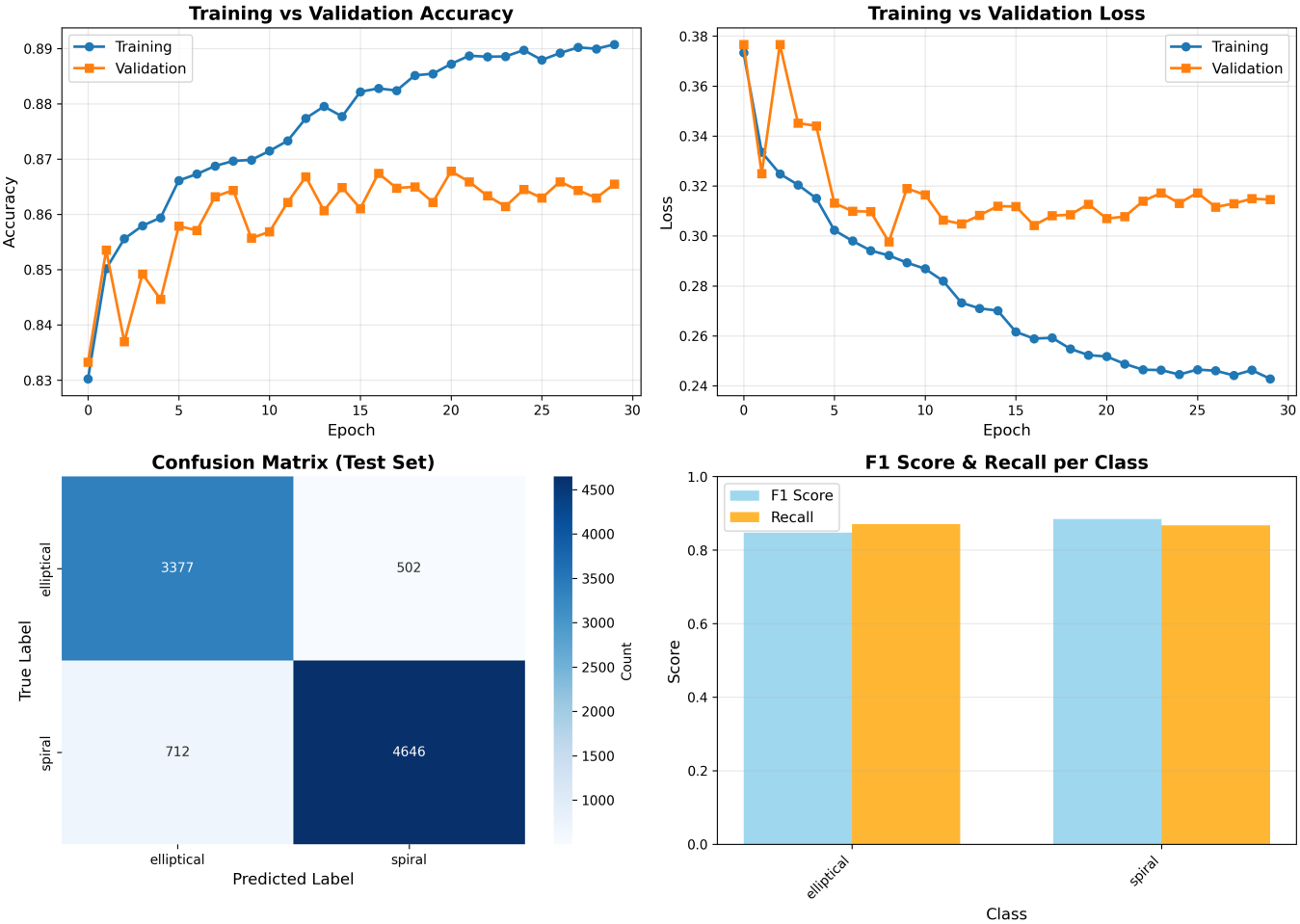


Figure 4.7: Galaxy Zoo ResNet18 training and validation accuracy, loss, test confusion matrix and per-class F1-score and recall

ResNet50

ResNet50 classified 7,939 test images correctly and needed 99.1 minutes of training on a Kaggle GPU. Its training accuracy (91.62%) was 5.7 points above its test accuracy, and its validation loss, after an early minimum, rose steadily. It recovered

3,382 of 3,879 elliptical galaxies (recall about 87%) but only 4,557 of 5,358 spirals (about 85%); spiral predictions were more precise (0.90) than elliptical ones (0.81) (Figure 4.8).

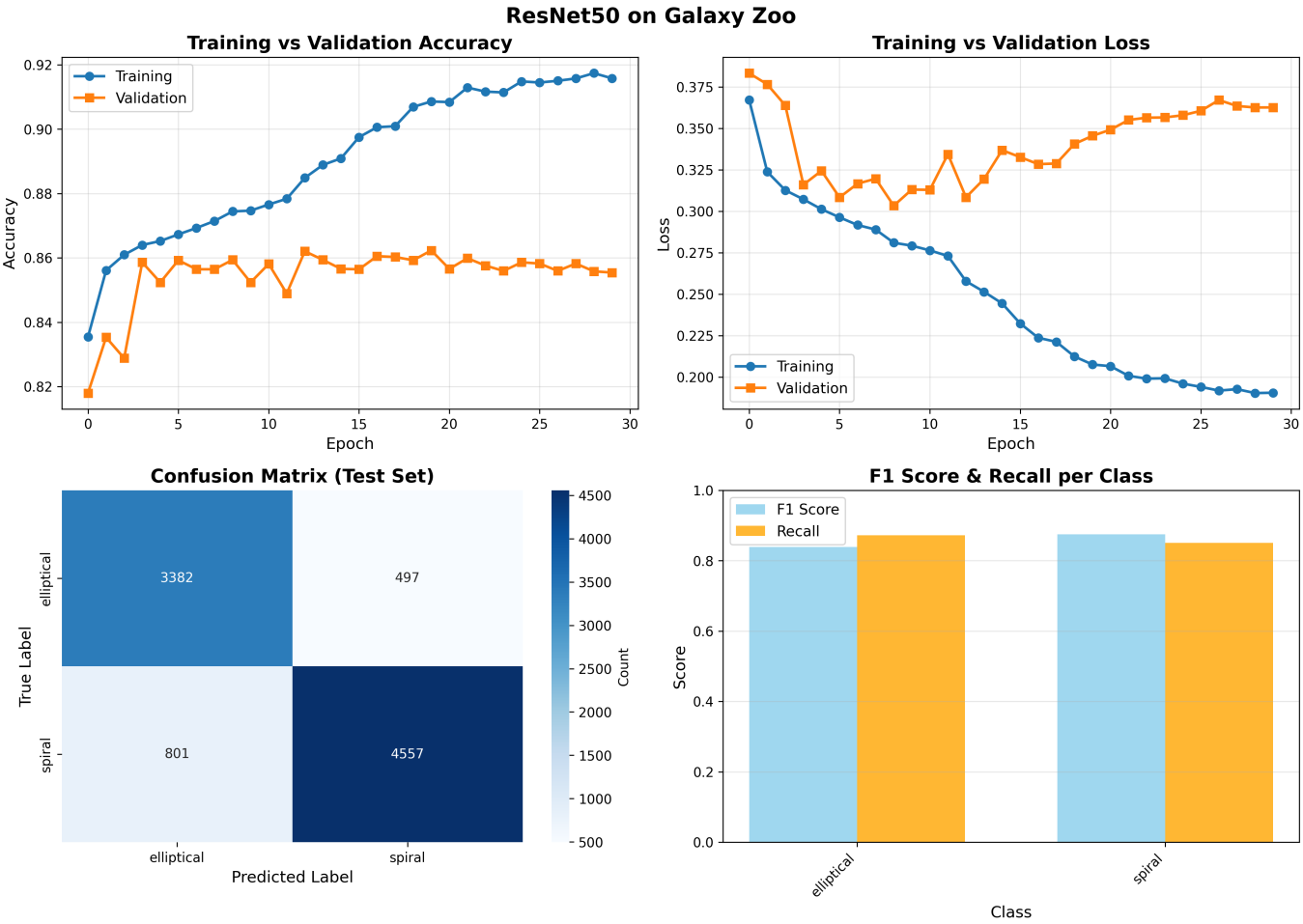


Figure 4.8: Galaxy Zoo ResNet50 training and validation accuracy, loss, test confusion matrix and per-class F1-score and recall

EfficientNet-B0

EfficientNet-B0 classified 7,967 test images correctly. Its best validation accuracy of 86.04% was reached at epoch 7 and the training took 169.8 minutes. The saved checkpoint is about 16 MB, against about 45 MB for ResNet18. It recovered 4,763 of 5,358 spiral galaxies (about 89%) but only 3,204 of 3,879 elliptical galaxies (about 83%), so it leans towards the majority class more than the two ResNets do (Figure 4.9 and Table 4.6).

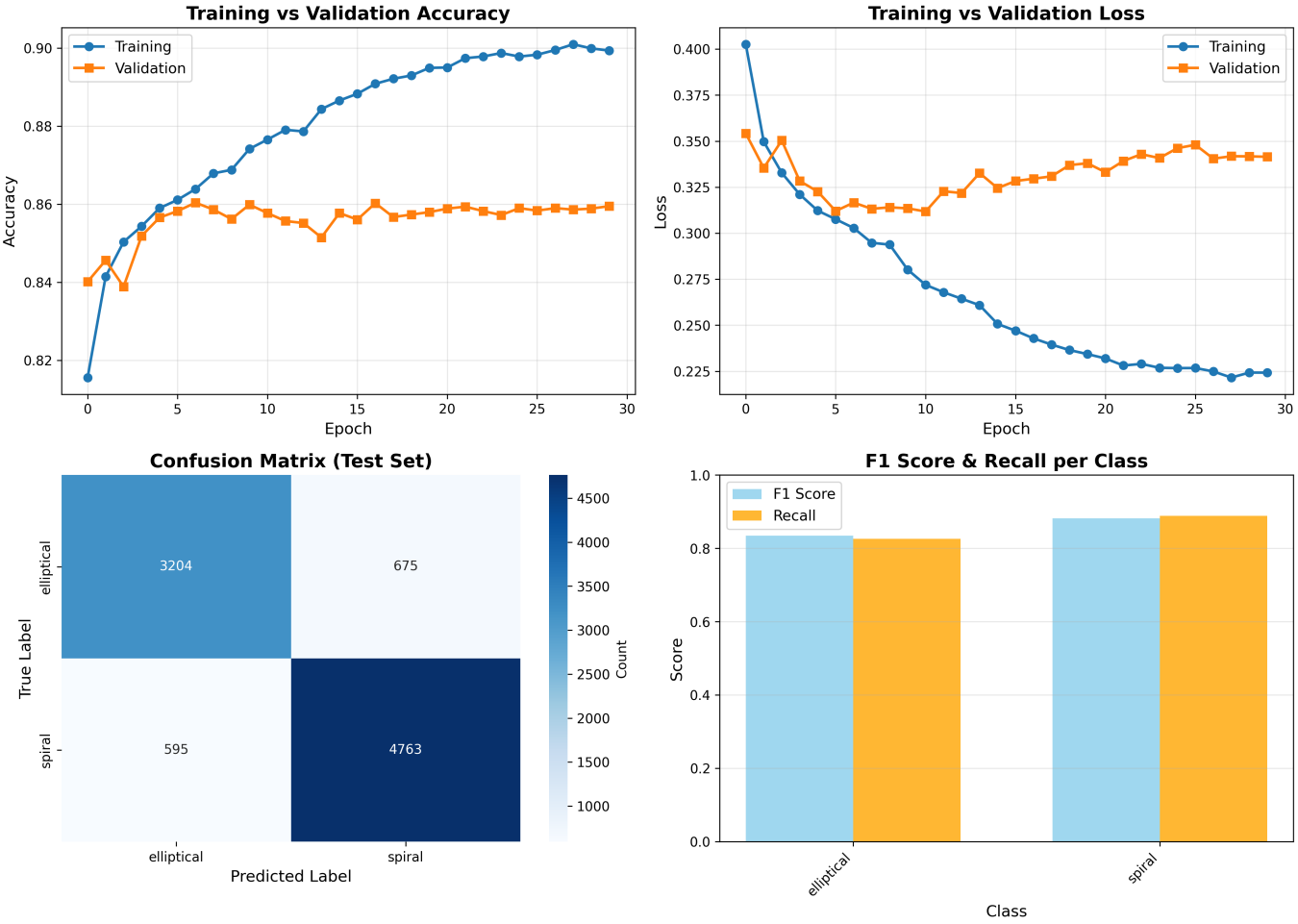


Figure 4.9: Galaxy Zoo EfficientNet-B0 training and validation accuracy, loss, test confusion matrix and per-class F1-score and recalll

Table 4.6: Galaxy Zoo per-class classification results on the test set

Model	**Class**	**Precision**	**Recall**	**F1-score**
ResNet18	Elliptical	0.83	0.87	0.85
ResNet18	Spiral	0.90	0.87	0.88
ResNet50	Elliptical	0.81	0.87	0.84
ResNet50	Spiral	0.90	0.85	0.88
EfficientNet-B0	Elliptical	0.84	0.83	0.83
EfficientNet-B0	Spiral	0.88	0.89	0.88

On both tasks the architectures differ by less than one percentage point in test accuracy. The statistical analysis of these differences and the reasons for selecting EfficientNet-B0 for the deployed system are given in Chapter 5.

4.5 Web Application Implementation

The trained models, the CLIP gate and the RAG stage are provided via a web application. This section demonstrates the implemented interface by using screen-

shots of a signed-in session. The examples were selected to represent the primary behaviours of the system: a confident result with morphology and cited explanation, an uncertain result, a refused input, and the supporting pages. They are examples of the design of Section 4.2 as it would look to the user, not a systematic test set.

4.5.1 Landing Page and Dashboard

The public landing page tells what the platform does and provides account creation and sign-in. It shows an example panel that displays a result: the most likely class, the confidence, the scores of the alternatives, and an explanation with statements that are cited with numbered references to Wikipedia articles. Accounts must have an email address and must be verified by email, and must be able to reset their password, delete their account, and opt-in to update emails.

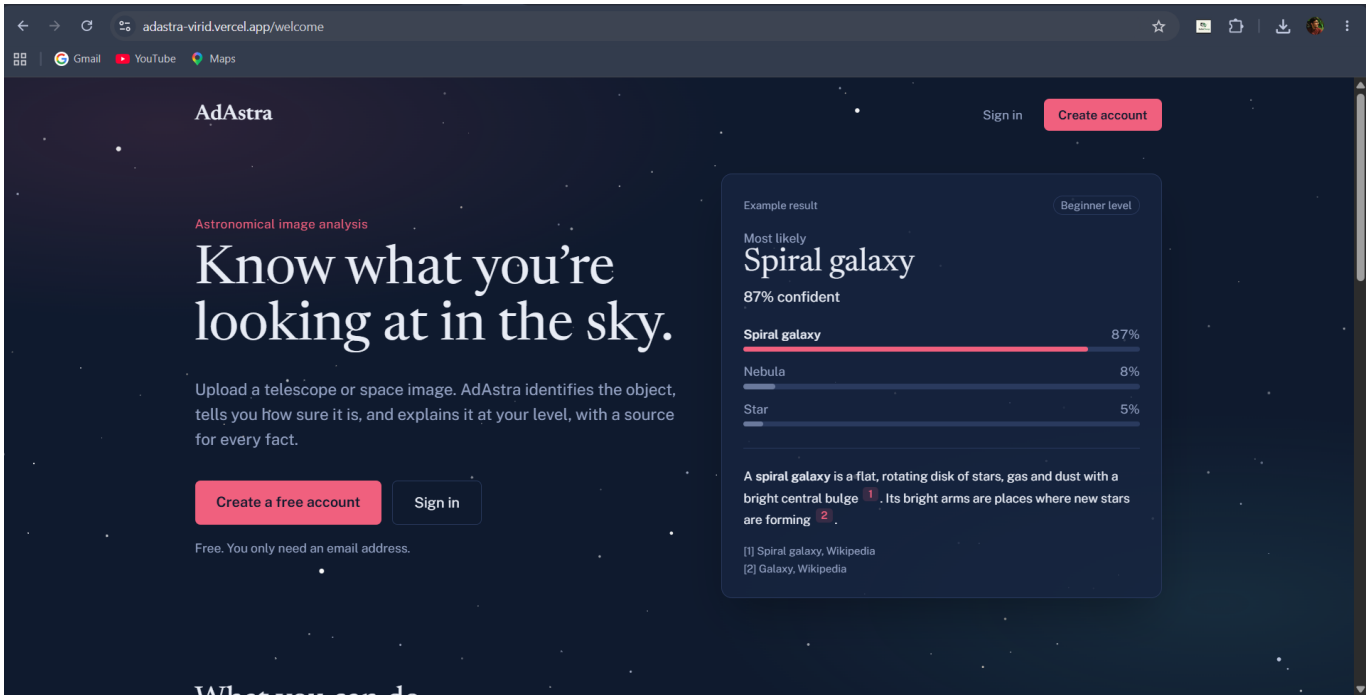


Figure 4.10: Landing page with an example result

An email varification is setup where a link is sent to the user via their email. Once they click the varification link, they are signed in.

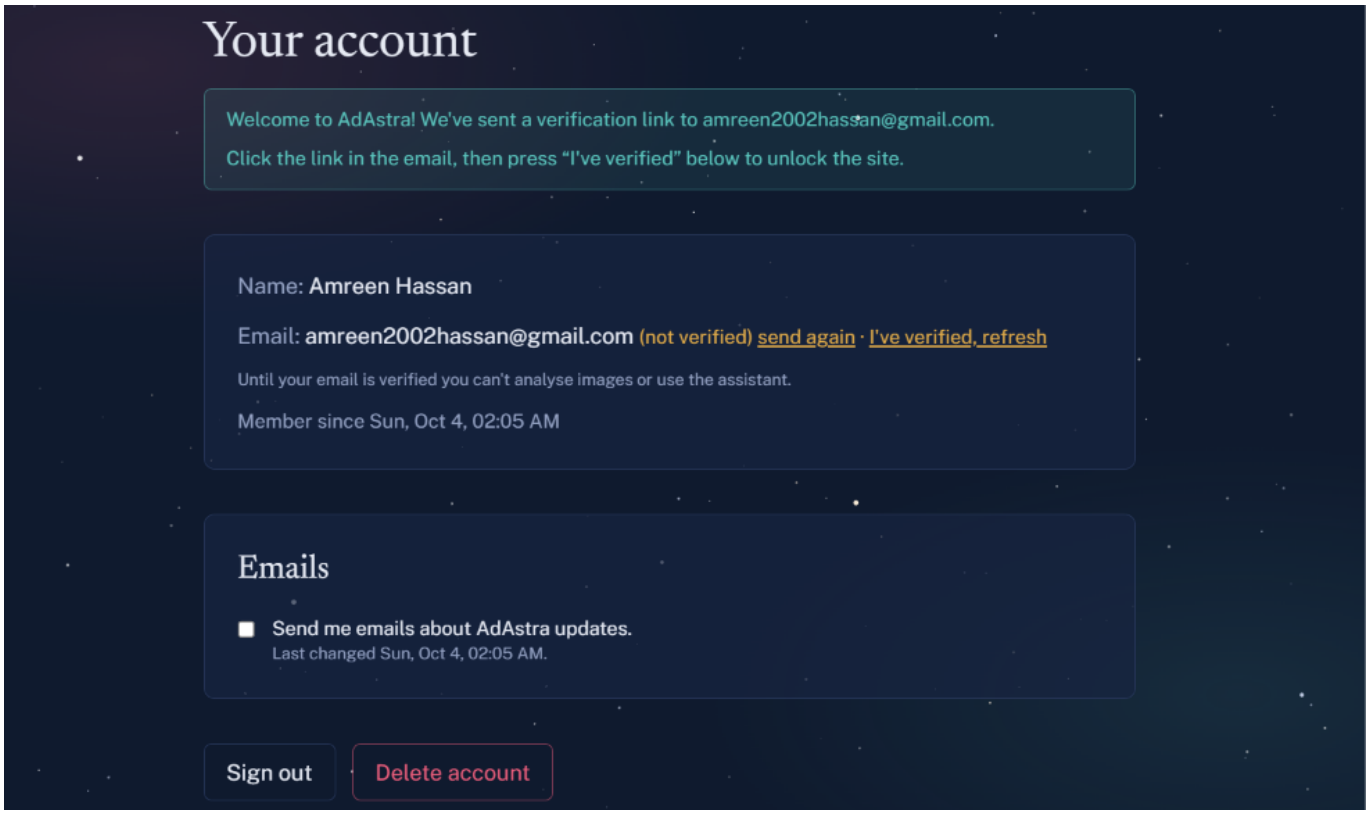


Figure 4.11: Email Varification

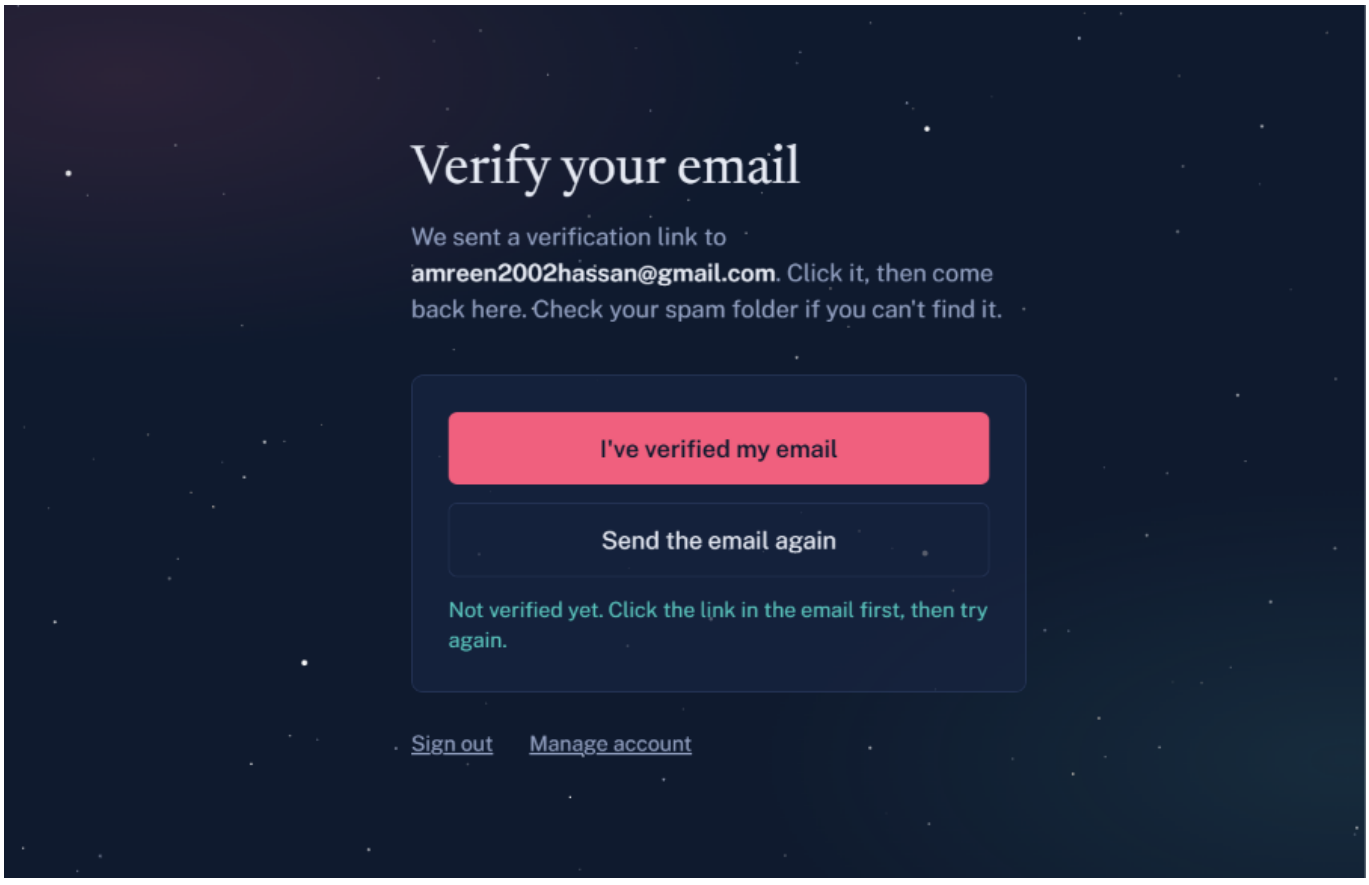


Figure 4.12: Email Varification

Once signed in, the user will see the recent analyses performed, the current phase of the moon, a shortcut to the astronomy assistant and a system-status panel. The status panel indicates that the object classifier, the galaxy-morphology classifier, CLIP, the knowledge base, accounts, the database and the Gemma explanation service are ready. The navigation bar provides access to the analysis, results, assistant, history, Moon and sky pages.

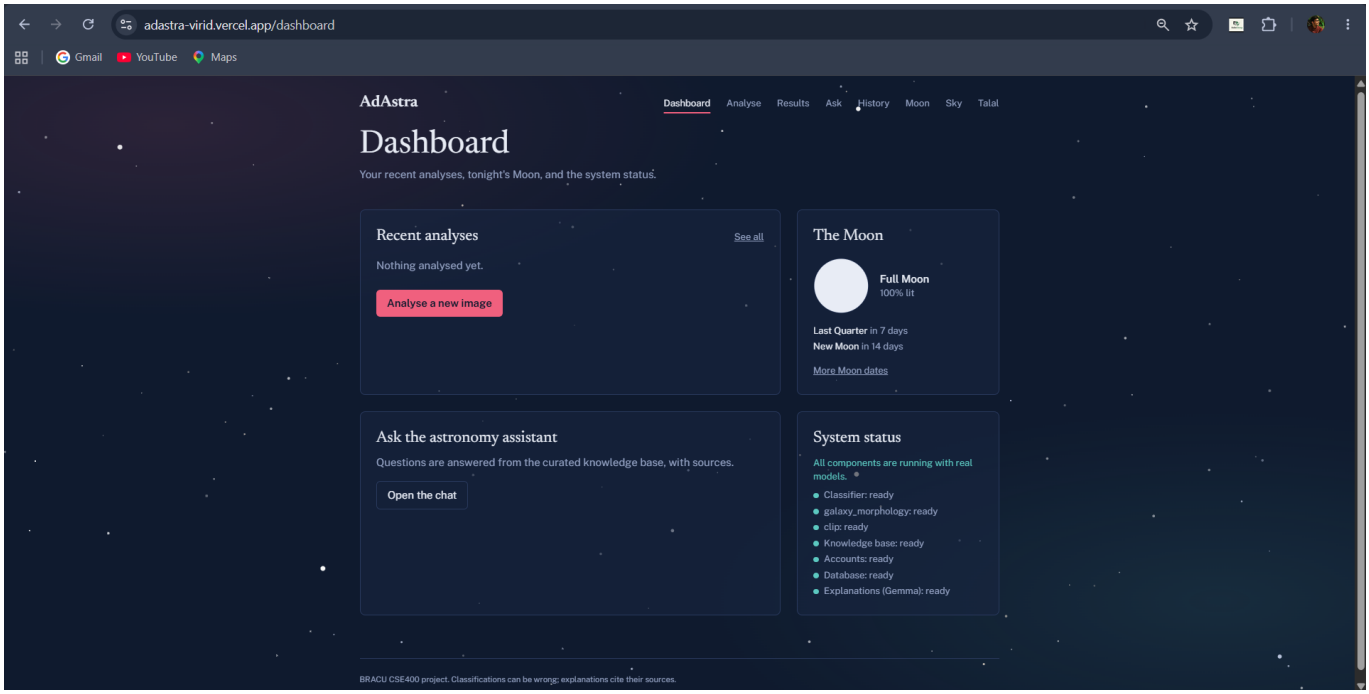


Figure 4.13: Dashboard with recent analyses, Moon summary and system status

4.5.2 Analysis Results

The outcome for an image of interacting galaxies is displayed in figure 4.15. The uploaded image will be displayed on the page, along with the most probable class (96% confident in this case, galaxy), and the scores of each of the five classes. The explanation can be switched between beginner, intermediate and advanced below. The text announces the classification and its level of confidence, mentions that a secondary model determined that the galaxy is spiral, and explains galaxies and spiral morphology using numbered citations that correspond to the source cards on the right. Passages which were recovered but not included in the text are indicated by a dimmed version and marked “Retrieved but not cited”.

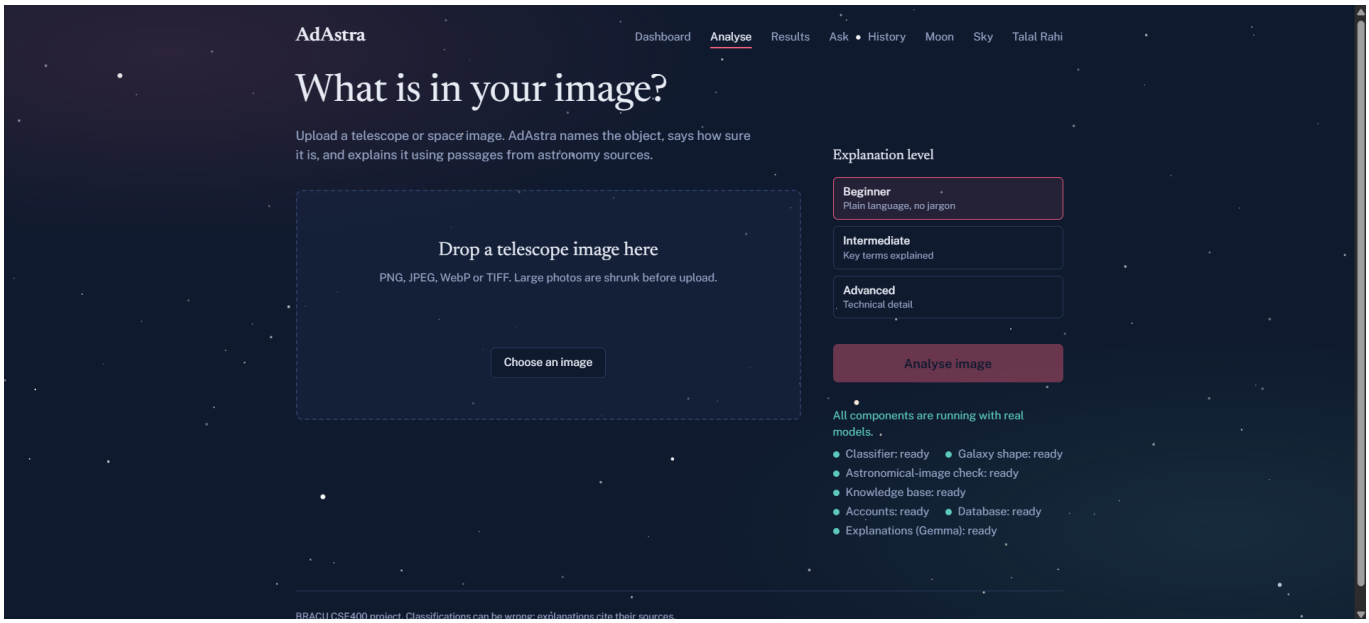


Figure 4.14: Analysis page

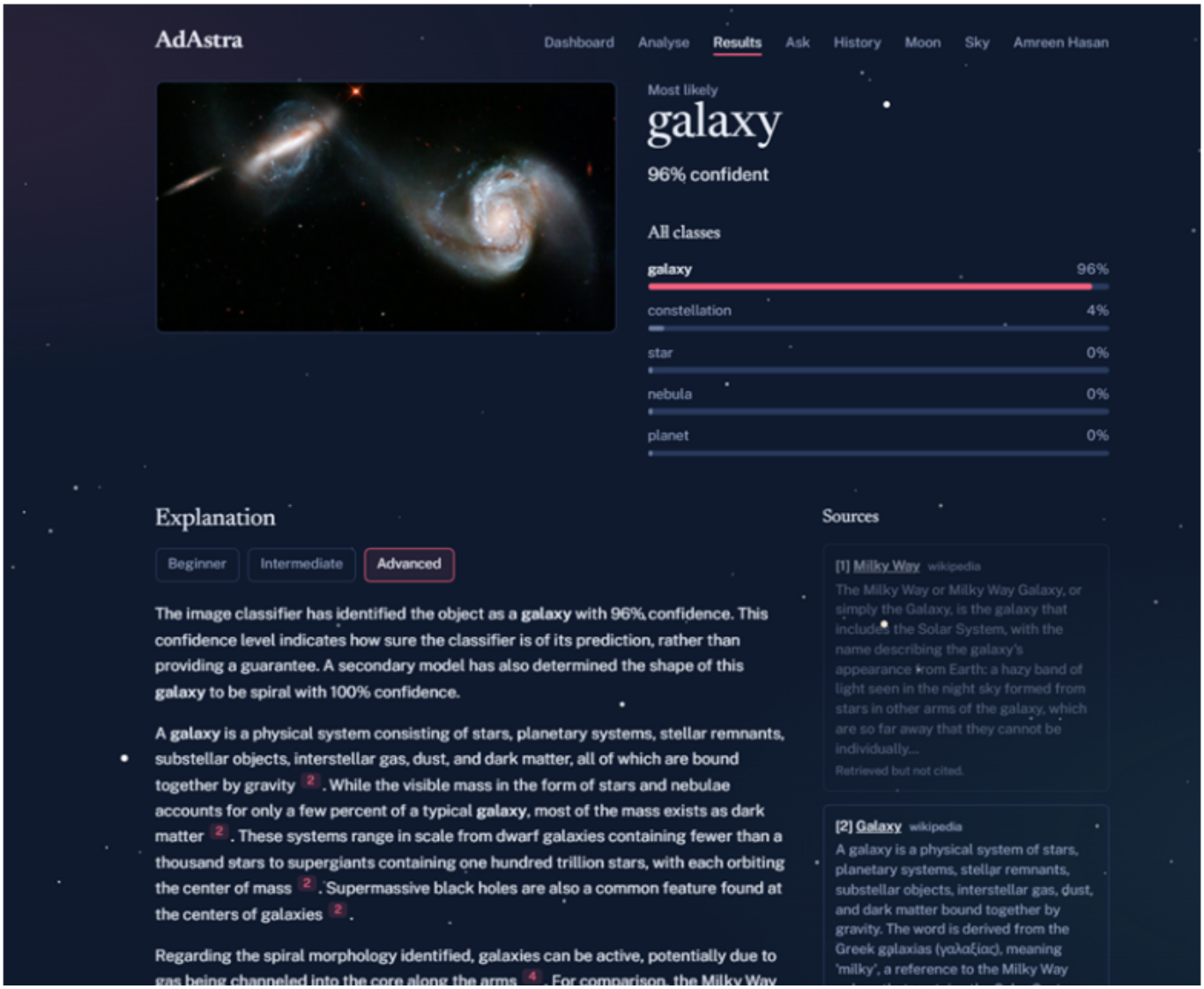


Figure 4.15: Result for a galaxy image with class scores, explanation levels and sources

The bottom of the same page (Figure 4.16) illustrates how the explanation deals with the boundaries of the sources. The advanced explanation concludes with a statement that there is no information in the retrieved passages about the distance, mass or exact type of this particular object, as stipulated by the prompt rules. The page also provides information on the processing time for each step of the request and provides the option to download the report or analyse another image.

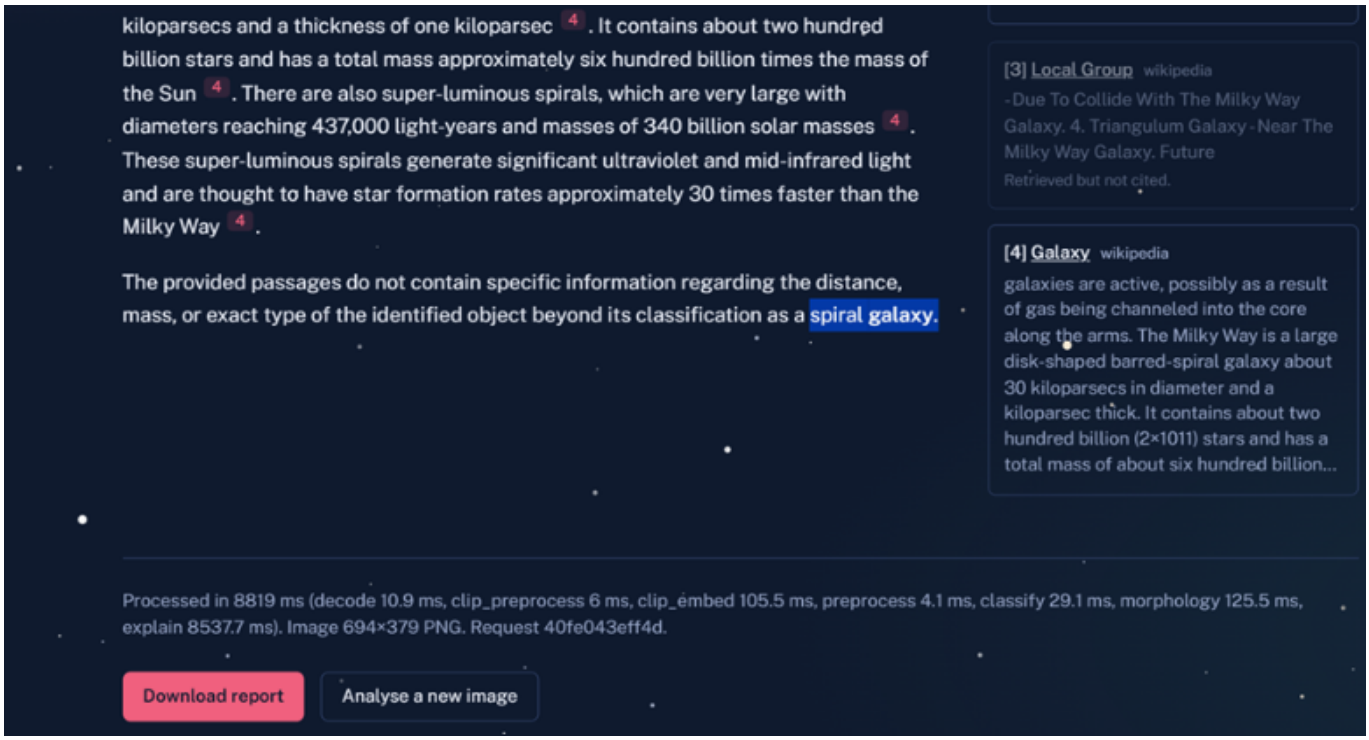


Figure 4.16: Remainder of the explanation, source cards and request timings

The timings printed on the page show where the time is spent (Table 4.7). Decoding, CLIP, classification and morphology together take about 280 ms, and the language-model call takes 8.5 s of the total 8.8 s. The explanation stage is therefore the part of the system that sets its response time.

Table 4.7: Processing time of the request shown in Figure 4.15 (single request, 694 × 379 px PNG)

Stage	**Time (ms)**
Image decoding	10.9
CLIP preprocessing	6.0
CLIP embedding	105.5
Classifier preprocessing	4.1
Object classification	29.1
Galaxy morphology	125.5
Explanation (retrieval and generation)	8,537.7
Total	8,819

The result is uncertain in Figure 4.17. The image of a nebula is classified as a galaxy with 47% confidence, followed by nebula (38%) and star (15%). The top score is less than 0.55, so the page shows “but uncertain”, and explains that the object may also be a nebula. The beginner’s explanation admits the uncertainty, talks about the runner-up class, and quotes passages about planetary nebulae and about galaxies. The purpose of the uncertainty rule is that the classifier’s error is not corrected, but the user is informed that the label should not be trusted.

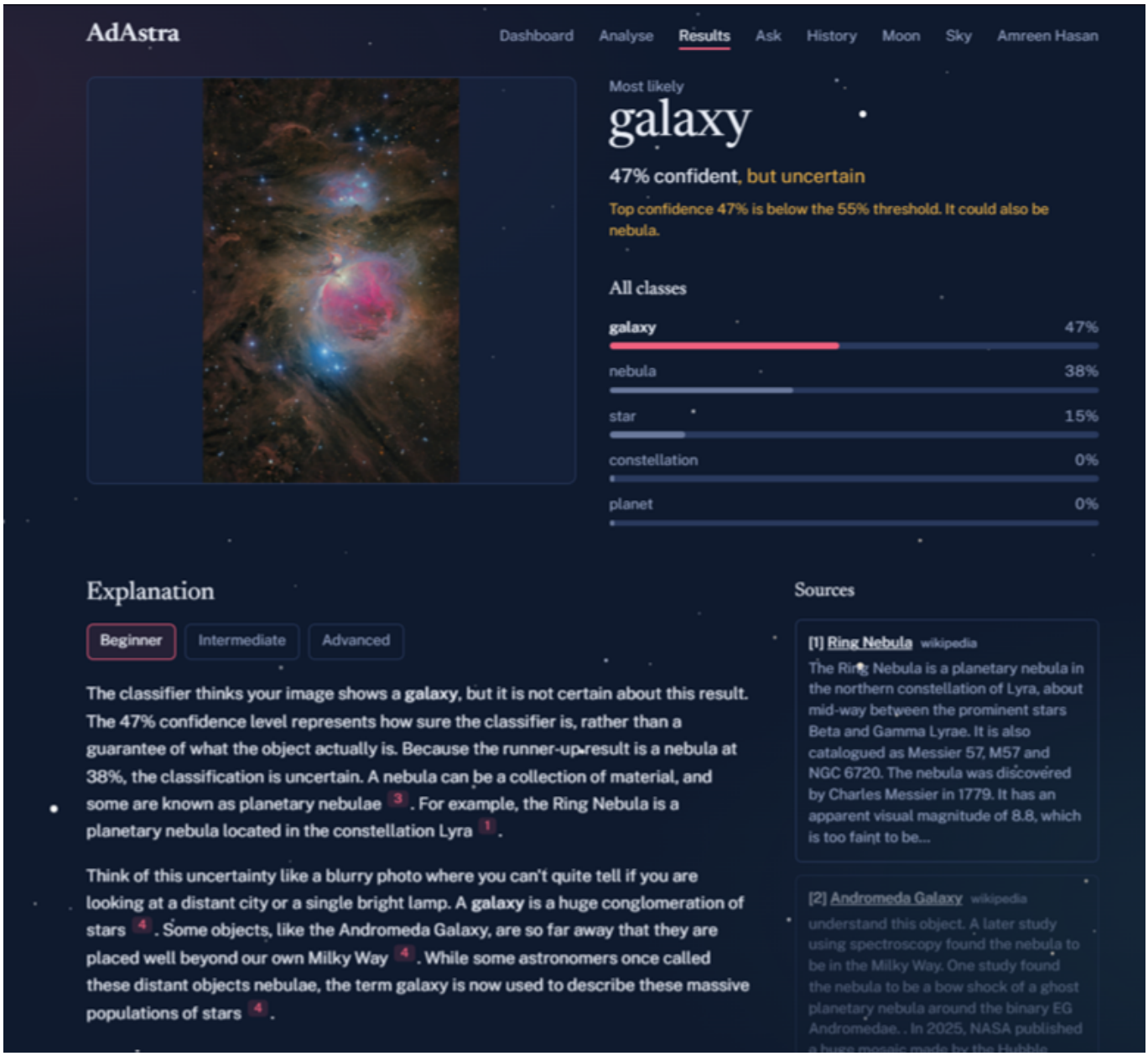


Figure 4.17: Uncertain result with the low-confidence warning and runner-up class

Figure 4.18 shows the CLIP gate at work. A photograph of an alarm clock is not classified. The page states that the image does not look astronomical and offers to try another image, and it also reports that the browser reduced the image before upload.

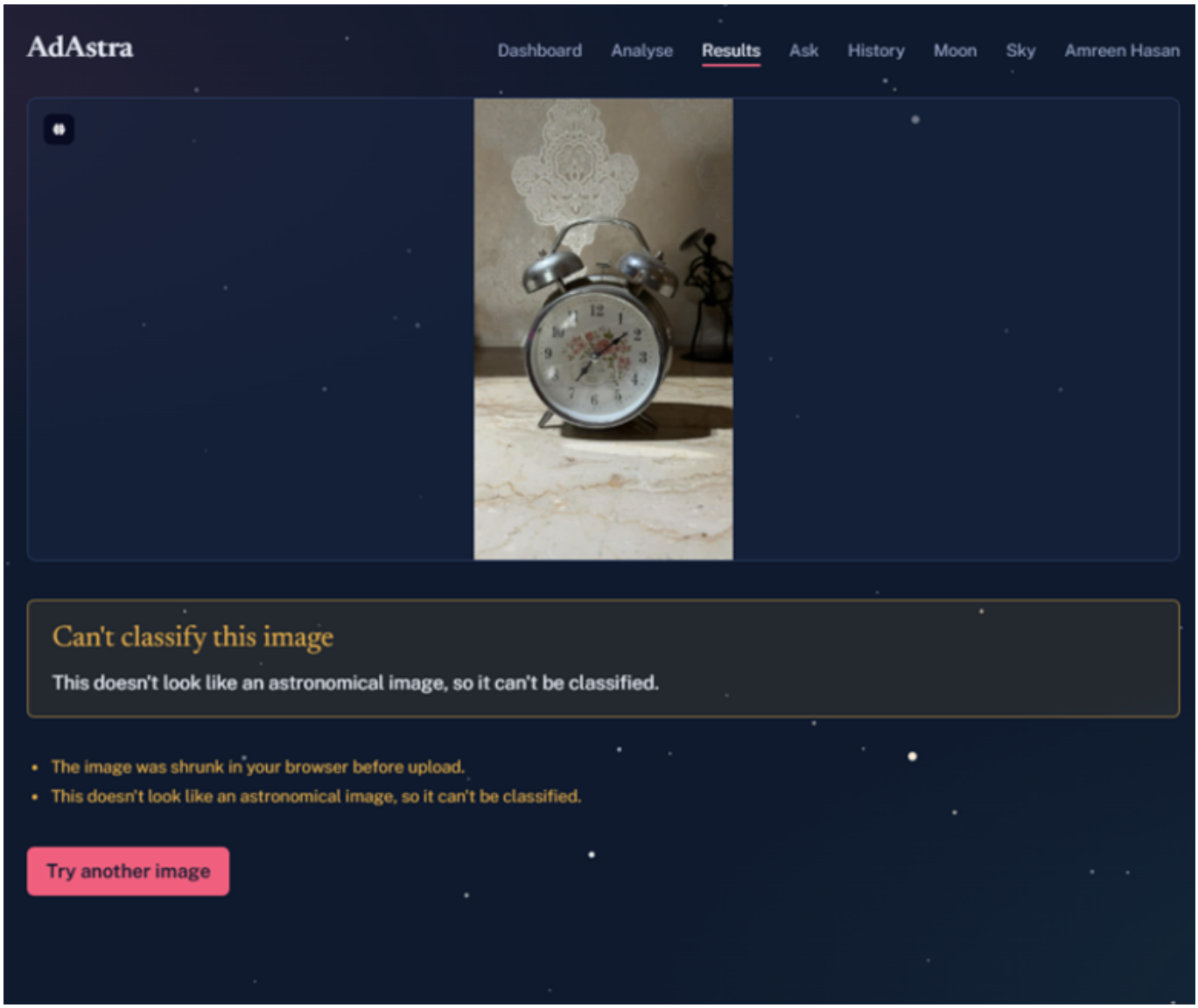


Figure 4.18: Refusal to classify a non-astronomical image

4.5.3 History and Astronomy Assistant

There is a history records with the image reference classification results and time that the image was uploaded.

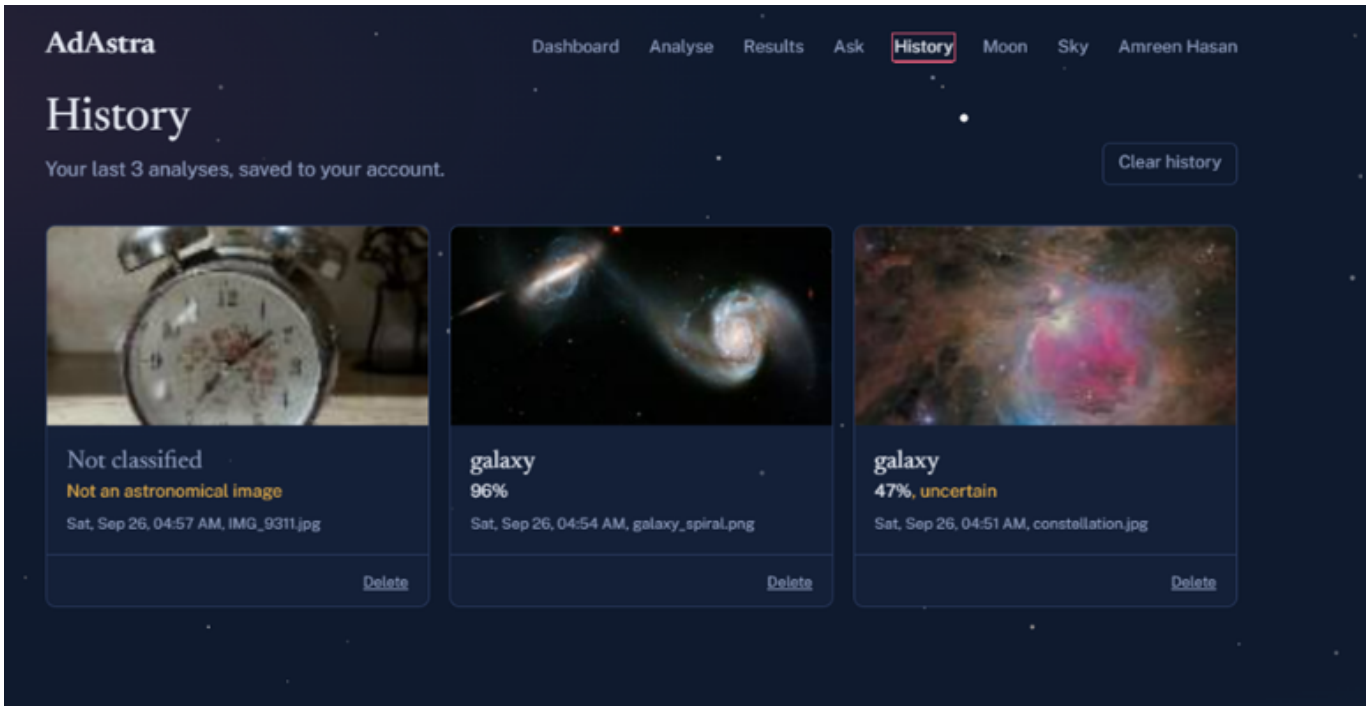


Figure 4.19: Previous Search History

The astronomy assistant is not a free chat, but rather an open book exam. You type a question and press Send, and it goes to the back end, along with your previous few messages (which is why follow-up questions are possible), and, if you've checked the box, your previous image result as additional context.

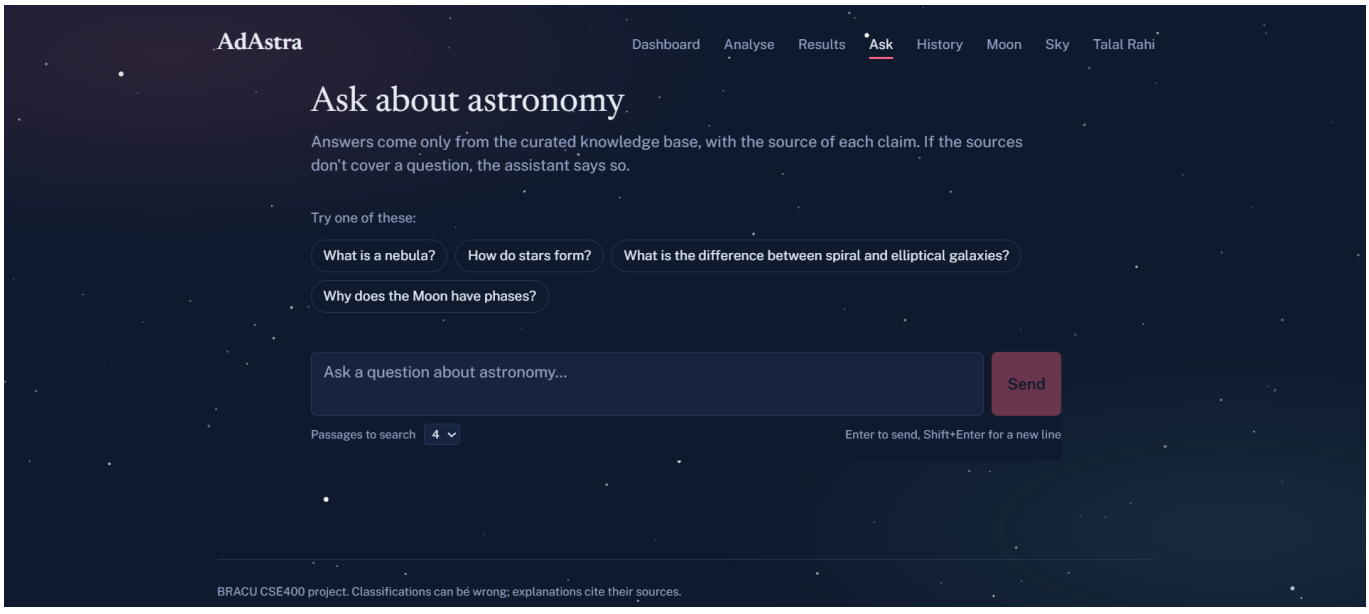


Figure 4.20: Ask Page

On the server, a small embedding model called MiniLM transforms your question into a list of numbers that represent what it means, and this is compared to a pre-built knowledge base which has 113 Wikipedia astronomy articles, chopped into

small paragraphs and transformed into their own number fingerprints ahead of time, and stored in a searchable index called FAISS (which is built once, offline, and never used while you're on the site). The 4 most similar real paragraphs are extracted and given to Gemma, Google's language model, in one very strict instruction: use only these paragraphs, do not use any other paragraphs you already know, and mark each fact with the number of the paragraph it is from. The entire retrieve then prompt then generate pipeline is connected together using LangChain's LCEL, which is essentially glue code that connects those steps together automatically, but doesn't do any thinking. Then Gemma writes the answer using only what it was given, referencing [1], [2] and so on, and before you can see it, any citation number it may have made that doesn't correspond to a real paragraph is removed. The answer is displayed on screen, followed by the actual source paragraphs, each of which is clickable, so you can jump directly to the source and the important thing to remember is that Gemma never searches anything or reads Wikipedia directly, it only ever sees the handful of paragraphs it's given each time, so every claim it makes can be traced back to a real source, not just trusted on faith.

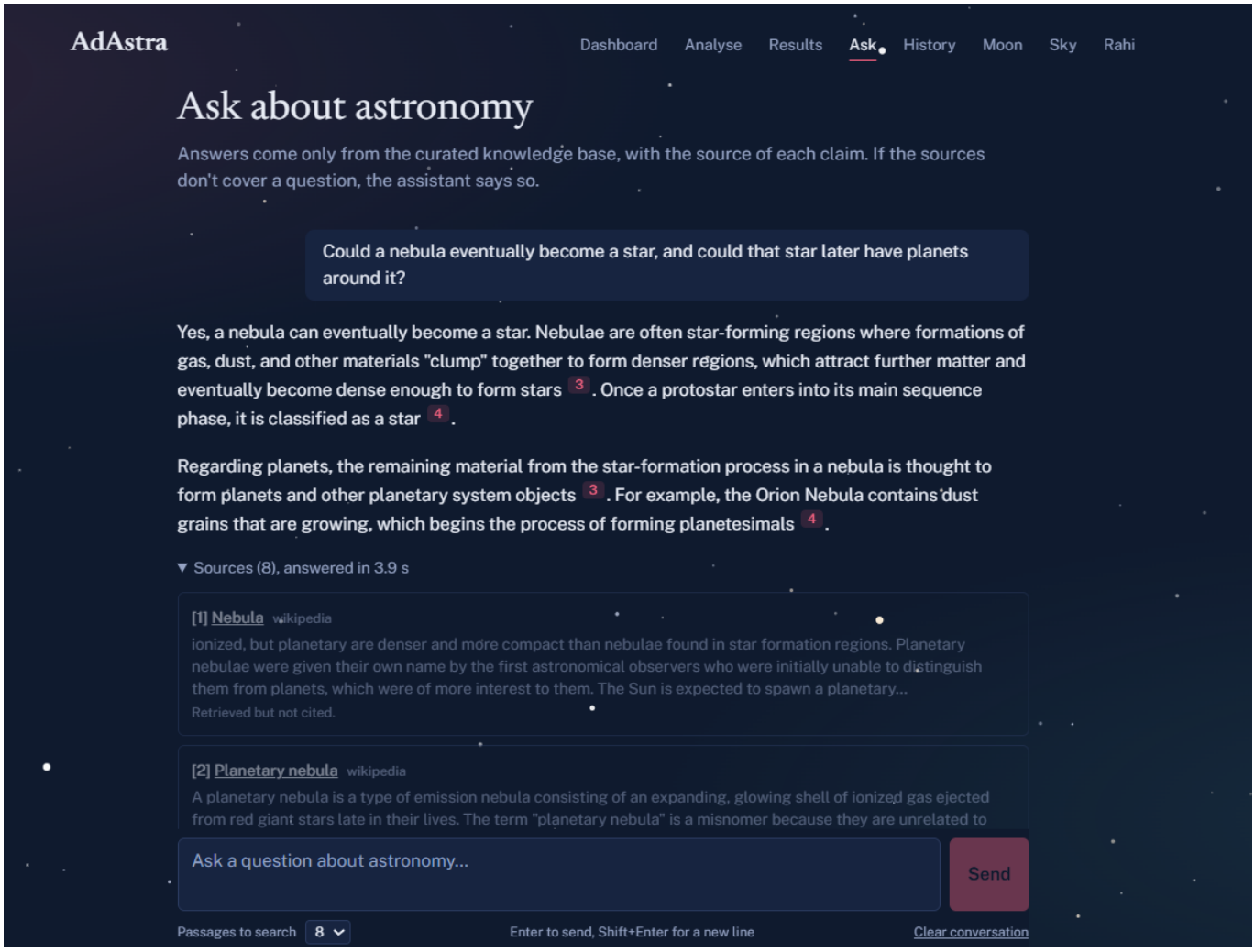


Figure 4.21: Astronomy assistant answering a question with cited sources

When presented with questions that are out of the knowledge base, it refrains from answering and gives the message, **I am sorry, but the knowledge base does not cover this question.**

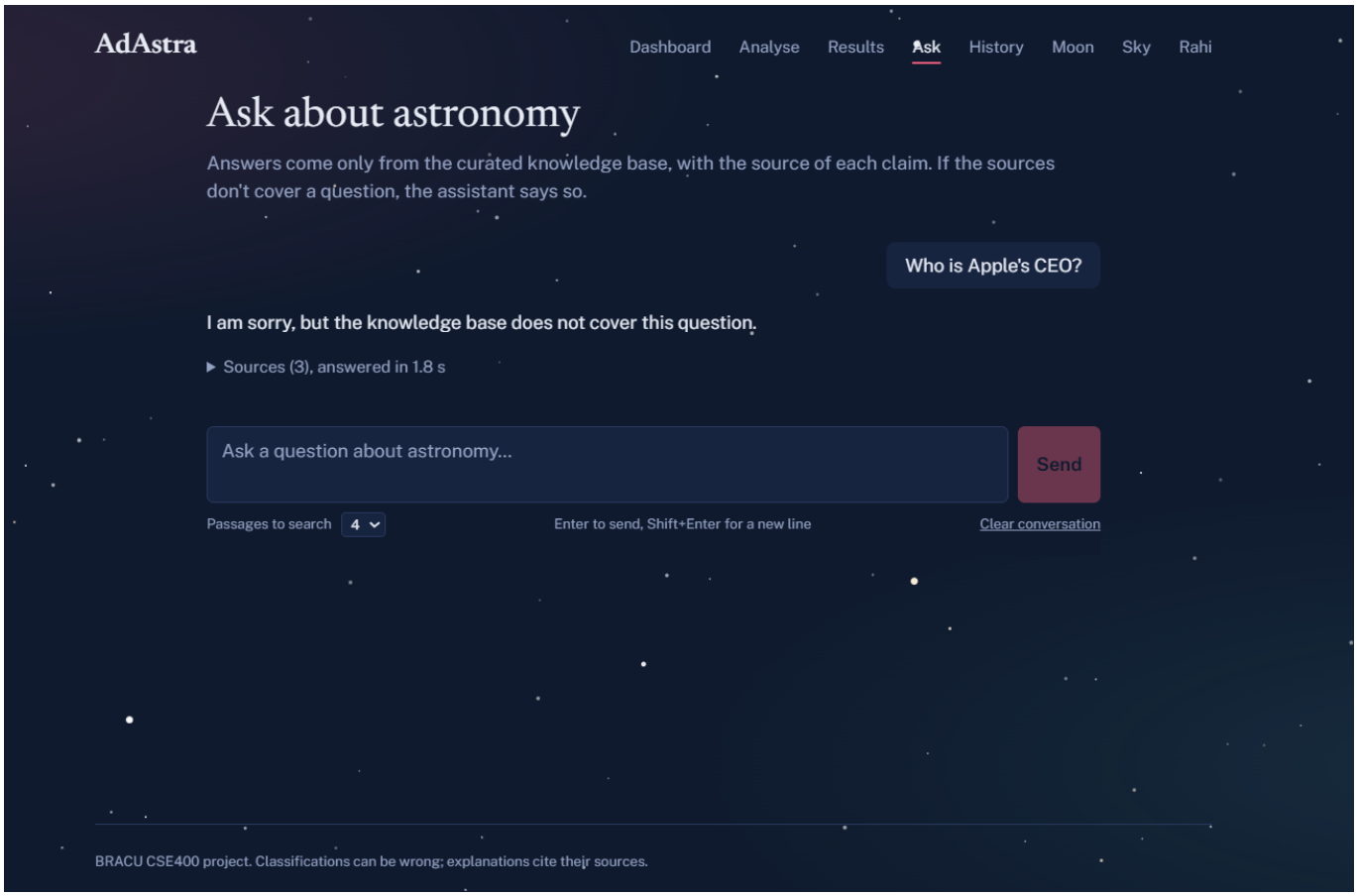


Figure 4.22: Astronomy assistant abstaining from answering

4.5.4 Sky Explorer and Moon Page

None of it is a photograph; each picture is pure SVG code, drawn live by the browser using shapes, gradients, and in the case of the galaxies, actual orbital math (a logarithmic spiral formula, used to place each star-dot along the arms). The facts (distance, diameter, a short blurb, a wikipedia link) are stored in one data file, and the drawing logic is stored in another, and the two are only linked by a shared ID string when the page requests "andromeda", the art file reads that ID from its own table to determine the tightness and tilt of the spiral, without consulting the facts file. Hovering over an object displays a brief tooltip, clicking it selects the object and opens a full details card containing the facts about that object, and an "Ask about this" button takes you directly to the Ask page with a question about that object already typed and sent for you.

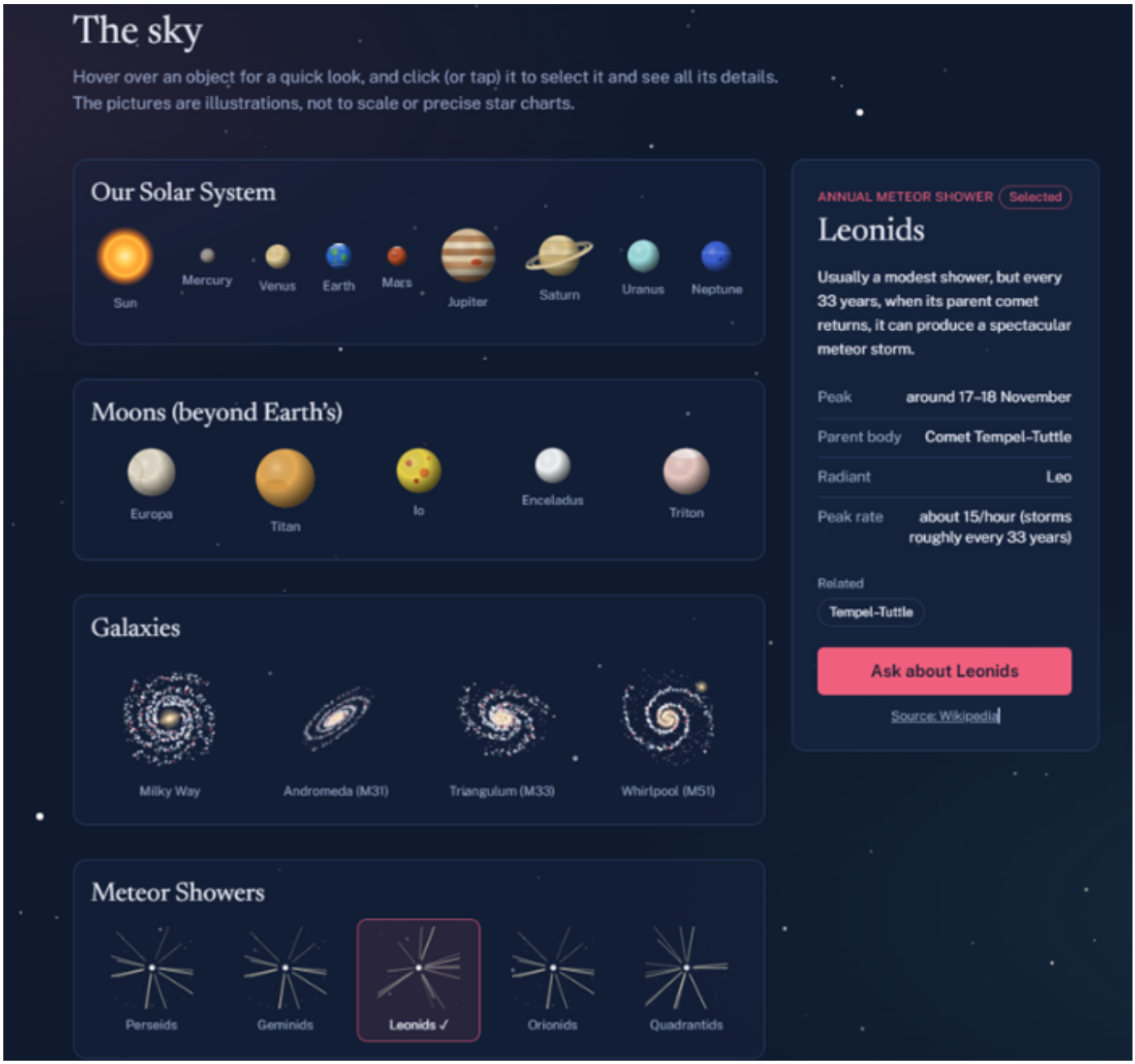


Figure 4.23: Sky explorer with the Leonids meteor shower selected

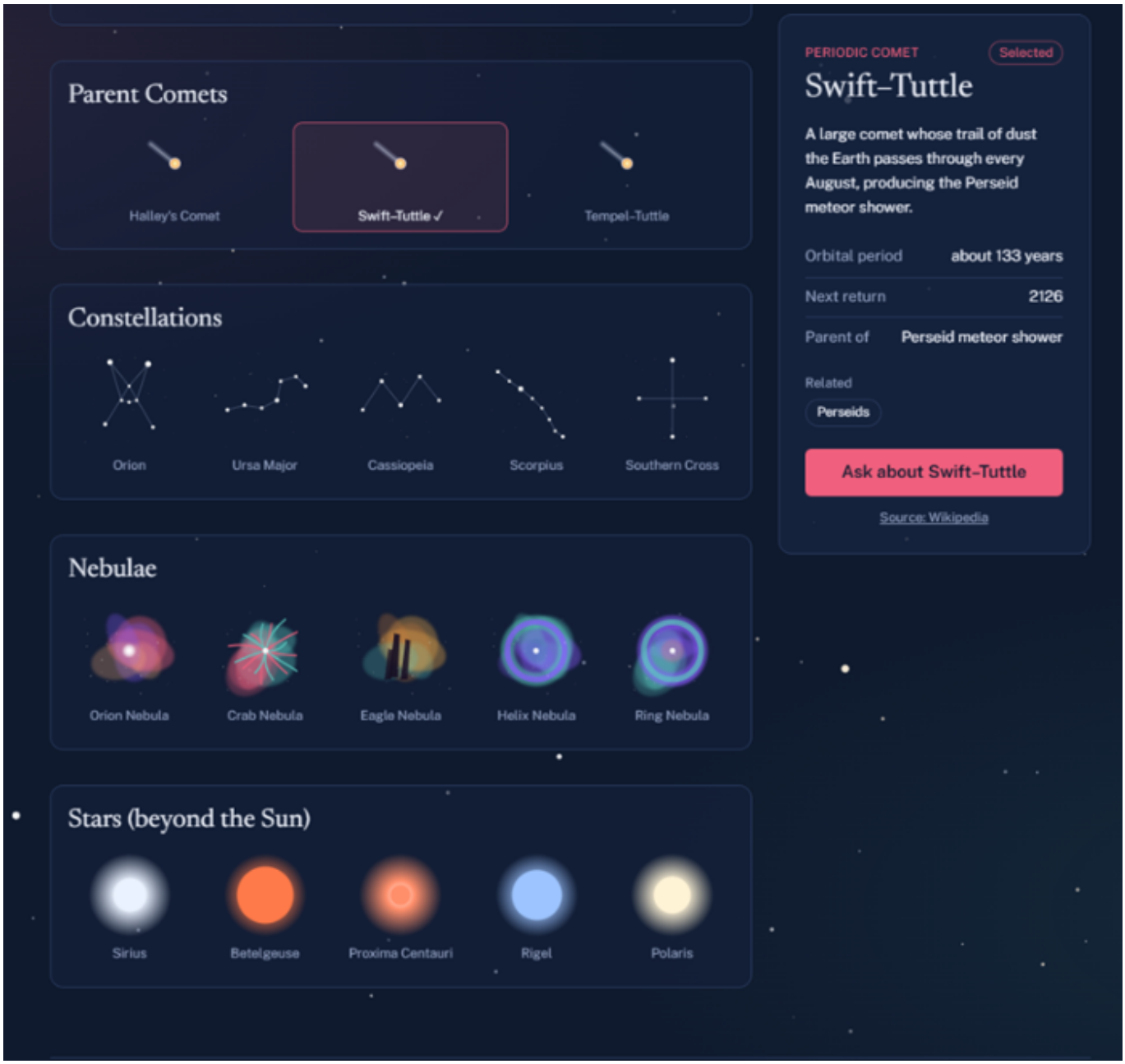


Figure 4.24: Sky explorer with the comet Swift–Tuttle selected

The **Moon page** (Figure 4.25 & 4.26) is completely different it is not an illustration, it is real astronomical calculation. It reads your device's own clock (the same one in your phone's status bar, which is already accurate), and feeds that exact moment into Jean Meeus's lunar phase algorithm: a rough formula for when the next New/Full/Quarter Moon falls, corrected by about 20 small adjustment terms that account for real wobbles in the Moon's orbit, bringing the prediction to within about 2 minutes of the true time (checked against a professional astronomy library, PyEphem, to confirm this). A simpler calculation, using the angle between the Sun and Moon as seen from Earth, works out exactly how much of the disc is lit at this moment. This math always uses the raw underlying timestamp of the device clock, never "what hour is it locally", so it always returns the same correct answer for all users on earth at the same real moment, and was directly tested by running the

calculation under 5 different timezones and getting the same answer to 10 decimal places, the page simply re-checks the clock every minute while it is open so the numbers stay current.

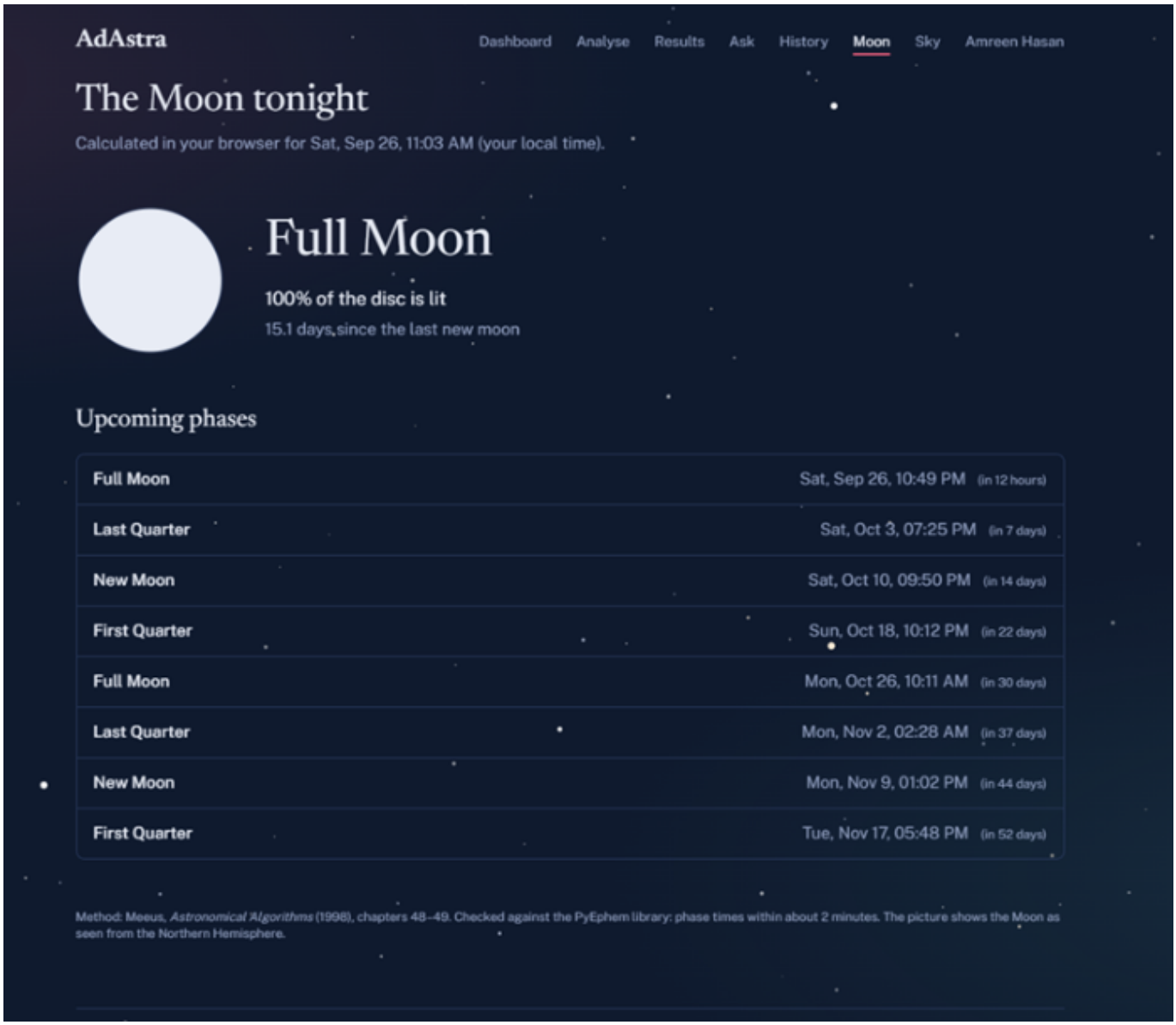


Figure 4.25: Moon phase and upcoming phase schedule on 26th September

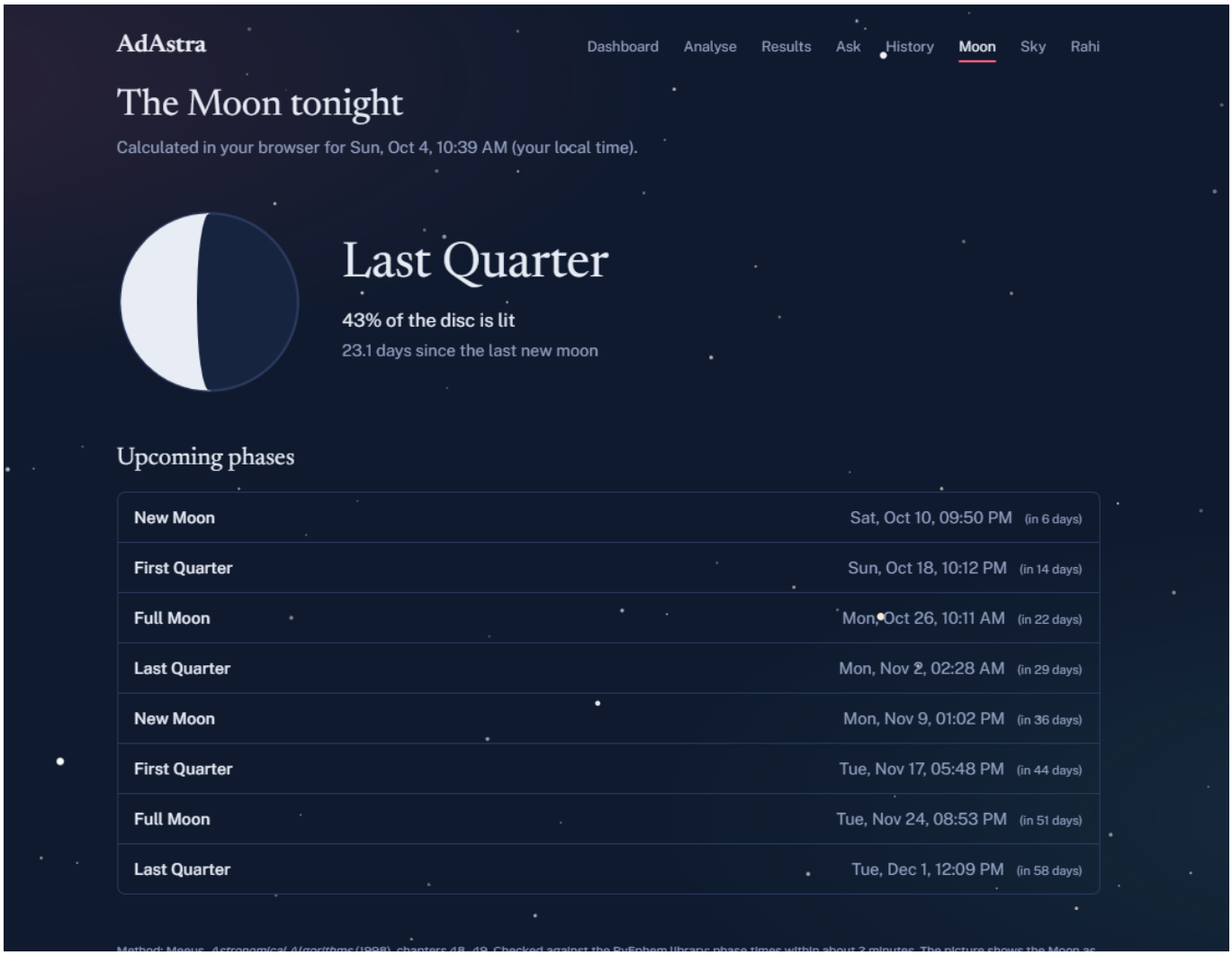


Figure 4.26: Moon phase and upcoming phase schedule on 4th October

4.5.5 System Diagrams

The activity diagram shows the whole interaction flow of the users with the application AdAstra, starting with the sign-in and upload of the image, followed by the validation process, preprocessing, verification of the image astronomically, classification of the objects, optional classification of the galaxy morphology, generation of explanations, displaying the results, and saving of the history. It also displays the system decision paths for invalid or non-astronomical inputs and whether or not the user wants to continue with another analysis.

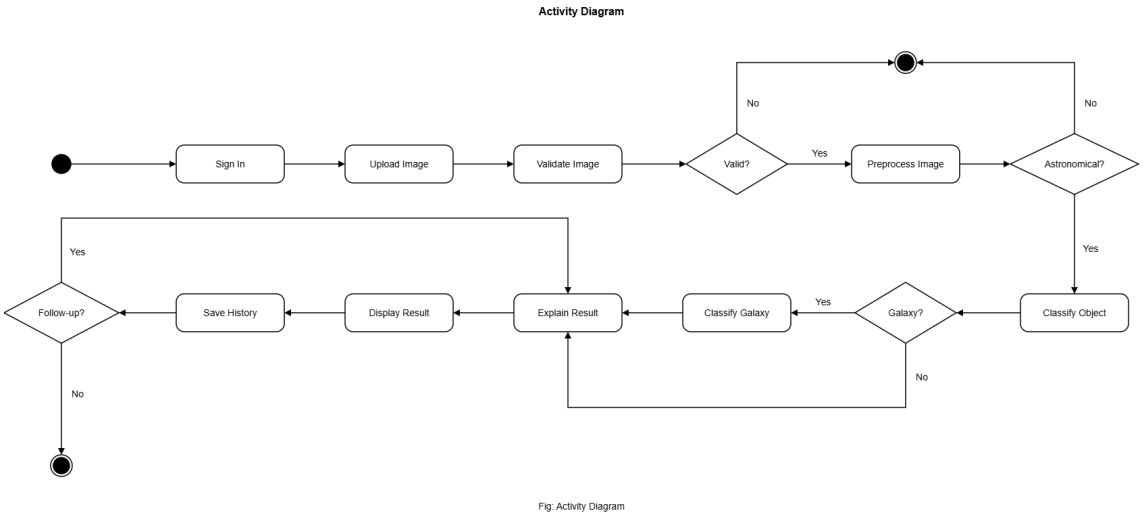


Figure 4.27: Activity Diagram

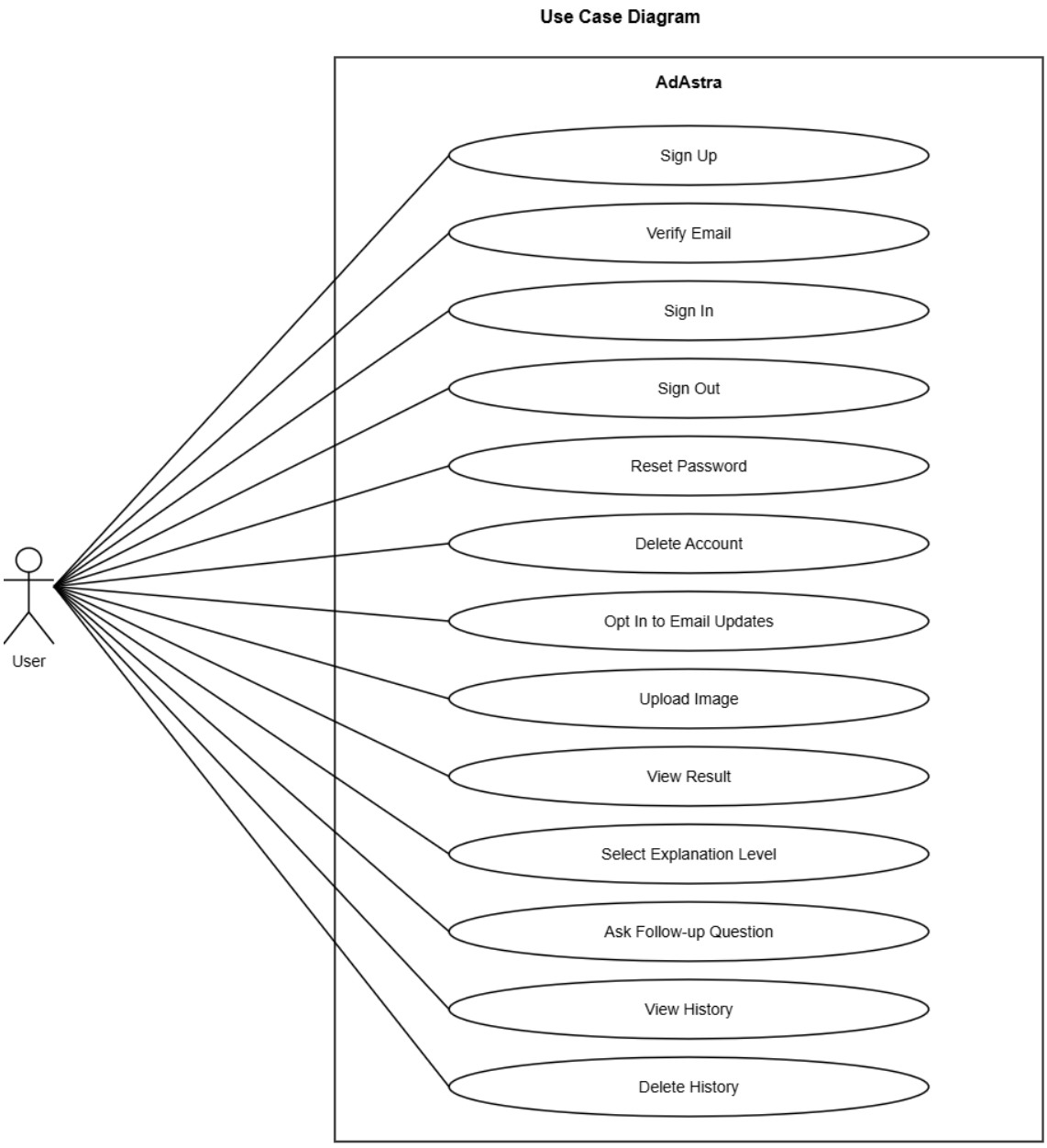


Figure 4.28: Use Case Diagram

There is a one-to-many relationship between User and History in the ER diagram below.

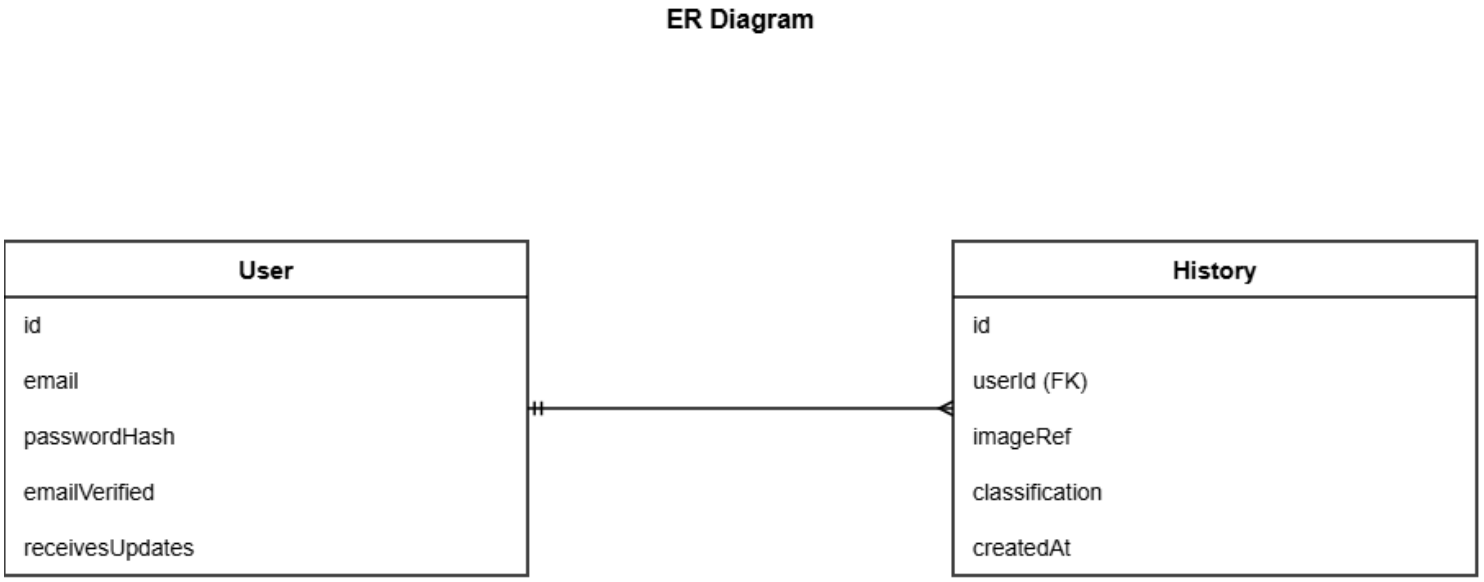


Fig: ER Diagram

Figure 4.29: ER Diagram

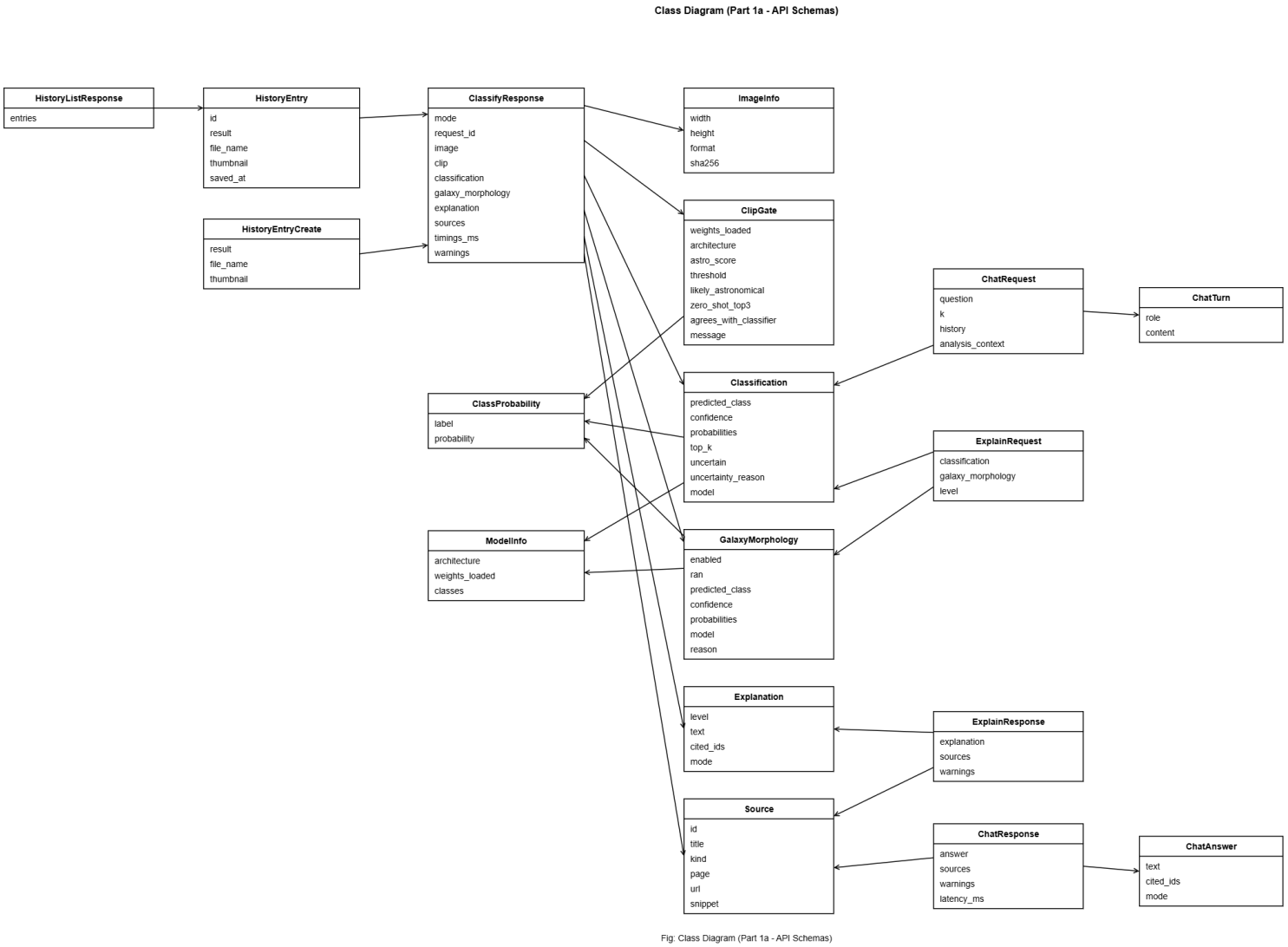


Figure 4.30: Class Diagram Part 1a

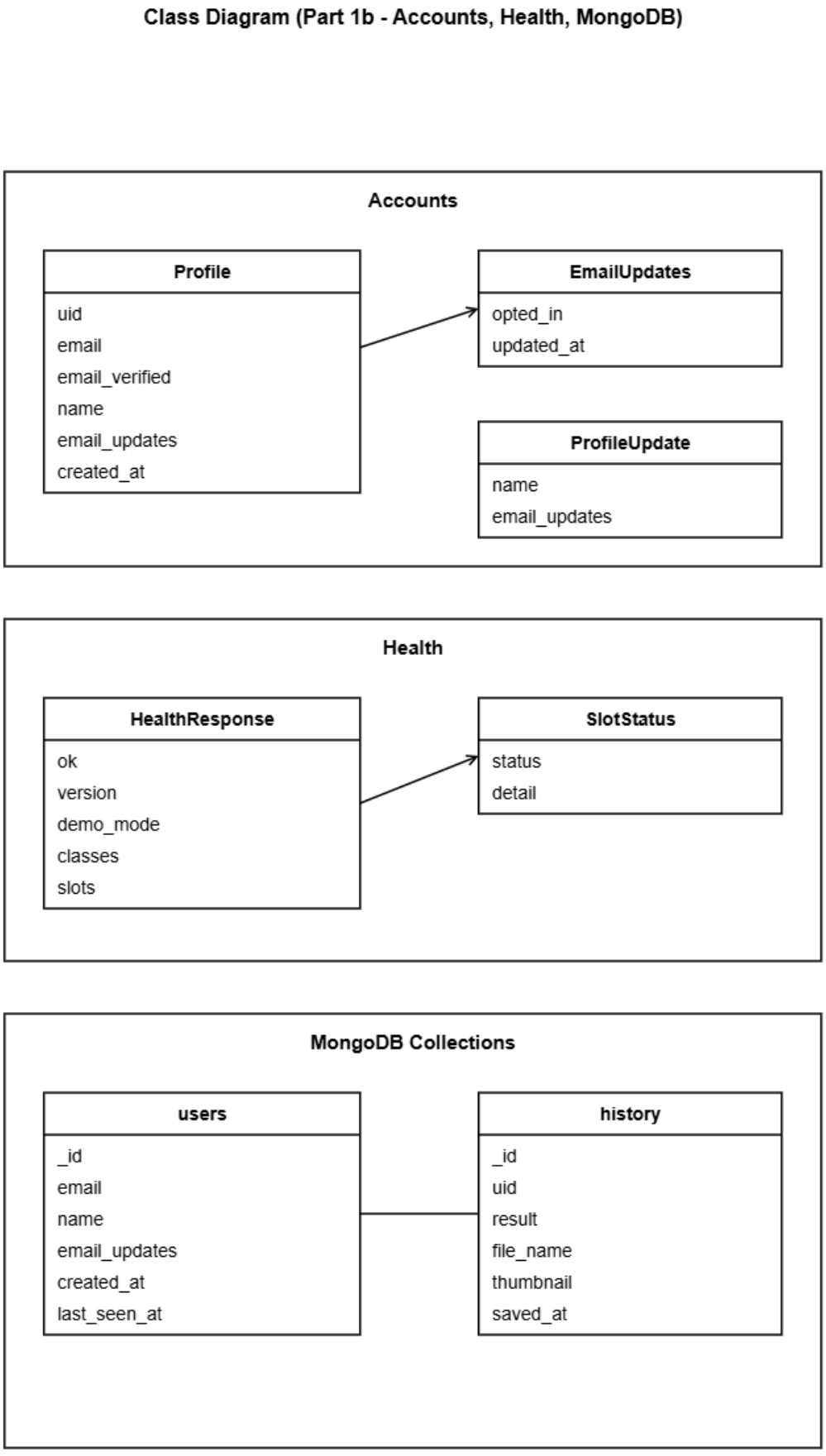


Figure 4.31: Class Diagram Part 1b

Class Diagram Part 1: The first part of the class diagram which contains information on the data structures and relationships of the AdAstra application, such as user accounts, history records, classifications results, model information, CLIP gate output, galaxy morphology prediction, explanations, chat requests, and source metadata.

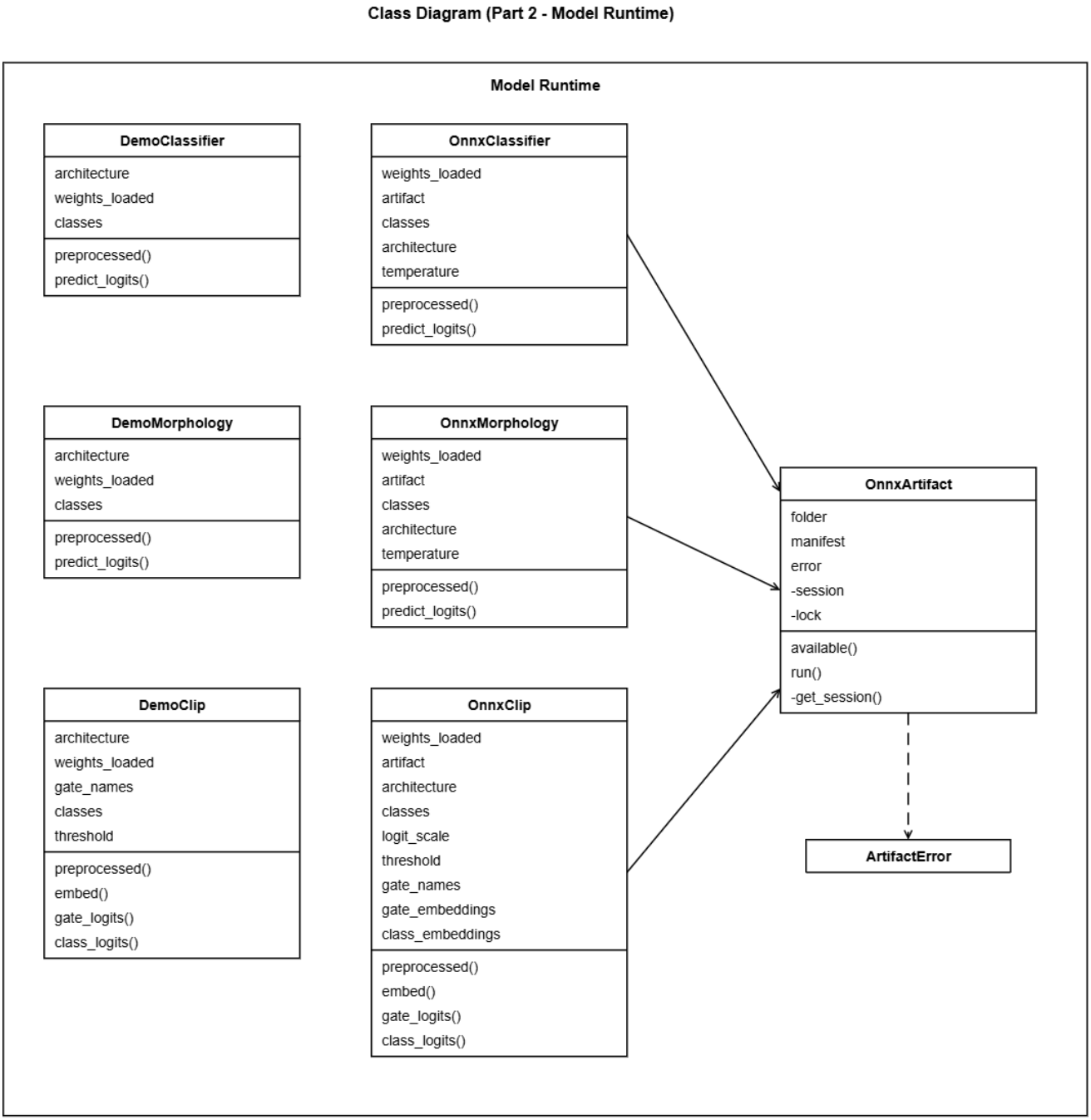


Figure 4.32: Class Diagram Part 2

Class Diagram Part 2: The second class diagram depicts the model runtime architecture, where the classifier, the morphology and the CLIP are implemented as ONNX-based runtime classes, along with the shared OnnxArtifact class that loads

and executes model artefacts, and manages runtime errors.

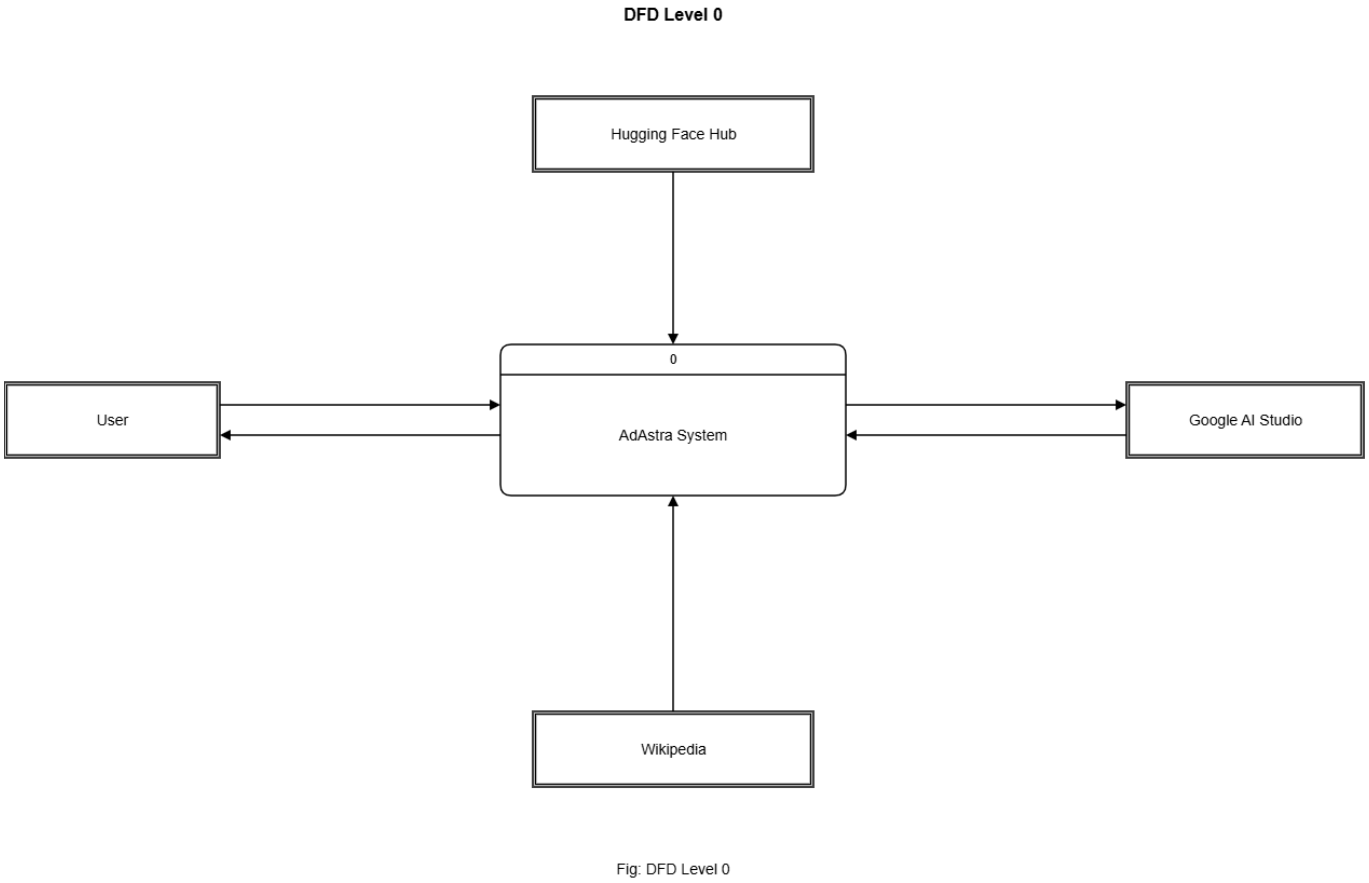


Figure 4.33: DFD Level 0

The Level 0 DFD depicts AdAstra as a single process that communicates with the primary external processes: the User, Hugging Face Hub, Google AI Studio, and Wikipedia, and displays the exchange of data between these external resources and AdAstra.

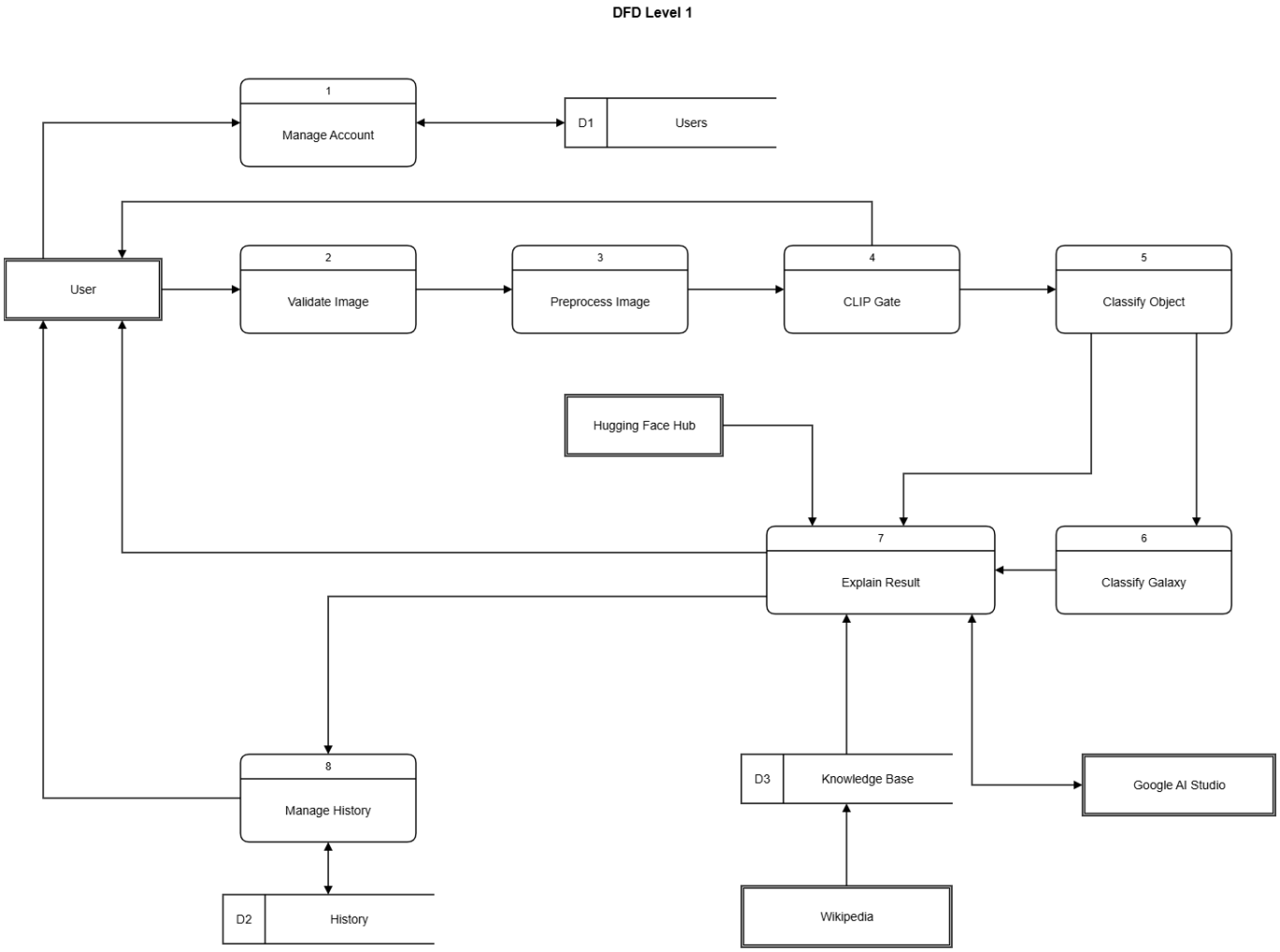


Figure 4.34: DFD Level 1

The Level 1 DFD breaks down the core processes of AdAstra, such as account management, image validation and preprocessing, CLIP-based checking, astronomical object classification, galaxy morphology classification, explanation generation, and history management, along with interactions with the user database, knowledge base, Hugging Face Hub, Google AI Studio, and Wikipedia.

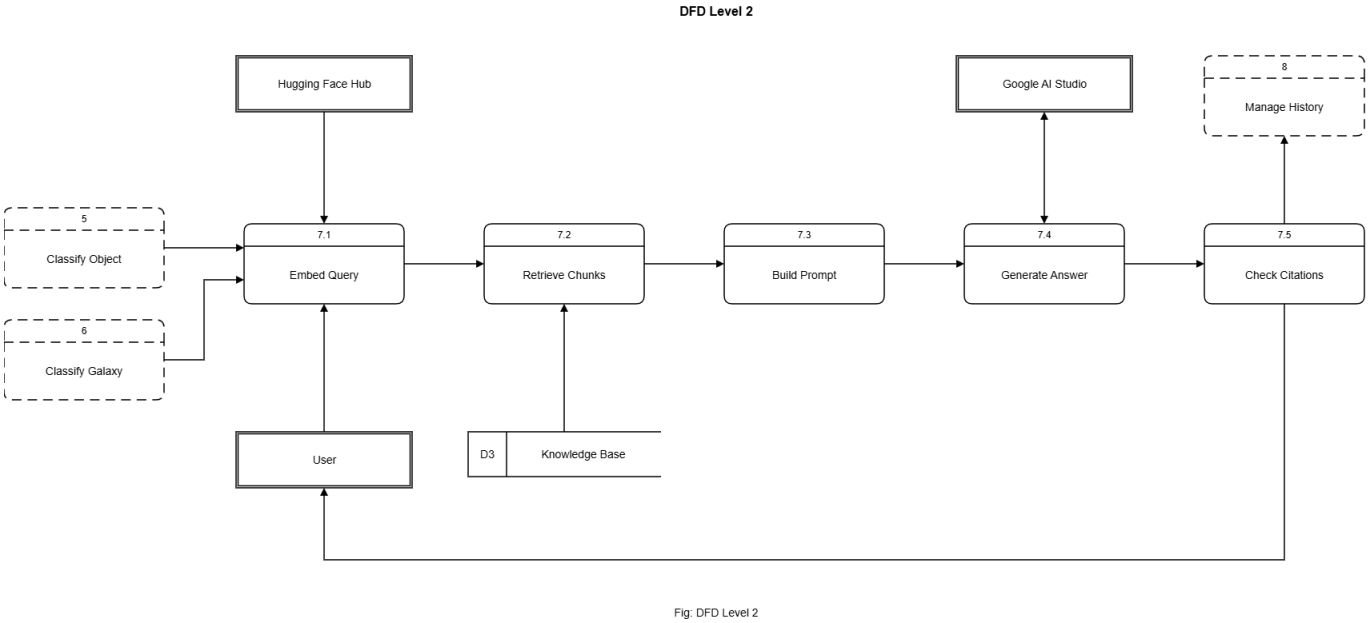


Figure 4.35: DFD Level 2

The Level 2 DFD shows the retrieval-augmented explanation process, where the classification output is incorporated in the user input as a query, relevant knowledge chunks are retrieved, a prompt is formulated, the answer is generated by Google AI Studio, citations are verified, and the result is fed back to the user and added to the analysis history.

4.6 Chapter Summary

In this chapter, the design and implementation of AdAstra was presented. Two public datasets were prepared and partitioned: SpaceNet, which was reduced to five object classes and 10,093 images, and Galaxy Zoo, which had 61,578 images labeled elliptical or spiral. Each dataset was fine-tuned using a single protocol with three ImageNet-pretrained CNNs. On SpaceNet, EfficientNet-B0, ResNet50 and ResNet18 reached test accuracies of 83.66%, 83.51% and 83.01%, and on Galaxy Zoo, ResNet18, EfficientNet-B0 and ResNet50 reached 86.86%, 86.25% and 85.95%. The differences within each task are small, and the common errors are between classes that are visually similar, particularly between star, constellation, galaxy and nebula on SpaceNet, and elliptical and spiral galaxies on Galaxy Zoo. The classifiers are embedded in a two-stage recognition pipeline that is guarded by a CLIP gate and an uncertainty rule, and are served via a web application that includes accounts, history, an assistant, a sky explorer and a Moon page, with checked citations provided as a retrieval-augmented explanation. The screenshots show that the safeguards work as expected in the tested examples: a confident galaxy prediction is explained with supporting sources, an uncertain nebula prediction is flagged, and a normal photograph is rejected. The trained models are statistically assessed in Chapter 5, the approaches are compared and the limitations of the results are discussed.

Chapter 5

Result Analysis

The results of the experiments described in Chapter 4 are reported and analysed in this chapter. The three convolutional neural network (CNN) architectures were trained on two datasets: the SpaceNet dataset for astronomical object classification (five classes: constellation, galaxy, nebula, planet and star) and the Galaxy Zoo dataset for galaxy morphology classification (two classes: elliptical and spiral). All the models in this chapter were trained for 30 epochs following the same procedure, and all the figures in this chapter are taken from the held-out test partition unless otherwise stated. The chapter first outlines the criteria and testing methods (Section 5.1), then examines each candidate solution in light of those criteria (Section 5.2), describes the design changes that ensued from the results (Section 5.3), applies statistical tests to the reported differences (Section 5.4), compares the approaches with one another and with previous work (Section 5.5), and discusses the implications of the results and the limits of their reliability (Section 5.6).

5.1 Performance Evaluation

5.1.1 Evaluation Criteria

The models were evaluated based on three sets of criteria: accuracy, efficiency and reliability.

Accuracy: The main measure is overall test accuracy. Both datasets are class-imbalanced (Table 5.1), so accuracy alone can mask poor performance on the minority class, and is reported alongside per-class precision, recall and F1-score, the macro-averaged versions of these, and the confusion matrix. In Section 5.4, two chance-corrected agreement measures, Cohen's kappa and the multiclass Matthews correlation coefficient (MCC), are included. When two models have nearly the same accuracy, the macro F1-score is used as a tie-breaker as it gives equal weight to each class. **Efficiency:** Efficiency includes the size of the trained model, the number of parameters and computational cost, the time it takes to train the model on the available GPU, and the time it takes for the deployed system to return a result. These are significant because the system is not intended to be a GPU server, but rather a web application on free tier.
Reliability: Reliability is the generalization power of a model and the stability of its learning. It can be evaluated as follows: Training accuracy - test accuracy, epoch-

to-epoch variance in the validation loss, epoch-to-epoch variance in the validation accuracy, at system level, whether or not the explanation phase is prevented from taking action on uncertain predictions.

5.1.2 Testing Methods

Both datasets were split once into disjoint partitions, and all three architectures were trained and tested on the same partitions so that the comparison is controlled. Table 5.1 summarises the data used in the final experiments.

Table 5.1: Datasets and partitions used in the final experiments

Property	**SpaceNet (object classification)**	**Galaxy Zoo (morphology)**
Classes	5: constellation, galaxy, nebula, planet, star	2: elliptical, spiral
Valid images after cleaning	10,093 (from 11,469 in the five kept classes)	61,578 (all used)
Training set	8,074	44,489
Validation set	Not separate (see Section 5.6)	7,852
Test set	2,019	9,237
Training class counts	constellation 1,241; galaxy 2,086; nebula 954; planet 1,178; star 2,615	spiral 25,806; elliptical 18,683
Test class counts	constellation 311; galaxy 522; nebula 238; planet 294; star 654	spiral 5,358; elliptical 3,879
Majority-class baseline	32.4% (star)	58.0% (spiral)

All images were resized to 224×224 pixels and normalised with ImageNet statistics. Training images were only subjected to random flips and small rotations. Both networks were initialized with weights from ImageNet and their last layer was replaced with the number of classes. All parameters were optimized. The AdamW optimiser (learning rate 1×10^{-4}, weight decay 1×10^{-4}), cross-entropy loss and mixed-precision arithmetic were used for 30 epochs of training. Galaxy Zoo had a batch size of 64 and SpaceNet had a batch size of 32. The learning rate was decreased automatically during training: in the Galaxy Zoo logs it halves at epochs 10, 14, 17, 20, 23, 26 and 29, dropping from 1×10^{-4} to 7.81×10^{-7} (below 1×10^{-6}) by epoch 30. The imbalanced classes were given the full training budget by effectively turning off early stopping by setting the patience to the number of epochs. The checkpoint that achieved the best validation accuracy was reloaded and tested again on the test set after training.

The accuracy of the training was evaluated after training on the original training images without augmentation, which is comparable to the test accuracy. The classification report, confusion matrix and accuracy, macro F1 and macro recall were the outputs of the test-set. Here all results are obtained from a single training run for each model and dataset.

5.1.3 Results of the Selected Model

The final system (Section 5.3) is chosen to be EfficientNet-B0. The results are presented first, while the other two architectures are presented in Section 5.2.

SpaceNet, EfficientNet-B0

Achieved an accuracy of 83.66%, macro F1 score of 0.8362 and macro recall of 0.8342 on the 2,019 test images. The training accuracy was 94.20%. The learning curves, the confusion matrix and the per-class scores are presented in figure 5.1. Table 5.2 gives the per-class results.

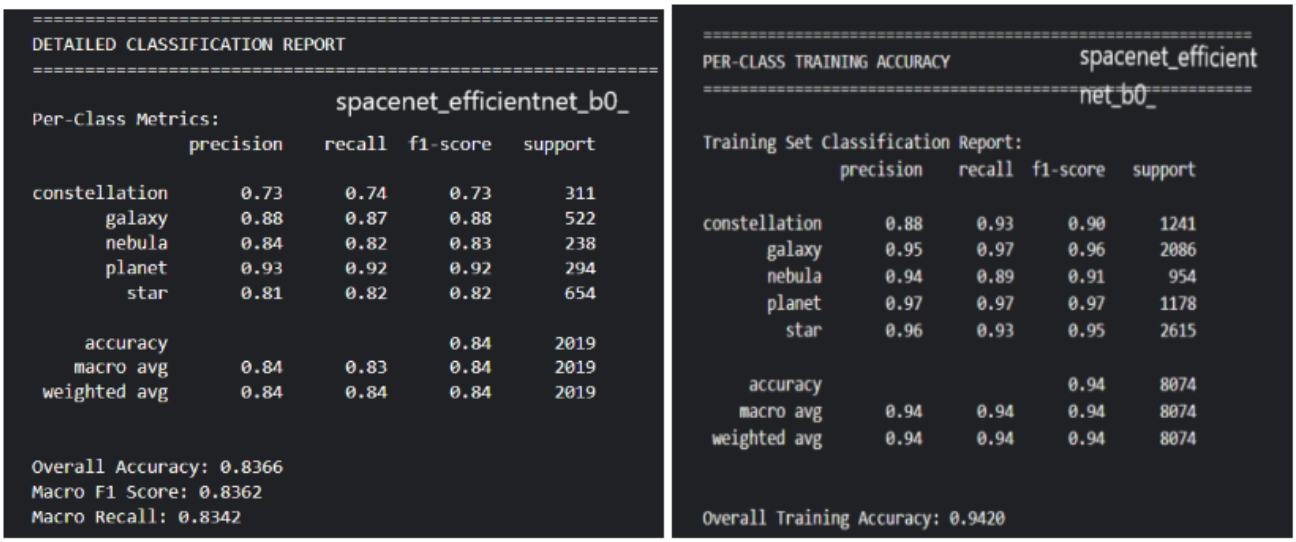


Figure 5.1: SpaceNet EfficientNet-B0 training and validation accuracy, loss, test confusion matrix and per-class F1-score and recall

Table 5.2: SpaceNet EfficientNet-B0 per-class test results

Class	Precision	Recall	F1-score	Support
constellation	0.73	0.74	0.73	311
galaxy	0.88	0.87	0.88	522
nebula	0.84	0.82	0.83	238
planet	0.93	0.92	0.92	294
star	0.81	0.82	0.82	654
Macro average	0.84	0.83	0.84	2,019

The easiest class was planet (F1 = 0.92) and the hardest was constellation (F1 = 0.73). The confusion matrix indicates that the majority of the errors are related to the star class. Of the 330 misclassified test images, 240 (72.7%) had either a true label or a predicted label of star, and the two largest single error cells are constellation predicted as star and star predicted as constellation (44 images each). Other errors were also concentrated on galaxy and nebula images: 34 galaxy images and 28 nebula images were predicted as star, and 25 galaxy images were predicted as

constellations. By contrast, Planet was seldom mistaken for anything other than a star (19 images).

Galaxy Zoo, EfficientNet-B0

On the 9,237 test images EfficientNet-B0 achieved an accuracy of 86.25%, a macro F1-score of 0.8585 and a macro recall of 0.8575, with a training accuracy of 87.12%. The highest validation accuracy of 86.04% was obtained at epoch 7 and did not improve in the remaining 23 epochs. The results are shown in Figure 5.2 and Table 5.3.

```
================================================================
STEP 9: TRAINING LOOP (30 EPOCHS - GPU)
================================================================

Epoch [ 1/30] | Train: L=0.4026 A=0.8156 | Val: L=0.3542 A=0.8402 | LR=1.00e-04 | 368s ✓ BEST
Epoch [ 2/30] | Train: L=0.3497 A=0.8414 | Val: L=0.3354 A=0.8456 | LR=1.00e-04 | 341s ✓ BEST
Epoch [ 3/30] | Train: L=0.3328 A=0.8504 | Val: L=0.3504 A=0.8389 | LR=1.00e-04 | 345s
Epoch [ 4/30] | Train: L=0.3210 A=0.8544 | Val: L=0.3284 A=0.8519 | LR=1.00e-04 | 343s ✓ BEST
Epoch [ 5/30] | Train: L=0.3122 A=0.8590 | Val: L=0.3226 A=0.8566 | LR=1.00e-04 | 344s ✓ BEST
Epoch [ 6/30] | Train: L=0.3076 A=0.8611 | Val: L=0.3120 A=0.8583 | LR=1.00e-04 | 345s ✓ BEST
Epoch [ 7/30] | Train: L=0.3028 A=0.8639 | Val: L=0.3166 A=0.8604 | LR=1.00e-04 | 344s ✓ BEST
Epoch [ 8/30] | Train: L=0.2948 A=0.8679 | Val: L=0.3132 A=0.8586 | LR=1.00e-04 | 346s
Epoch [ 9/30] | Train: L=0.2938 A=0.8688 | Val: L=0.3141 A=0.8562 | LR=5.00e-05 | 342s
Epoch [10/30] | Train: L=0.2802 A=0.8742 | Val: L=0.3135 A=0.8599 | LR=5.00e-05 | 342s
Epoch [11/30] | Train: L=0.2719 A=0.8766 | Val: L=0.3118 A=0.8577 | LR=5.00e-05 | 338s
Epoch [12/30] | Train: L=0.2679 A=0.8791 | Val: L=0.3228 A=0.8557 | LR=5.00e-05 | 335s
Epoch [13/30] | Train: L=0.2644 A=0.8787 | Val: L=0.3218 A=0.8552 | LR=5.00e-05 | 336s
Epoch [14/30] | Train: L=0.2609 A=0.8844 | Val: L=0.3327 A=0.8515 | LR=2.50e-05 | 337s
Epoch [15/30] | Train: L=0.2507 A=0.8866 | Val: L=0.3245 A=0.8577 | LR=2.50e-05 | 335s
Epoch [16/30] | Train: L=0.2470 A=0.8883 | Val: L=0.3284 A=0.8561 | LR=2.50e-05 | 337s
Epoch [17/30] | Train: L=0.2429 A=0.8909 | Val: L=0.3296 A=0.8603 | LR=1.25e-05 | 335s
Epoch [18/30] | Train: L=0.2395 A=0.8922 | Val: L=0.3309 A=0.8567 | LR=1.25e-05 | 338s
Epoch [19/30] | Train: L=0.2365 A=0.8930 | Val: L=0.3370 A=0.8574 | LR=1.25e-05 | 337s
Epoch [20/30] | Train: L=0.2343 A=0.8949 | Val: L=0.3380 A=0.8580 | LR=6.25e-06 | 336s
Epoch [21/30] | Train: L=0.2320 A=0.8950 | Val: L=0.3331 A=0.8589 | LR=6.25e-06 | 339s
Epoch [22/30] | Train: L=0.2281 A=0.8974 | Val: L=0.3392 A=0.8594 | LR=6.25e-06 | 336s
Epoch [23/30] | Train: L=0.2290 A=0.8979 | Val: L=0.3429 A=0.8583 | LR=3.13e-06 | 335s
Epoch [24/30] | Train: L=0.2269 A=0.8987 | Val: L=0.3408 A=0.8572 | LR=3.13e-06 | 341s
Epoch [25/30] | Train: L=0.2267 A=0.8978 | Val: L=0.3462 A=0.8590 | LR=3.13e-06 | 340s
Epoch [26/30] | Train: L=0.2267 A=0.8983 | Val: L=0.3480 A=0.8584 | LR=1.56e-06 | 340s
Epoch [27/30] | Train: L=0.2249 A=0.8995 | Val: L=0.3405 A=0.8590 | LR=1.56e-06 | 336s
Epoch [28/30] | Train: L=0.2216 A=0.9010 | Val: L=0.3418 A=0.8586 | LR=1.56e-06 | 334s
Epoch [29/30] | Train: L=0.2243 A=0.8999 | Val: L=0.3417 A=0.8589 | LR=7.81e-07 | 331s
Epoch [30/30] | Train: L=0.2242 A=0.8994 | Val: L=0.3415 A=0.8595 | LR=7.81e-07 | 333s

✓ Training completed in 169.8 minutes
✓ Loaded best model (val acc = 0.8604) from /content/output/efficientnet_b0_galaxyzoo_best.pth
✓ Best model saved in: /content/output/efficientnet_b0_galaxyzoo_best.pth
```

Figure 5.2

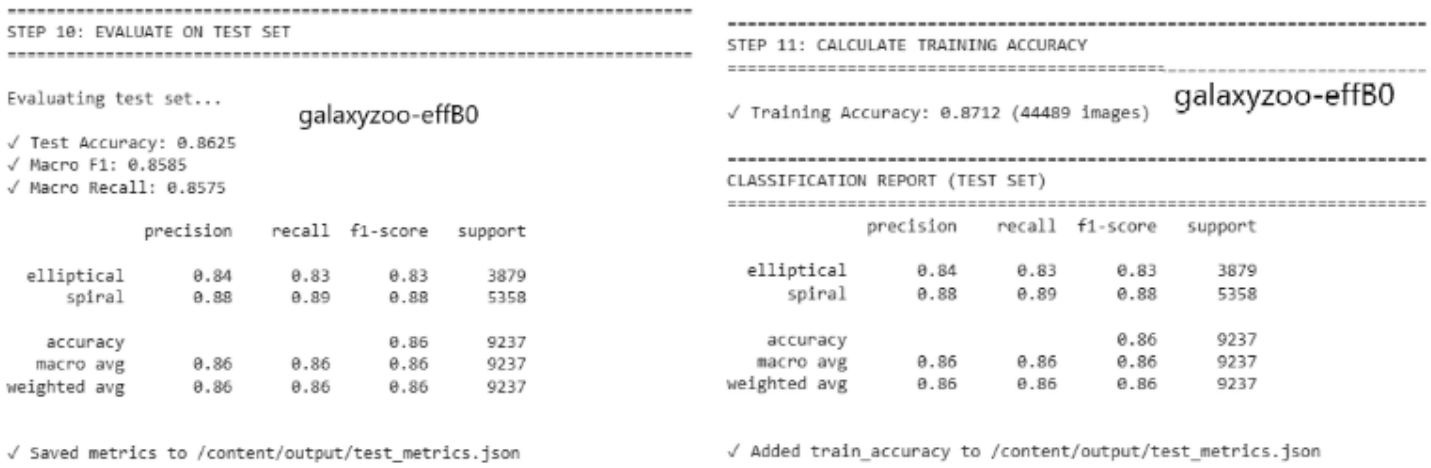


Figure 5.3: Galaxy Zoo EfficientNet-B0 training and validation accuracy, loss, test confusion matrix and per-class F1-score and recall

Table 5.3: Galaxy Zoo EfficientNet-B0 per-class test results

Class	**Precision**	**Recall**	**F1-score**	**Support**
elliptical	0.84	0.83	0.83	3,879
spiral	0.88	0.89	0.88	5,358
Macro average	0.86	0.86	0.86	9,237

Of the 3,879 elliptical galaxies, 675 (17.4%) were classified as spiral, and only 595 of the 5,358 spiral galaxies (11.1%) were classified as elliptical. The model thus tends to the majority class, as the training data has a 58%/42% ratio of spiral-to-elliptical.

5.2 Analysis of Design Solutions

5.2.1 Description of the Candidate Solutions

For each task, three CNN classifiers were evaluated and all were paired with the same retrieval-augmented explanation stage. ResNet18 is an 18-layer residual network and was used as a lightweight baseline. ResNet50 is a 50-layer network constructed using bottleneck residual blocks, and was added to see if more layers are beneficial for similar classes in the visual domain. EfficientNet-B0 is a model that scales depth, width and input resolution in a compound manner to achieve competitive accuracy with significantly fewer parameters. Their static costs are shown in Table 5.4.

Table 5.4: Size and compute of the candidate architectures (parameters and FLOPs are the standard published figures for the 1000-class ImageNet versions; checkpoint sizes are from the training logs, except ResNet50, which is estimated from its parameter count)

Model	**Parameters (approx.)**	**Compute per 224×224 image (approx.)**	**Saved checkpoint**
ResNet18	11.7 M	1.8 GFLOPs	about 45 MB (measured)
ResNet50	25.6 M	4.1 GFLOPs	roughly 90–100 MB
EfficientNet-B0	5.3 M	0.39 GFLOPs	about 16 MB (measured)

5.2.2 Accuracy

The test results of all six trained models are shown in Table 5.5. On SpaceNet the three architectures are almost indistinguishable: EfficientNet-B0 (83.66%) is ahead of ResNet50 (83.51%) and ResNet18 (83.01%) by less than one percentage point. The order is reversed on Galaxy Zoo: ResNet18 is best (86.86%), followed by EfficientNet-B0 (86.25%) and ResNet50 (85.95%) within one percentage point.

Table 5.5: Test results of all models (* the SpaceNet validation set is the same as the test set; see Section 5.6)

Dataset	**Model**	**Training acc.**	**Best val. acc.**	**Test acc.**	**Macro F1**	**Macro recall**
SpaceNet	ResNet18	0.9370	0.8301*	0.8301	0.8322	0.8344
SpaceNet	ResNet50	0.9500	0.8351*	0.8351	0.8348	0.8326
SpaceNet	EfficientNet-B0	0.9420	n/a	0.8366	0.8362	0.8342
Galaxy Zoo	ResNet18	0.8907	0.8678	0.8686	0.8660	0.8688
Galaxy Zoo	ResNet50	0.9162	0.8623	0.8595	0.8572	0.8612
Galaxy Zoo	EfficientNet-B0	0.8712	0.8604	0.8625	0.8585	0.8575

The learning curves, confusion matrices and per-class scores of the four models not presented in Section 5.1.3 are presented in Figures 5.4 to 5.10. The SpaceNet models have the same weaknesses: constellations (F1 0.73-0.74 for all three models), star and nebula, with planet being the best class (F1 0.92 for all three). The per-class F1-scores of SpaceNet are directly compared in Figure 5.11.

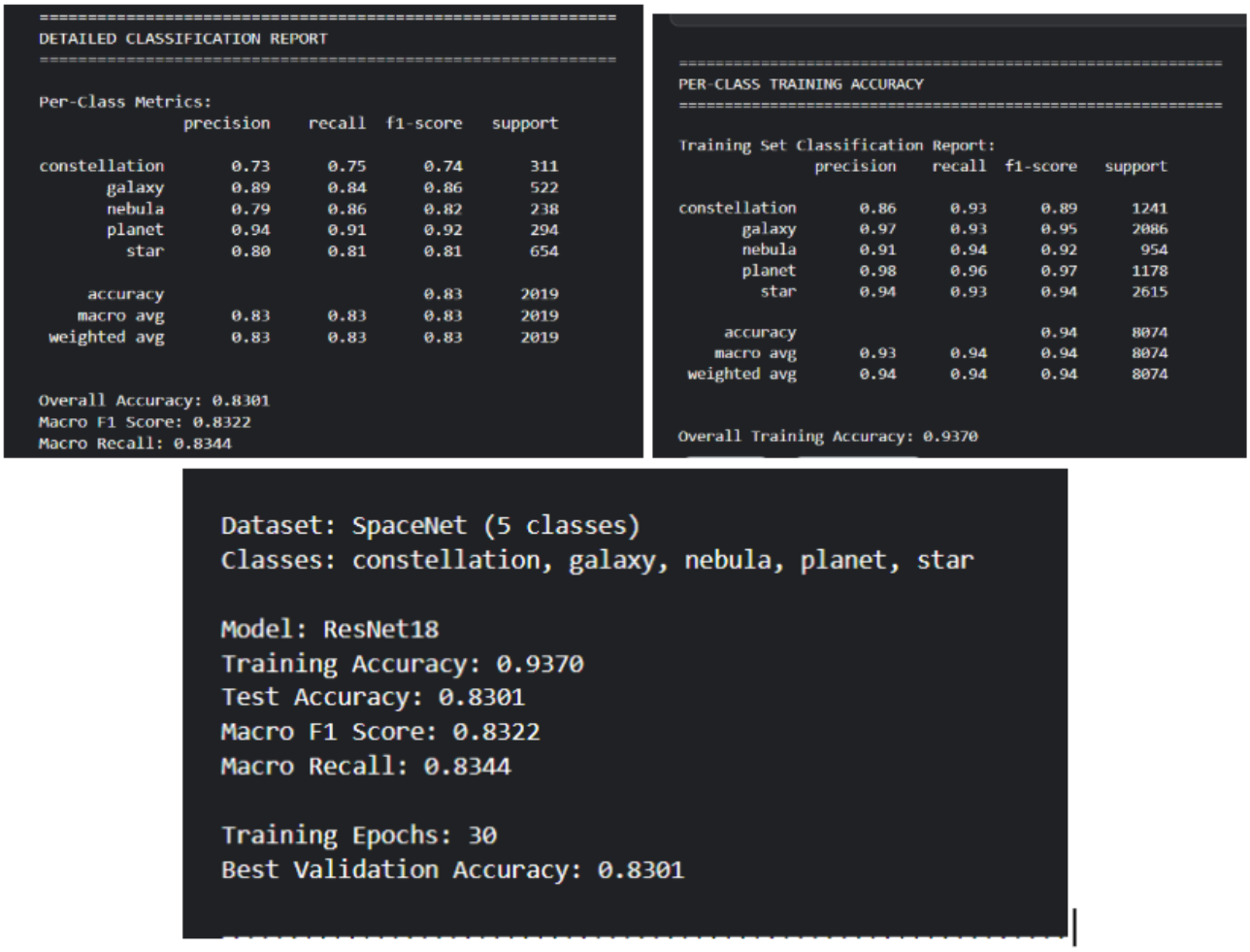


Figure 5.4: SpaceNet ResNet18 learning curves, test confusion matrix and per-class F1-score and recall

```
Epoch [ 1/30] | Train: L=0.7831 A=0.7083 | Val: L=0.5138 A=0.7949 ✓ BEST
Epoch [ 2/30] | Train: L=0.4791 A=0.8085 | Val: L=0.4638 A=0.8073 ✓ BEST
Epoch [ 3/30] | Train: L=0.4047 A=0.8330 | Val: L=0.4562 A=0.8039
Epoch [ 4/30] | Train: L=0.3664 A=0.8498 | Val: L=0.4566 A=0.8113 ✓ BEST
Epoch [ 5/30] | Train: L=0.3142 A=0.8671 | Val: L=0.4310 A=0.8123 ✓ BEST
Epoch [ 6/30] | Train: L=0.2857 A=0.8791 | Val: L=0.3981 A=0.8286 ✓ BEST
Epoch [ 7/30] | Train: L=0.2608 A=0.8887 | Val: L=0.4284 A=0.8148
Epoch [ 8/30] | Train: L=0.2374 A=0.8965 | Val: L=0.4228 A=0.8276
Epoch [ 9/30] | Train: L=0.2286 A=0.9015 | Val: L=0.4505 A=0.8182
Epoch [10/30] | Train: L=0.2132 A=0.9054 | Val: L=0.4284 A=0.8291 ✓ BEST
Epoch [11/30] | Train: L=0.1943 A=0.9112 | Val: L=0.4350 A=0.8306 ✓ BEST
Epoch [12/30] | Train: L=0.1819 A=0.9163 | Val: L=0.4463 A=0.8237
Epoch [13/30] | Train: L=0.1813 A=0.9175 | Val: L=0.4664 A=0.8301
Epoch [14/30] | Train: L=0.1791 A=0.9165 | Val: L=0.5076 A=0.8207
Epoch [15/30] | Train: L=0.1703 A=0.9217 | Val: L=0.4965 A=0.8247
Epoch [16/30] | Train: L=0.1699 A=0.9214 | Val: L=0.4913 A=0.8306
Epoch [17/30] | Train: L=0.1491 A=0.9289 | Val: L=0.5365 A=0.8247
Epoch [18/30] | Train: L=0.1509 A=0.9244 | Val: L=0.5152 A=0.8281
Epoch [19/30] | Train: L=0.1446 A=0.9314 | Val: L=0.5138 A=0.8311 ✓ BEST
Epoch [20/30] | Train: L=0.1454 A=0.9290 | Val: L=0.5254 A=0.8316 ✓ BEST
Epoch [21/30] | Train: L=0.1443 A=0.9304 | Val: L=0.5361 A=0.8192
Epoch [22/30] | Train: L=0.1530 A=0.9257 | Val: L=0.5609 A=0.8242
Epoch [23/30] | Train: L=0.1409 A=0.9314 | Val: L=0.5013 A=0.8276
Epoch [24/30] | Train: L=0.1275 A=0.9361 | Val: L=0.5788 A=0.8177
Epoch [25/30] | Train: L=0.1402 A=0.9321 | Val: L=0.5548 A=0.8162
Epoch [26/30] | Train: L=0.1420 A=0.9314 | Val: L=0.5397 A=0.8217
Epoch [27/30] | Train: L=0.1290 A=0.9349 | Val: L=0.5394 A=0.8351 ✓ BEST
Epoch [28/30] | Train: L=0.1263 A=0.9340 | Val: L=0.5478 A=0.8262
Epoch [29/30] | Train: L=0.1213 A=0.9387 | Val: L=0.5881 A=0.8281
Epoch [30/30] | Train: L=0.1247 A=0.9367 | Val: L=0.5966 A=0.8222

✓ Training completed in 248.1 minutes
```

Figure 5.5

```
======================================
STEP 10: EVALUATE ON TEST SET
======================================

Evaluating test set...

✓ Test Accuracy: 0.8351
✓ Macro F1: 0.8348
✓ Macro Recall: 0.8326
```

```
======================================================================
STEP 11: CALCULATE TRAINING ACCURACY
======================================================================

✓ Training Accuracy: 0.9500

======================================================================
CLASSIFICATION REPORT (TEST SET)
======================================================================
              precision    recall  f1-score   support

constellation       0.74      0.73      0.73       311
      galaxy        0.88      0.85      0.86       522
      nebula        0.83      0.83      0.83       238
      planet        0.93      0.91      0.92       294
        star        0.81      0.84      0.82       654

    accuracy                            0.84      2019
   macro avg        0.84      0.83      0.83      2019
weighted avg        0.84      0.84      0.84      2019
```

Figure 5.6: SpaceNet ResNet50 learning curves, test confusion matrix and per-class F1-score and recall

```
======================================================================
STEP 9: TRAINING LOOP (30 EPOCHS - GPU)
======================================================================

Epoch [ 1/30] | Train: L=0.3734 A=0.8303 | Val: L=0.3767 A=0.8333 | LR=1.00e-04 | 323s ✓ BEST
Epoch [ 2/30] | Train: L=0.3335 A=0.8502 | Val: L=0.3250 A=0.8535 | LR=1.00e-04 | 325s ✓ BEST
Epoch [ 3/30] | Train: L=0.3248 A=0.8556 | Val: L=0.3766 A=0.8370 | LR=1.00e-04 | 326s
Epoch [ 4/30] | Train: L=0.3204 A=0.8579 | Val: L=0.3451 A=0.8492 | LR=1.00e-04 | 324s
Epoch [ 5/30] | Train: L=0.3151 A=0.8594 | Val: L=0.3441 A=0.8446 | LR=5.00e-05 | 323s
Epoch [ 6/30] | Train: L=0.3023 A=0.8661 | Val: L=0.3131 A=0.8579 | LR=5.00e-05 | 326s ✓ BEST
Epoch [ 7/30] | Train: L=0.2980 A=0.8673 | Val: L=0.3099 A=0.8571 | LR=5.00e-05 | 327s
Epoch [ 8/30] | Train: L=0.2941 A=0.8688 | Val: L=0.3097 A=0.8632 | LR=5.00e-05 | 325s ✓ BEST
Epoch [ 9/30] | Train: L=0.2922 A=0.8697 | Val: L=0.2976 A=0.8644 | LR=5.00e-05 | 324s ✓ BEST
Epoch [10/30] | Train: L=0.2893 A=0.8699 | Val: L=0.3190 A=0.8557 | LR=5.00e-05 | 325s
Epoch [11/30] | Train: L=0.2869 A=0.8715 | Val: L=0.3164 A=0.8569 | LR=5.00e-05 | 324s
Epoch [12/30] | Train: L=0.2819 A=0.8733 | Val: L=0.3063 A=0.8622 | LR=2.50e-05 | 324s
Epoch [13/30] | Train: L=0.2733 A=0.8774 | Val: L=0.3048 A=0.8668 | LR=2.50e-05 | 323s ✓ BEST
Epoch [14/30] | Train: L=0.2710 A=0.8795 | Val: L=0.3083 A=0.8607 | LR=2.50e-05 | 319s
Epoch [15/30] | Train: L=0.2701 A=0.8777 | Val: L=0.3119 A=0.8649 | LR=1.25e-05 | 323s
Epoch [16/30] | Train: L=0.2616 A=0.8822 | Val: L=0.3118 A=0.8611 | LR=1.25e-05 | 323s
Epoch [17/30] | Train: L=0.2589 A=0.8828 | Val: L=0.3042 A=0.8674 | LR=1.25e-05 | 333s ✓ BEST
Epoch [18/30] | Train: L=0.2592 A=0.8824 | Val: L=0.3081 A=0.8647 | LR=6.25e-06 | 325s
Epoch [19/30] | Train: L=0.2548 A=0.8851 | Val: L=0.3085 A=0.8650 | LR=6.25e-06 | 325s
Epoch [20/30] | Train: L=0.2523 A=0.8854 | Val: L=0.3127 A=0.8622 | LR=6.25e-06 | 327s
Epoch [21/30] | Train: L=0.2517 A=0.8872 | Val: L=0.3069 A=0.8678 | LR=3.13e-06 | 327s ✓ BEST
Epoch [22/30] | Train: L=0.2488 A=0.8887 | Val: L=0.3077 A=0.8659 | LR=3.13e-06 | 333s
Epoch [23/30] | Train: L=0.2464 A=0.8885 | Val: L=0.3139 A=0.8633 | LR=3.13e-06 | 328s
Epoch [24/30] | Train: L=0.2463 A=0.8886 | Val: L=0.3172 A=0.8614 | LR=1.56e-06 | 324s
Epoch [25/30] | Train: L=0.2445 A=0.8897 | Val: L=0.3131 A=0.8645 | LR=1.56e-06 | 319s
Epoch [26/30] | Train: L=0.2465 A=0.8879 | Val: L=0.3173 A=0.8630 | LR=1.56e-06 | 319s
Epoch [27/30] | Train: L=0.2460 A=0.8892 | Val: L=0.3116 A=0.8659 | LR=7.81e-07 | 319s
Epoch [28/30] | Train: L=0.2442 A=0.8902 | Val: L=0.3129 A=0.8644 | LR=7.81e-07 | 317s
Epoch [29/30] | Train: L=0.2463 A=0.8900 | Val: L=0.3149 A=0.8630 | LR=7.81e-07 | 321s
Epoch [30/30] | Train: L=0.2428 A=0.8907 | Val: L=0.3145 A=0.8655 | LR=3.91e-07 | 323s
```

Figure 5.7

```
----------------------------------------------------------------------
STEP 11: CALCULATE TRAINING ACCURACY
----------------------------------------------------------------------

✓ Training Accuracy: 0.8907

----------------------------------------------------------------------
CLASSIFICATION REPORT (TEST SET)
----------------------------------------------------------------------
              precision    recall  f1-score   support

  elliptical       0.83      0.87      0.85      3879
      spiral       0.90      0.87      0.88      5358

    accuracy                           0.87      9237
   macro avg       0.86      0.87      0.87      9237
weighted avg       0.87      0.87      0.87      9237
```

```
======================================================================
STEP 10: EVALUATE ON TEST SET
======================================================================

Evaluating test set...

✓ Test Accuracy: 0.8686
✓ Macro F1: 0.8660
✓ Macro Recall: 0.8688

              precision    recall  f1-score   support

  elliptical       0.83      0.87      0.85      3879
      spiral       0.90      0.87      0.88      5358

    accuracy                           0.87      9237
   macro avg       0.86      0.87      0.87      9237
weighted avg       0.87      0.87      0.87      9237
```

Figure 5.8: Galaxy Zoo ResNet18 training and validation accuracy, loss, test confusion matrix and per-class F1-score and recall

```
Epoch [ 1/30] | Train: L=0.3673 A=0.8355 | Val: L=0.3834 A=0.8180 | LR=1.00e-04 | 197s ✓ BEST
Epoch [ 2/30] | Train: L=0.3238 A=0.8561 | Val: L=0.3766 A=0.8353 | LR=1.00e-04 | 199s ✓ BEST
Epoch [ 3/30] | Train: L=0.3127 A=0.8610 | Val: L=0.3639 A=0.8288 | LR=1.00e-04 | 199s
Epoch [ 4/30] | Train: L=0.3072 A=0.8640 | Val: L=0.3161 A=0.8586 | LR=1.00e-04 | 198s ✓ BEST
Epoch [ 5/30] | Train: L=0.3013 A=0.8652 | Val: L=0.3245 A=0.8524 | LR=1.00e-04 | 198s
Epoch [ 6/30] | Train: L=0.2964 A=0.8673 | Val: L=0.3084 A=0.8593 | LR=1.00e-04 | 199s ✓ BEST
Epoch [ 7/30] | Train: L=0.2917 A=0.8693 | Val: L=0.3167 A=0.8565 | LR=1.00e-04 | 198s
Epoch [ 8/30] | Train: L=0.2890 A=0.8715 | Val: L=0.3197 A=0.8565 | LR=1.00e-04 | 198s
Epoch [ 9/30] | Train: L=0.2810 A=0.8744 | Val: L=0.3034 A=0.8594 | LR=1.00e-04 | 198s ✓ BEST
Epoch [10/30] | Train: L=0.2793 A=0.8747 | Val: L=0.3131 A=0.8524 | LR=1.00e-04 | 199s
Epoch [11/30] | Train: L=0.2764 A=0.8766 | Val: L=0.3130 A=0.8581 | LR=1.00e-04 | 199s
Epoch [12/30] | Train: L=0.2731 A=0.8784 | Val: L=0.3345 A=0.8490 | LR=5.00e-05 | 198s
Epoch [13/30] | Train: L=0.2578 A=0.8849 | Val: L=0.3085 A=0.8621 | LR=5.00e-05 | 198s ✓ BEST
Epoch [14/30] | Train: L=0.2514 A=0.8889 | Val: L=0.3195 A=0.8594 | LR=5.00e-05 | 198s
Epoch [15/30] | Train: L=0.2444 A=0.8909 | Val: L=0.3369 A=0.8566 | LR=2.50e-05 | 198s
Epoch [16/30] | Train: L=0.2323 A=0.8975 | Val: L=0.3327 A=0.8565 | LR=2.50e-05 | 198s
Epoch [17/30] | Train: L=0.2237 A=0.9006 | Val: L=0.3285 A=0.8605 | LR=2.50e-05 | 198s
Epoch [18/30] | Train: L=0.2212 A=0.9009 | Val: L=0.3289 A=0.8603 | LR=1.25e-05 | 198s
Epoch [19/30] | Train: L=0.2124 A=0.9069 | Val: L=0.3407 A=0.8593 | LR=1.25e-05 | 198s
Epoch [20/30] | Train: L=0.2075 A=0.9086 | Val: L=0.3456 A=0.8623 | LR=1.25e-05 | 198s ✓ BEST
Epoch [21/30] | Train: L=0.2065 A=0.9084 | Val: L=0.3493 A=0.8566 | LR=6.25e-06 | 198s
Epoch [22/30] | Train: L=0.2007 A=0.9129 | Val: L=0.3552 A=0.8599 | LR=6.25e-06 | 199s
Epoch [23/30] | Train: L=0.1990 A=0.9117 | Val: L=0.3565 A=0.8576 | LR=6.25e-06 | 199s
Epoch [24/30] | Train: L=0.1992 A=0.9114 | Val: L=0.3567 A=0.8560 | LR=3.13e-06 | 198s
Epoch [25/30] | Train: L=0.1960 A=0.9148 | Val: L=0.3580 A=0.8586 | LR=3.13e-06 | 198s
Epoch [26/30] | Train: L=0.1940 A=0.9145 | Val: L=0.3607 A=0.8583 | LR=3.13e-06 | 198s
Epoch [27/30] | Train: L=0.1918 A=0.9150 | Val: L=0.3672 A=0.8560 | LR=1.56e-06 | 198s
Epoch [28/30] | Train: L=0.1927 A=0.9158 | Val: L=0.3636 A=0.8583 | LR=1.56e-06 | 198s
Epoch [29/30] | Train: L=0.1903 A=0.9174 | Val: L=0.3627 A=0.8558 | LR=1.56e-06 | 197s
Epoch [30/30] | Train: L=0.1905 A=0.9158 | Val: L=0.3627 A=0.8555 | LR=7.81e-07 | 197s

✓ Training completed in 99.1 minutes
```

Figure 5.9

```
================================================================
STEP 10: EVALUATE ON TEST SET - ResNet50
================================================================

Evaluating test set...

✓ Test Accuracy: 0.8595
✓ Macro F1: 0.8572
✓ Macro Recall: 0.8612

              precision    recall  f1-score   support

  elliptical       0.81      0.87      0.84      3879
      spiral       0.90      0.85      0.88      5358

    accuracy                           0.86      9237
   macro avg       0.86      0.86      0.86      9237
weighted avg       0.86      0.86      0.86      9237

Confusion matrix (rows = true, cols = predicted):
[[3382  497]
 [ 801 4557]]
```

```
================================================================
STEP 11: CALCULATE TRAINING ACCURACY - ResNet50
================================================================

✓ Training Accuracy (clean images): 0.9162

================================================================
CLASSIFICATION REPORT (TEST SET)
================================================================
              precision    recall  f1-score   support

  elliptical       0.81      0.87      0.84      3879
      spiral       0.90      0.85      0.88      5358

    accuracy                           0.86      9237
   macro avg       0.86      0.86      0.86      9237
weighted avg       0.86      0.86      0.86      9237
```

Figure 5.10: Galaxy Zoo ResNet50 training and validation accuracy, loss, test confusion matrix and per-class F1-score and recall

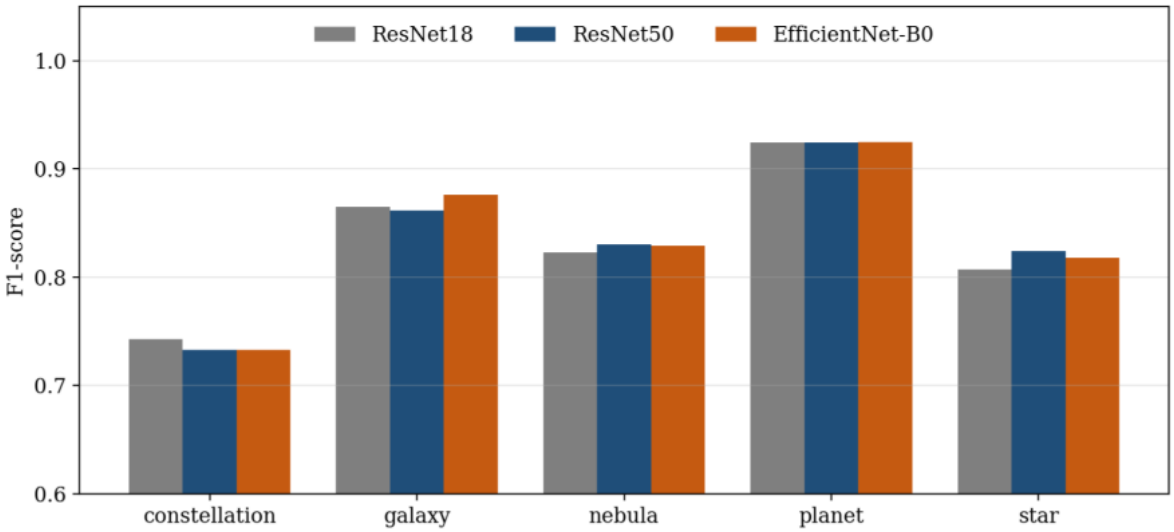


Figure 5.11: SpaceNet per-class test F1-score of the three architectures

The learning curves are the same for all six runs. The accuracy of the training increases steadily over the 30 epochs, while the accuracy of the validation does not increase after a few epochs and then oscillates within a narrow range. On Galaxy Zoo, EfficientNet-B0 achieved the highest validation accuracy at epoch 7, ResNet50 at epoch 20, and ResNet18 at epoch 21, with the validation accuracy of the last 20 epochs of each model remaining within about 1 percentage point of the best accuracy. The validation loss of each model drops to a minimum early on and then increases (e.g. ResNet50 on Galaxy Zoo has a minimum of 0.303 at epoch 9 and ends at 0.363), which is the hallmark of over-fitting to the training set.

5.2.3 Efficiency, Feasibility and Cost

The three architectures are very similar in terms of accuracy, so efficiency is the key factor. EfficientNet-B0 has approximately 1/5 of the parameters of ResNet50 and approximately 1/10 of the floating-point operations (Table 5.4), and its checkpoint (approx. 16 MB) is roughly 1/3 of the size of ResNet18's (approx. 45 MB). This is a practical benefit that is not reflected in the accuracy comparison for a system deployed on a free-tier serverless platform, where package size, memory and start-up time are limited.

The amount of training time was heavily reliant on the hardware. The training times are not comparable as the runs were not performed on the same accelerator: the Galaxy Zoo runs took 169.8 minutes for EfficientNet-B0 and 162.1 minutes for ResNet18 on a Google Colab GPU, while ResNet50 took 99.1 minutes on a Kaggle GPU. What can be said is that all runs were completed with free GPU allowances, meaning that the training cost was zero in monetary terms and the experiments can be reproduced by other people without paying for infrastructure.

At inference time the classifier is not the bottleneck. In the deployed pipeline, the image decoding, preprocessing and classification are stored separately from the explanation stage, and the call to the language model (Gemma 4 via Google AI Studio) is the primary factor in the response time. The default reasoning setting of the model used more than 4,000 hidden tokens; in measurements taken on 23 September 2026 it ran for more than 90 seconds without returning an answer and the request ended in a gateway time-out (HTTP 504). The reasoning level was set to minimal, so no hidden tokens were used and a cited answer was returned well within the timeout. The total time for the whole request was 8.8 seconds, of which 8.5 seconds were spent on the explanation, and the time spent on image decoding, the CLIP gate, object classification and galaxy morphology was about 0.28 seconds. The retrieval part of the explanation stage is relatively inexpensive: The knowledge base contains 6,222 text chunks from 113 Wikipedia articles, which are embedded using a 384-dimensional MiniLM model and searched exactly with FAISS. Explanations are also only created for the level the user requests, and only once per level, to avoid paying that latency three times.

5.2.4 Strengths and Weaknesses

Table 5.6: Strengths and weaknesses of the candidate solutions

Solution	**Strengths**	**Weaknesses**
ResNet18	Best accuracy and macro F1 on Galaxy Zoo (86.86%, 0.866); most balanced recall between elliptical (0.871) and spiral (0.867); simple and fast to train.	Larger checkpoint than EfficientNet-B0 (about 45 MB against 16 MB); lowest SpaceNet accuracy (83.01%); training accuracy still 10.7 points above test accuracy on SpaceNet.
ResNet50	Highest training accuracy on both datasets (95.00% and 91.62%), showing the greatest capacity; competitive SpaceNet accuracy (83.51%).	Largest and most expensive model; lowest Galaxy Zoo accuracy (85.95%); strongest over-fitting on Galaxy Zoo (5.7-point gap; validation loss rising from 0.303 to 0.363); no accuracy benefit from its extra depth.
EfficientNet-B0	Best SpaceNet accuracy and macro F1 (83.66%, 0.8362); smallest Galaxy Zoo train-test gap (0.9 points); about 5.3 M parameters and 16 MB checkpoint, easiest to deploy.	Lowest Galaxy Zoo recall for the minority class (elliptical 0.826) and a bias towards spiral; validation loss on Galaxy Zoo rises after epoch 11; constellation remains weak (F1 0.73).

5.3 Final Design Adjustments

The experiments led to a number of changes to the design. Some are related to the data and the training protocol, while others are related to the deployed system.

Reducing the SpaceNet label space from eight to five classes: Earlier eight-class experiments (not reported in this thesis) reached about 80% accuracy. The raw SpaceNet class counts were very uneven: asteroid 283, black hole 656 and comet 416 images, against 3,984 for galaxy and 3,269 for star. These three small classes were also the most difficult to distinguish visually from the other classes, and were removed, with the models retrained on the five classes that were better represented (constellation, galaxy, nebula, planet and star). The accuracy of the five-class models in Table 5.5 is therefore not directly comparable with the eight-class figure, since the task is easier, and Section 5.5 treats it as such.

Removing corrupted images: Validation of the images before training found 1,376 unreadable images, all in the galaxy class, reducing it from 3,984 to 2,608 images. There were no corrupted files in the other four classes. The final SpaceNet set is thus composed of 10,093 images (Table 5.1). The corrupted files are flagged but not deleted automatically. If a user re-running this pipeline on their own copy of the data does not remove them first, the pipeline does not fail with a clear error; it silently treats the class as if it had fewer images than it really does.

Training protocol: Both datasets were retrained for 30 epochs with patience of 30, which means that the training would not stop before the minority classes were learned, but the best validation checkpoint was still used for the final evaluation. The learning rate was automatically decreased in steps during training.

Choice of EfficientNet-B0 as the deployed classifier: EfficientNet-B0 is the backbone already used in Chapter 4's deployed system; this section sets out why it was chosen over the other two candidates compared in Section 5.2. There is no accuracy advantage for ResNet50 on either dataset (Table 5.5), it over-fits most on Galaxy Zoo, and it is the most expensive to deploy. EfficientNet-B0 has the best SpaceNet result, a Galaxy Zoo result that is statistically indistinguishable from the best model (Section 5.4), the smallest train-test gap on Galaxy Zoo and the smallest footprint. It was therefore chosen as the final model for both tasks.

Guarding the explanation stage against wrong or irrelevant inputs: Chapter 4 showed that the explanation layer will fluently explain a wrong prediction, since the retrieval and generation stages act on the predicted class whether or not it is correct. The final system thus provides three safeguards. The first is a CLIP-based gate (ViT-B/32, exported to ONNX and quantised) that filters out images that are not astronomical, returning no classification. Secondly, if the highest probability is less than 0.55, or the difference between the two highest probabilities is less than 0.15, the prediction is marked as 'uncertain'. If so, then the retrieval step will also search for the runner-up class, and the language-model prompt will indicate that it is not certain. Third, the number of citations in the generated text is validated against the retrieved text and invalid citations are deleted.

Upgrading the retrieval and generation components: The TF-IDF retrieval over a small hand-written knowledge base was replaced by dense retrieval: a knowledge base built offline from 113 Wikipedia articles (6,222 chunks) embedded with MiniLM and searched with FAISS, followed by generation with Gemma 4. The

passages retrieved are numbered, allowing each statement in the explanation to be traced back to a source. The user can select the level of explanation (beginner, intermediate or advanced).

Persistence and accounts: The final system needs sign-in for most of its functions and keeps the analysis history of each user in a database (MongoDB Atlas) rather than in the browser only, so that previous results can be viewed and assessed.

5.4 Statistical Analysis

The statistical analysis is based on the counts in the confusion matrices, not on the predictions made in the test sets, as only the confusion matrices were saved. The following techniques were used.

Confidence intervals: For each model, a 95% Wilson score interval was calculated for the test accuracy, assuming each test image was an independent Bernoulli trial. Table 5.7 shows the intervals, along with Cohen's kappa and the multiclass MCC, that correct the raw accuracy for the agreement that would be expected by chance.

Table 5.7: Test accuracy with 95% Wilson confidence intervals, Cohen's kappa and MCC

Dataset	**Model**	**Accuracy**	**95% CI**	**Cohen's κ**	**MCC**
SpaceNet	ResNet18	83.01%	81.31 – 84.59%	0.779	0.780
SpaceNet	ResNet50	83.51%	81.82 – 85.06%	0.785	0.785
SpaceNet	EfficientNet-B0	83.66%	81.98 – 85.20%	0.787	0.787
Galaxy Zoo	ResNet18	86.86%	86.15 – 87.53%	0.732	0.733
Galaxy Zoo	ResNet50	85.95%	85.22 – 86.64%	0.715	0.716
Galaxy Zoo	EfficientNet-B0	86.25%	85.53 – 86.94%	0.717	0.717

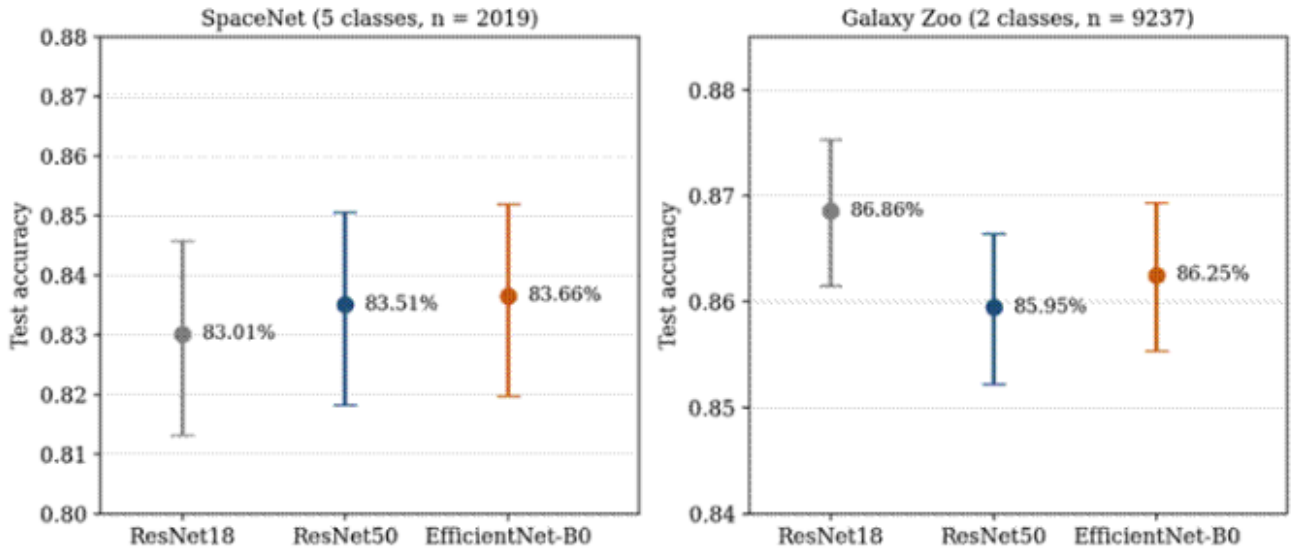


Figure 5.12: Test accuracy of the three architectures with 95% confidence intervals

The three models have almost identical intervals on both data sets (Figure 5.12). The width of an interval is approximately ±1.6 points on SpaceNet and ±0.7 points on Galaxy Zoo, which is larger than or similar to the differences between the models.

Significance of the differences between models: Significance of the differences between models: A two-proportion z-test was used to compare pairs of models on accuracy (Table 5.8). None of the differences is significant at the 5% level. The closest to significance is ResNet18 vs ResNet50 on Galaxy Zoo (0.91 points, p = 0.071). This is an unpaired test, but a paired test like McNemar's would be more sensitive, as it would have been possible here if per-image predictions had been saved. The reported p-values are therefore conservative, but the differences are so small that a paired test is unlikely to change the conclusion for the SpaceNet models, whose accuracies differ by only 3 to 13 test images out of 2,019.

Table 5.8: Two-proportion z-tests on test accuracy between pairs of models (test images out of 2,019)

Dataset	**Comparison**	**Difference (points)**	**z**	**p-value**
SpaceNet	EfficientNet-B0 vs ResNet50	+0.15	0.13	0.90
SpaceNet	EfficientNet-B0 vs ResNet18	+0.64	0.55	0.58
SpaceNet	ResNet50 vs ResNet18	+0.50	0.42	0.67
Galaxy Zoo	ResNet18 vs EfficientNet-B0	+0.61	1.21	0.23
Galaxy Zoo	ResNet18 vs ResNet50	+0.91	1.80	0.071
Galaxy Zoo	EfficientNet-B0 vs ResNet50	+0.30	0.60	0.55

Tests against chance: For all six models, the hypothesis that predictions and true labels are independent is rejected by a chi-square test of independence (p ¡ 0.001 for each model). The six accuracy intervals are well above the majority-class baselines of 32.4% for SpaceNet and 58.0% for Galaxy Zoo, and the kappa values (0.71 to

0.79) show good to very good agreement. The models have thus learned real structure, but the tests do not distinguish between the models.

Generalisation gap: The difference between training and test accuracy (Figure 5.13) is more informative than the test accuracy alone. All three models over-fit by a similar amount (10.5 to 11.5 points) on SpaceNet. The difference increases as the model size increases: 0.9 points for EfficientNet-B0, 2.2 for ResNet18 and 5.7 for ResNet50 on Galaxy Zoo. The test accuracy of each model is estimated with a margin of approximately ±0.7 points, which is quite large for the Galaxy Zoo test set (9,237 images), and the 4.8-point difference in gap between EfficientNet-B0 and ResNet50 is well outside the sampling error.

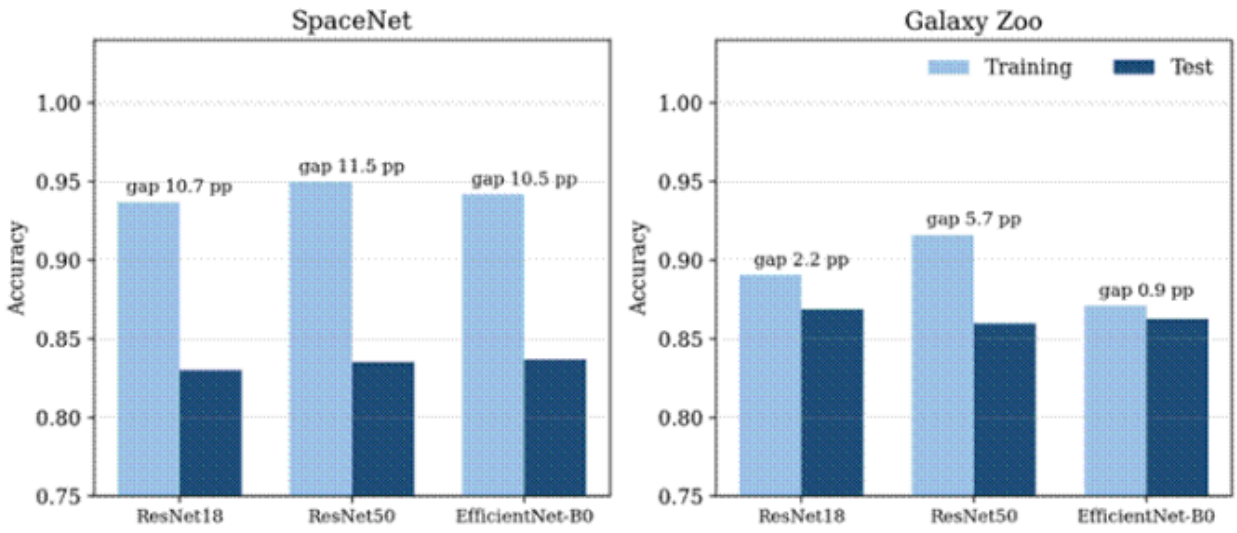


Figure 5.13: Training and test accuracy of the three architectures, with the generalisation gap in percentage points

Class size and class performance: For the SpaceNet EfficientNet-B0 model the Spearman rank correlation between the number of training images per class and the class F1-score is $\rho = -0.30$ (n = 5 classes), which is not significant. The relationship is not monotonic: planet (1,178 training images) is the best class, and constellation (1,241 images) is the worst, while star, with the largest class (2,615 images), has a mid-range F1-score of 0.82. The differences per class are thus not primarily due to the size of the class; the similarity of the classes visually appears to be more important (Section 5.6).

5.5 Comparisons and Relationships

5.5.1 Comparison Between the Approaches

The three architectures reveal three findings. First, there was no benefit in increasing the depth: ResNet50 has approximately 5 times the number of parameters as EfficientNet-B0, but is not more accurate on either dataset and is the least accurate model on Galaxy Zoo. Second, the ranking of the models is dependent on

the dataset, with EfficientNet-B0 nominally first on SpaceNet and ResNet18 nominally first on Galaxy Zoo, but none of these rankings is statistically significant, and should be interpreted as a tie. Third, the models vary in their trade-off between precision and recall for the minority class. Galaxy Zoo: ResNet18 and ResNet50 recover more elliptical galaxies (recall of 0.871 and 0.872, respectively, compared to 0.826 for EfficientNet-B0), but ResNet50 does not do as well on spiral recall (0.851), and EfficientNet-B0 does better on the majority class. Class weighting or a lower decision threshold for the elliptical class would be required if the class of interest were elliptical galaxies.

5.5.2 Comparison Between the Two Tasks

The Galaxy Zoo task has two classes and approximately five and a half times more training data, but its accuracy is only approximately three points higher than the SpaceNet five-class accuracy (about 83%). The difference is better illustrated by the chance-corrected measures: kappa is 0.71–0.73 on Galaxy Zoo and 0.78–0.79 on SpaceNet, which is indeed higher than chance, as the majority-class baseline is much lower on SpaceNet (32.4% vs. 58.0%). The two-class problem is a limited problem due to the quality of the labels. The Galaxy Zoo labels were generated from the vote fractions of volunteers, and galaxies with vote fractions near the decision boundary are ambiguous even for humans, which sets a limit on the accuracy any architecture can achieve on this task. All three architectures perform very similarly (85.95% to 86.86%) and this is consistent with such a ceiling.

5.5.3 Comparison with Earlier Results and Existing Work

The best accuracy obtained with the earlier eight-class SpaceNet experiments was 80.09% using ResNet50. The five-class models here achieve 83.0% to 83.7%. The increase of three points is not to be interpreted as an improvement of the models, because the task was easier due to the elimination of the three smallest and most ambiguous classes, and the label spaces are different. The advantage is that the resulting classifier is more appropriate for the deployed system, where the confident errors are more important than the rare categories.

The literature reviewed in Chapter 2 does not report results for the same datasets and splits, so a numerical comparison with published accuracies would be misleading, and none is claimed. The findings, however, are consistent with the qualitative findings of that literature. Fine-tuning an ImageNet-pretrained CNN on a relatively small astronomical dataset yields competitive results (Dalvi, 2022), errors focus on visually ambiguous and under-represented classes (Ma et al., 2022), and the networks become over-confident and over-fit their training data (Guo et al., 2017), as illustrated by the gap analysis in Section 5.4 and the discussion of confident-but-wrong predictions in Chapter 4. The major difference between this design and the previous one of this project is that the explanation stage now accesses a larger, cited knowledge base, rejects non-astronomical inputs, and identifies uncertain predictions.

5.5.4 Relationships Observed in the Results

There are a number of relationships that can be made from the results. The more capacity a model has, the more accurate it is at training time (ResNet50 is the best on both datasets), but not at test time. This means that the extra capacity is used to memorise the training data. As the model size grows, the difference between training and test accuracy grows on Galaxy Zoo. Class difficulty is determined by the visual distinctiveness of a class, not by the number of images: planets are clearly bounded and compact, and are separated well by all models; constellations, stars and nebulae are all composed of bright points on a dark background, and are confused with each other. Finally, the quality of the explanation is limited by the quality of the classification: retrieval is based on the predicted class, and if the class is wrong, then the explanation is well written but wrong, so the uncertainty flag and the CLIP gate are placed before the explanation stage.

5.6 Discussions

5.6.1 Interpretation of the Results

The primary finding is that the three CNN architectures are all comparable in the accuracy that the test sets can measure, with an accuracy of approximately 83-84% on the five SpaceNet classes and approximately 86-87% on the two Galaxy Zoo classes. The architecture should be chosen based on cost and reliability, not accuracy, and on these metrics EfficientNet-B0 is the best: it is the smallest and cheapest model, it has the smallest over-fitting on the larger dataset and it is nominally the best on SpaceNet. The observation that the accuracy on the validation set reaches a maximum after a few epochs, while the accuracy on the training set continues to improve, indicates that the models learned most of the transferable information from the ImageNet initialisation in the first few epochs and that the remaining epochs mostly decreased the training loss.

The error analysis shows that the limits are set by the data. The most common errors on SpaceNet are between the categories of star, constellation, galaxy and nebula, which share visual characteristics (points of light, diffuse glow, dark backgrounds) and may also include some mislabelled or ambiguous images in a publicly collected dataset. On Galaxy Zoo the elliptical class is the one that is lost, as a result of both the class imbalance and the overlap between smooth spirals and ellipticals.

5.6.2 Implications

The results suggest that a lightweight classifier is enough for the AdAstra platform and that it is better to invest in the protection of the platform than in finding a larger backbone. The classification accuracy is approximately 84%, which translates to about one misclassification every six images like SpaceNet, so it is important that

the interface displays the confidence and the uncertainty flag, as well as the result, and that the explanation references its sources. The system is most useful for the categories that the models are best suited to (planets, galaxies) and least reliable for constellations and stars, and this should be made clear to users.

5.6.3 Limitations

The results have the following limitations, which should be taken into account when interpreting them.

Single run per model: The variance between runs is not known for each model as they were trained with only one random seed and one data split. When comparing the results across epochs (around ± 1 percentage point on both datasets), the results are as big as the differences among models, suggesting that the models cannot be distinguished in terms of accuracy. This would require repeated runs using different seeds (or k-fold cross validation).

No independent validation set for SpaceNet: The SpaceNet data were split into 8,074 training images and 2,019 test images, and the best accuracy recorded during training under the label "validation accuracy" is identical to the test accuracy (0.8301 for ResNet18 and 0.8351 for ResNet50) to 4 decimal places, because that "validation set" is the 2,019-image test set itself. The reported SpaceNet test accuracies are thus a little bit optimistic. This is not the case with the Galaxy Zoo experiments, which were validated with another set of 7,852 images. In subsequent SpaceNet experiments, it is recommended to use a separate validation partition.

Data quality and dataset shift: The SpaceNet images were taken from public sources and 1,376 of its galaxy pictures were corrupted from a public Kaggle repository. The models were tested only on images from the same collections and the ability of the models to classify images from other surveys and/or observing conditions is a known limitation of astronomical classifiers (Chapter 2).

Calibration was not measured: Uncertainty flag was based on predicted probabilities with fixed thresholds (0.55 and 0.15) without evaluation of the calibration of the models (e.g. expected calibration error) or tuning of the thresholds on held-out data.

The explanation stage was not quantitatively evaluated: The retrieval augmented explanations data includes the correct and incorrect examples described in Chapter 4, and the system latency tests and citation checks. No labelled set of examples have been used as a measure of faithfulness and correctness of the explanations, and a study with users has not been conducted.

Hardware and comparability: The training times were collected on different GPU platforms, are not suitable to rank the architectures, and the parameter counts and FLOPs of Table 5.4 are published values, not measured in this work.

While these compromises must be accepted, the results paint a consistent picture:

in terms of accuracy, the three architectures are all the same within the margin of error of the test sets; in terms of performance, selecting the small architecture will incur no cost; and the rest of the errors are all clustered in a few visually similar classes, a region where future work with respect to data quality, class balancing and uncertainty handling should focus.

Chapter 6

Conclusion

6.1 Summary of Findings

The main goal of this thesis was to create an astronomical image analysis system that can automatically classify images and give understandable and knowledge-based explanations for its predictions. The resulting system, AdAstra, combines the image recognition capabilities of a convolutional neural network with the retrieval-augmented explanation generation in a web-based platform. The system is designed to perform two related classification tasks: 5-class astronomical object recognition with SpaceNet and galaxy morphology classification with Galaxy Zoo. The overall design goes beyond the traditional image classification, which only provides a predicted label and a confidence score, by linking the prediction to the relevant astronomical knowledge that can be presented at various levels of technical complexity.

Three CNN architectures trained on ImageNet were fine-tuned and tested using a consistent experimental protocol.Three CNN architectures trained on ImageNet were fine-tuned and tested following a common experimental procedure. EfficientNet-B0 had the highest test accuracy of 83.66% on 2,019 test images for the five-class SpaceNet task, followed by ResNet50 with 83.51% and ResNet18 with 83.01%. ResNet18 performed at 86.86% test accuracy, EfficientNet-B0 at 86.25% and ResNet50 at 85.95% on 9,237 test images for the Galaxy Zoo morphology task (elliptical and spiral galaxies). The results show that the three architectures achieved similar predictive performance under the experimental conditions, and that the test accuracy did not improve with increasing network depth.

The differences in predictive performance were relatively small, so the final choice of model was made based on the overall balance between classification performance, computational requirements, model size and generalisation behaviour, not just test accuracy. EfficientNet-B0 was chosen as the backbone for the production due to its competitive classification performance with significantly reduced model storage and computation compared to ResNet50. It deployed around 16 MB of checkpoints and had a smaller generalisation gap on the larger Galaxy Zoo dataset as well. The features of EfficientNet-B0 were well suited for the interactive web application, where inference efficiency and responsiveness are crucial.

The error analysis also showed that the classifiers' weaknesses were not evenly dis-

tributed among the datasets but were focused in specific classes that are visually ambiguous. In SpaceNet, one of the most significant confusion patterns was star and constellation, and confusion about the star class was a significant part of the errors observed. The highest F1 score was obtained for the SpaceNet category planet (0.92), while the lowest was for the SpaceNet category constellation (0.73). In Galaxy Zoo, the classification accuracy was lower for elliptical galaxies, with EfficientNet-B0 achieving a recall of 0.83 for elliptical galaxies and 0.89 for spiral galaxies. The results show that the class imbalance and overlapping visual features are still significant constraints regardless of the specific CNN architecture chosen.

An important design principle of AdAstra is that the reliability of an explanation is fundamentally dependent on the classification on which it is based. The retrieval-augmented explanation stage does not make its own prediction of the correctness of a CNN prediction. Rather, it fetches knowledge related to the predicted class and constructs an explanation based on that knowledge. Thus, a misclassification can still lead to a coherent and scientifically plausible explanation of the misclassified class. This behaviour is one of the most significant restrictions found in the development of a system.

The system has several reliability mechanisms around the classification and explanation pipeline to minimize the risk of uncertain predictions being presented as unquestionable results. Ambiguous classifications are identified using confidence and margin-based uncertainty rules, and a CLIP-based gates check whether uploaded image looks astronomical. The language-generation part is asked to explain the retrieved and numbered knowledge passages, and generated citations are then verified to enhance traceability. These mechanisms are designed to minimize unsupported or misleading outputs, but they are not meant to ensure that the original CNN classification is accurate.

The web-based implementation showed that the entire classification and explanation process could be implemented in an interactive application. The total web response time in the recorded example was about 8.8 seconds, with about 0.28 seconds being spent in the image-classification stages. This suggests that the overall response time was mainly due to the language-model generation, not CNN inference, which further validates the use of a lightweight classification backbone for deployment.

The study achieved the following goals: It successfully created an automated CNN-based astronomical image-classification system, it performed a controlled comparison of three candidate architectures, it evaluated the models using accuracy, precision, recall, F1-score, and confusion matrices, it integrated retrieval-based explanation generation, and it analysed representative correct and incorrect predictions. The original SpaceNet task consisted of 8 classes, but the asteroid, black hole and comet classes had much fewer samples available for use than the other classes, so the final task is based on 5 classes. The final model-selection strategy was also adapted to take into account deployment efficiency and generalisation in addition to predictive accuracy.

However, there are some restrictions that need to be noted. The CNN architectures

were trained only once with the final experimental setup, which restricted the quantification of the run-to-run variation. A completely independent validation partition was not used, predictive calibration was not formally assessed, and the explanation system was not assessed using a labelled quantitative evaluation, but rather through representative examples. Therefore, the numerical results presented here should not be taken as a general performance of the algorithm for all astronomical surveys, but should be considered in the context of the datasets, training protocol and computing conditions employed in this study.

6.2 Contributions to the Field

6.2.1 Controlled Comparison of CNN Architectures

This thesis makes a contribution by comparing the three CNN architectures with different computational characteristics: ResNet18, ResNet50, and EfficientNet-B0 in a controlled manner. The models were tested on two complementary astronomical classification tasks with similar data preparation, input resolution, evaluation metrics and training conditions. This allowed the study to investigate if there was a practical benefit to the increased network depth for the chosen datasets.

The results indicated that the three architectures performed similarly on the test set, despite the significant differences in model complexity. The results of this study did not show a clear generalisation advantage for the deeper ResNet50 under the experimental conditions used. EfficientNet-B0 thus offered a good engineering compromise between prediction accuracy and deployment efficiency. The comparison highlights the need to take into account the computational cost and generalisation of the model, in addition to the accuracy, when choosing a model for an operational machine-learning system.

6.2.2 Classification with Reliability-Aware Safeguards

The second contribution is the creation of an image-analysis pipeline that takes into account the conditions under which the classification results may be unreliable, rather than assuming that all model predictions are equally reliable. The system integrates astronomical object classification and galaxy morphology classification, and includes mechanisms for determining potentially ambiguous or inappropriate inputs.

The confidence and prediction-margin criteria explicitly mark predictions that are ambiguous, and the CLIP-based validation component offers an extra semantic mechanism to differentiate astronomical from potentially unrelated inputs. These protections are especially important because if a closed-set classifier will always assign one pf it's known classes, even when the image matches none of them.

The system thus shows a practical way to add reliability checks to an astronomical CNN classifier, instead of using the CNN as a standalone prediction tool. The analysis of confident, uncertain, incorrect, and rejected cases also reveals the limitations of the system, rather than hiding them behind a single prediction score.

6.2.3 Retrieval-Augmented, Source-Grounded Explanation

The astronomical image classification and retrieval-augmented explanation is one of the major contributions of AdAstra. The final knowledge base is not only the previous prototype based on keywords, but also a dense retrieval collection with about 6,222 chunks indexed from 113 astronomy-related Wikipedia articles. Contextual evidence is used to explain the text at the beginner, intermediate, and advanced reading levels. The numbered source references are retained so that the system can match the explanations with the knowledge retrieved.

This design is a solution to a practical problem of the conventional image classifiers. A CNN prediction and confidence score tells the user what the network predicts, but does not provide much context to a non-specialist user. AdAstra adds to that prediction with pertinent domain knowledge, enabling the user to learn the properties of the predicted astronomical object or morphology.

The explanation mechanism should not, however, be interpreted as an insight into the internal visual reasoning of the CNN. It provides a knowledge-based explanation of the predicted category, but not the actual image features that led the CNN to make its prediction. This distinction is important because a source-supported explanation may be a description of an incorrectly predicted category.

6.2.4 Development of an Integrated Web Platform

AdAstra was more than an experimental notebook series. It was a prototype website. The site included image classification, confidence indicators, retrieval-augmented explanations, user accounts saved analysis history, a question and answer system focused on astronomy, an illustrated sky explorer and a Moon-phase feature. This prototype shows that combining CNN inference, knowledge retrieval and language model-based explanations can be done within an educational interface that is easy to use. It was built using relatively low-cost development and deployment resources.

This combination is important for the study's goals. The research gap wasn't about creating an accurate classifier. The real aim was to connect automated recognition with an interpretation layer. This allows users to interact with the results, not just receive a class label. The web version proves that this unified approach—classification and explanation together is feasible.

6.2.5 Transparent Reporting of Limitations and Negative Outcomes

The study also makes a contribution by explicitly documenting decisions made in the pre-processing, limitations of the model, and unsuccessful or ambiguous outcomes. For instance, 1,376 corrupted galaxy images were removed during the preparation of SpaceNet, and the reduced number of images for the asteroid, black hole, and comet classes resulted in their exclusion from the final five-class task. The experiments also revealed situations where the validation performance remained constant but the training accuracy kept improving, suggesting overfitting.

It is important to report these outcomes for reproducibility and to correctly interpret the reported classification results. The thesis does not only show successful examples, but also limitations in the data, errors that can be made by the models, uncertainty in predictions, and situations where the explanation system can reinforce an incorrect classification. The observations serve as a basis for future research on more powerful astronomical image-analysis systems.

6.3 Recommendations for Future Work

6.3.1 Strengthening Experimental Evaluation

The limitations of the present evaluation procedure should be addressed first in future experiments. To assess the variation in performance across different runs, each architecture should be trained on several random seeds. If enough computational resources are available, however, k-fold cross-validation might offer a more reliable estimate of model behaviour.

SpaceNet should also be split into training, validation, and test partitions, with the test partition being completely separated from the training and validation partitions until the final evaluation. Test predictions should be stored at the individual-image level to allow for paired statistical comparison of architectures. Then, using techniques like McNemar's test and bootstrap confidence intervals, it would be possible to decide if the observed differences are due to sampling variation or systematic improvement of the model. Claims of cross-survey generalisation or robustness to domain shift would also require external evaluation based on images from other astronomical surveys.

6.3.2 Improving Performance on Difficult Classes

Specific attention should be paid to classes that caused most of the errors in the current model in the future model development. Confusion between images of star, constellation, galaxy and nebula is particularly problematic for SpaceNet, as is the recognition of elliptical galaxies for Galaxy Zoo.

Potential solutions such as class-weighted loss functions, focal loss, oversampling the minority classes, and a careful review of the ambiguous or incorrect labels. Additional data augmentation and increased image resolution could also be considered. In the Galaxy Zoo task, instead of assigning a single hard category, the vote fractions of the volunteers could be retained as soft probabilistic classification, which would allow to keep some of the information which is related to the ambiguity of the morphology. If a significant number of more reliable training examples become available for asteroid, black hole and comet classification, the original SpaceNet classification problem might also be revisited.

6.3.3 Calibration and Uncertainty Estimation

The current system's confidence and margin thresholds were manually determined. Future research should therefore examine the accuracy of the predicted probabili-

ties in terms of the empirical correctness. Metrics such as expected calibration error could be used to quantify calibration, while techniques such as temperature scaling could be investigated to improve probability reliability.

The confidence and abstention thresholds should then be determined based on held-out validation data, not on manually set thresholds. The astronomy validation part of CLIP should also be tested against a labelled benchmark with astronomical and non-astronomical images. Examples of borderline cases, e.g., telescopes, observatories, illustrations, night sky photos, and astrophotography with terrestrial foreground objects would be especially helpful in quantifying false rejection and missed out-of-domain detection.

6.3.4 Formal Evaluation of the Explanation System

The retrieval-augmented explanation component was mainly evaluated using representative examples. Future work should include a structured evaluation framework that includes retrieval and generation quality.

The evaluation should determine if retrieved passages are relevant to the predicted class or user query, if individual statements generated are actually supported by the retrieved passages, if the astronomical information presented at each reading level is correct, and how the system performs when it does not have enough information in its knowledge base to answer a request. A user study with students or general astronomy learners would also yield evidence on whether there are meaningful differences between beginner, intermediate and advanced explanations that are understood and whether the source attribution and uncertainty warnings are appropriately influencing user trust.

6.3.5 Expansion of the Knowledge Base

The current dense retrieval collection is created from 113 Wikipedia articles about astronomy. This knowledge base is suitable for the retrieval-augmented explanation pipeline, but is not yet large enough for more complex scientific applications. We think adding peer-reviewed research papers, astronomy textbooks, survey documentation and authoritative scientific archives to versions would be very helpful. Peer-reviewed research papers, astronomy textbooks, survey documentation and authoritative scientific archives would give the work a foundation. Adding these resources will provide readers with comprehensive information. Another enhancement to retrieval would be to create queries using more than the last class label. For instance, if the classifier is not sure about the class, the information from the top two or three probable classes can be added, and semantic information from the uploaded image can be added to the text query. This could enable the explanation stage to express uncertainty more accurately than just a single categorical prediction. If the language-model service is not available or is too slow, then a fallback explanation mechanism should also be considered.

6.3.6 More Advanced and Visually Explainable Models

Future research could involve comparing the existing CNN architectures to vision transformers and self-supervised models trained specifically on astronomical images. These methods could provide better representations for classes that have only slight differences in morphology, but would require similar computational costs and data efficiency as the CNN models.

In addition to the retrieval-based textual explanation, the visual explainability methods like Grad-CAM could be used. Grad-CAM would enable the system to identify areas of the image that were most significant for a CNN prediction, which would be a different aspect of interpretability than RAG. In this setup, visual explanation might be used to show the focus of the classifier, and retrieval-augmented explanation might be used to explain the meaning of the predicted astronomical category. Additionally, ensembles of the evaluated architectures could be explored, as disagreement between models could be another useful signal for uncertainty estimation.

6.3.7 Extension of the Web Platform

Additional file formats that are common in professional astronomy could also be included on the platform. Other potential features that future versions could have are FITS and multiband support, batch processing of images, look-up of catalogue objects by their coordinates, and connection with other astronomical archives. Future development may include interface and explanation translation, better mobile support, and batch-upload for classroom activities for educational accessibility. The analytical part of AdAstra could also be tied to the real world, by enabling users to explore astronomical objects visible from their current location. The extensions would take the platform from its current experimental and educational prototype status to a more general astronomical analysis environment.

6.4 Concluding Remarks

This project demonstrates the possibility of building an efficient CNN-based astronomical image classifier that incorporates safeguards of uncertainty, and that provides retrieval-augmented, source-grounded explanations through an accessible web platform. The experiments demonstrated that increasing the depth of the model did not confer a clear benefit in terms of predictive accuracy under the tested conditions, and so EfficientNet-B0 was chosen as the deployment backbone based on its predictive performance, generalisation behaviour, model size, and computational efficiency.

At the same time, the results demonstrate that classification accuracy alone is insufficient to support an interpretable astronomical image-analysis system: visually-similar and imbalanced classes can be significant sources of error, confidence scores do not necessarily imply correctness, and a retrieval-augmented explanation can offer a convincing explanation of a prediction even if the underlying classifier is wrong. These results suggest that conveying uncertainty is essential, and that distinguishing between providing an explanation and verifying a prediction should be preserved.

The key contribution of AdAstra therefore, is the combination of three different aspects that are often considered in isolation: comparative astronomical image classification, deployment-oriented model choice, and knowledge-grounded natural-language explanation. Together, these enable the system to provide end-users with not just an astronomical prediction and confidence score, but with additional information that can be traced back to retrieved sources. Under the constraints of the current data sets and evaluation procedure, this project builds a foundation that enables further work on more robust, calibrated and scientifically-founded, interpretable astronomical image-analysis systems.

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